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PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The agricultural sector continues to function as a fundamental economic safety net for rural households, acting as a primary engine for livelihood and economic sustenance throughout the archipelago. Within this landscape, livestock farming emerges as a vital component for augmenting household incomes and providing localized financial resilience. In the municipality of Oton, Iloilo, cattle raising operates as an essential pillar of the rural economy, contributing to the community's supplemental revenue streams and smallholder asset security. For local raisers, cattle assets represent a key mechanism for risk management; they function as high-value, long-term investments that provide an essential economic cushion during financial shocks. The escalating market demand for animal products underscores the urgent necessity for optimized management frameworks to ensure the long-term sustainability and profitability of these localized agricultural operations.

Data from the Philippine Statistics Authority (PSA, 2023) indicates that the livestock sub-sector contains highly specialized segments, though cattle production remains a structural component often overshadowed by dominant industries such as swine and poultry. Specialized breeds, including Brahman, Holstein, and indigenous crossbreeds, are managed locally due to their biological adaptability and reliable asset value. These assets are utilized for localized breeding, meat supply, and small-scale dairy production, offering raisers a viable pathway toward improved household economic status. However, maximizing these biological yields necessitates rigorous surveillance of reproductive cycles, pathological conditions, immunization timelines, and comprehensive herd synchronization.

Despite the immense economic value generated by the cattle industry in Oton, Iloilo, current operational paradigms continue to rely on antiquated manual recording and monitoring methodologies. The existing livestock insemination and breeding workflows are largely facilitated by field technicians under the Municipal Agriculture Office. These personnel manage a complex array of responsibilities vital to herd productivity, including the surveillance of artificial insemination windows, estrus cycle tracking, rebreeding coordination for failed cycles, pregnancy monitoring, vaccination logistics, and the documentation of mortality and health metrics.

Currently, these critical datasets are manually encoded into paper-based logbooks and physical ledgers. This traditional approach, while historically persistent, introduces severe operational bottlenecks that compromise the integrity and efficiency of municipal livestock management. Physical records are highly susceptible to environmental degradation from moisture, thermal exposure, fire, and biological pests. Furthermore, manual systems are prone to misplacement and physical loss, while the retrieval and updating of records are characterized by significant labor inefficiencies. As livestock inventories expand, the cumulative volume of paper records makes the retrieval of historical breeding patterns and the generation of administrative reports an increasingly arduous and error-prone endeavor.

The Food and Agriculture Organization (FAO, 2021) asserts that digital information management systems offer a superior and more resilient framework for agricultural data, enhancing accessibility and long-term preservation. In contrast to manual methods, digital infrastructures allow for the secure storage of high-density datasets while enabling real-time data retrieval and updates. Implementing such technological capabilities is essential for enhancing the operational throughput of local agricultural agencies and specialized livestock programs.

A secondary challenge involves the persistent communication gap and lack of synchronized information transit between producers and agricultural personnel. A significant portion of the livestock raisers in rural sectors belong to older demographics with limited digital literacy, relying on legacy communication channels such as personal visits, phone calls, or SMS. This decentralization often results in the delayed reporting of critical events, such as parturition, disease outbreaks, animal mortality, and vaccination needs. Consequently, livestock technicians may only receive field data days or weeks after an occurrence, hindering the synchronization of municipal databases.

These communication barriers introduce profound inaccuracies into the livestock management lifecycle. Technicians are frequently forced to utilize approximations for insemination dates, pregnancy durations, and estrus cycles due to fragmented field reporting. These estimations degrade the quality of official records, potentially reducing reproductive success rates and skewing municipal statistics. Moreover, delayed reporting impedes the deployment of time-sensitive veterinary interventions and epidemiological surveillance, heightening the risk of contagion and financial loss within farming clusters.

The persistent reliance on manual documentation and fragmented communication ultimately degrades the analytical quality of reports submitted by the Municipal Agriculture Office to provincial oversight bodies. High-fidelity livestock data is paramount for strategic planning, resource distribution, disease monitoring, and evidence-based policy formulation. As emphasized by the World Bank (2019), the integration of digital agricultural technologies is essential for improving sector productivity through enhanced data accuracy and streamlined stakeholder coordination.

The rapid evolution of information and communication technology provides a transformative opportunity to address these systematic vulnerabilities. Web and mobile-based infrastructures offer innovative frameworks for managing agricultural data and facilitating real-time stakeholder interaction. These platforms enable automated notifications, centralized data repositories, and precision monitoring of field activities. By automating these workflows, digital systems reduce administrative burdens, minimize human error, and ensure the consistent availability of actionable information for municipal livestock management.

In response to these challenges, We propose the development of **BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System**. This platform is engineered to modernize existing manual workflows by providing a centralized, user-friendly interface for technicians, raisers, and administrators. By leveraging distributed technologies, BreedSmart seeks to automate breeding cycle computations using breed-specific reproductive logic, facilitating precise surveillance of insemination schedules, expected delivery dates, and overall reproductive efficiency.

Beyond reproductive monitoring, the system implements a robust digital framework for comprehensive health surveillance. Features including vaccination tracking, disease reporting, clinical assessments, and mortality logging enable the maintenance of an organized digital registry. The cloud-synchronized database ensures that these records remain secure, auditable, and readily accessible for administrative reporting and data-driven decision-making.

The system further aims to bridge the communication gap between producers and the Municipal Agriculture Office. Through the mobile client, farmers can electronically submit service requests, report calving events, and notify technicians of health concerns while receiving automated alerts regarding breeding milestones and immunization windows. These features are designed to eliminate communication latency and improve field data fidelity. Simultaneously, technicians and office administrators gain real-time access to synchronized records, facilitating more effective service delivery and operational oversight.

Additionally, BreedSmart is anticipated to optimize reporting efficiency within the municipal agricultural sector. By generating standardized digital datasets, the platform streamlines the preparation of analytical reports for provincial and national agencies. This capability enhances the reliability of regional livestock statistics, supports agricultural policy development, and ensures the efficient implementation of government programs targeted at local farming communities.

Ultimately, the realization of **BreedSmart** marks a pivotal transition toward the digitalization of livestock management in Oton, Iloilo. By replacing vulnerable manual processes with a synchronized digital ecosystem, the study aims to strengthen record integrity, enhance stakeholder coordination, and promote evidence-based agricultural management. These systemic improvements are intended to support the long-term competitiveness and sustainability of the local cattle industry, aligning with broader objectives of regional food security and agricultural modernization.

1.2 Statement of the Problem

As technology continues to evolve, many government agencies are adopting digital solutions to improve the efficiency and accuracy of their operations. However, the livestock monitoring and artificial insemination services of the Municipal Agriculture Office still rely primarily on manual processes and paper-based records for managing livestock information, breeding schedules, animal health treatment records, pregnancy monitoring, and other related data.

The use of manual documentation makes livestock records susceptible to loss, damage, and misplacement. Retrieving and updating records also requires considerable time and effort, especially when historical information is needed during field operations. In addition, livestock-related reports from farmers are commonly communicated through phone calls or personal visits, since most farmers proceeding technicians are commonly senior citizens making it difficult to maintain a centralized and updated record of livestock activities.

Agricultural technicians also manually monitor breeding schedules and reproductive cycles for different cattle and carabao breeds. As the number of livestock records increases, managing these schedules becomes more challenging and may result in delays or inconsistencies in recording and reporting information. Furthermore, records collected during field activities are consolidated manually before being reflected in municipal reports, making it difficult to maintain updated information for monitoring and decision-making.

These existing challenges affect the efficiency of livestock monitoring, record management, communication between farmers and agricultural technicians, and the preparation of accurate municipal livestock reports.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton, Iloilo. Specifically, the study aims to improve the efficiency and accuracy of livestock record management, automate cattle breeding cycle monitoring, and strengthen communication between farmers, livestock technicians, and the Municipal Agriculture Office through the implementation of a centralized digital platform.

1.3.2 Specific Objectives

To achieve this general objective, the study pursues several specific objectives, each of which contributes to the overall efficiency, usability, and reliability of the proposed livestock management system.

1. **Analyze the current challenges faced by livestock technicians and cattle farmers regarding manual livestock monitoring, breeding management, and record-keeping processes.**

This objective ensures that the proposed system is grounded in the actual operational realities experienced within the Municipality of Oton. By examining the existing manual practices used by farmers and livestock technicians, the study identifies key problems such as data vulnerability, delayed reporting, communication gaps, and inaccurate livestock records. Understanding these challenges provides the foundation for designing a system that addresses local agricultural needs rather than implementing generalized technological solutions that may not fit the municipality's operational environment.

2. **Design a user-friendly and responsive web and mobile interface that allows farmers to request insemination services, report animal diseases, and monitor livestock activities efficiently.**

User interface (UI) and user experience (UX) design are essential factors influencing technology adoption, especially among rural users with varying levels of digital literacy. A responsive and accessible interface ensures that both farmers and agricultural personnel can effectively use the system across different devices, particularly smartphones, which are the most accessible digital tools in rural communities. By providing simplified navigation and organized features, the system enables users to efficiently manage livestock-related activities without requiring advanced technical knowledge.

3. **Develop an automated breeding cycle tracking module capable of calculating reheating schedules, insemination intervals, pregnancy timelines, and expected calf drop dates based on breed-specific reproductive logic.**

Livestock breeding management involves complex reproductive timelines that vary according to cattle breed characteristics. This objective addresses the limitations of manual calculations currently performed by livestock technicians. By automating breeding cycle tracking, the system minimizes human error and improves monitoring accuracy for breeds such as Brahman, Holstein, and local Bisaya cattle. Automated scheduling also enhances breeding efficiency by ensuring timely insemination procedures and accurate reproductive monitoring.

4. **Implement a real-time notification and reminder system that alerts farmers and livestock technicians regarding veterinary schedules, vaccination activities, breeding milestones, and animal health concerns.**

Timely communication is essential in livestock management because breeding schedules, disease reporting, and vaccination programs are highly time-sensitive. This objective introduces automation features that provide instant notifications and reminders to users regarding important livestock events. Through automated alerts, the system reduces missed schedules, improves response times, and strengthens coordination between farmers and agricultural personnel. The notification system also minimizes the risk of delayed reporting and enhances overall livestock monitoring efficiency.

5. **Create a centralized administrative dashboard for the Municipal Agriculture Office to monitor livestock records, manage farmer requests, and generate accurate reports for submission to the Provincial Agriculture Office.**

Administrative dashboards are essential tools for efficient agricultural management and decision-making. This objective focuses on providing the Municipal Agriculture Office with a centralized platform capable of organizing livestock data, monitoring field activities, and generating official reports. Through digital dashboards, administrators can easily access livestock statistics, breeding records, disease reports, and vaccination histories. The dashboard also improves data synchronization between field operations and administrative offices, reducing inconsistencies and improving report reliability

6. **Store and manage comprehensive livestock records, including animal profiles, breeding histories, vaccination records, disease monitoring data, and mortality reports using a secure database management system.**

Effective data management is critical for ensuring the reliability, accessibility, and security of livestock information. This objective emphasizes the development of a centralized database capable of securely storing and organizing large volumes of agricultural data. By maintaining digital livestock records, the system minimizes the risk of data loss associated with paper-based documentation. Secure database practices also protect sensitive agricultural information while improving long-term record accessibility for technicians and administrators.

7. **Improve communication and coordination between farmers, livestock technicians, and agricultural administrators through a centralized digital platform.**

Communication gaps remain one of the major challenges in the current livestock management process within the municipality. This objective seeks to establish a unified communication system where farmers can directly submit requests, report animal conditions, and receive updates from agricultural personnel in real-time. Improved coordination reduces delays in service delivery, enhances livestock monitoring, and strengthens the relationship between farmers and the Municipal Agriculture Office.

8. **Test and evaluate the system’s functionality, usability, reliability, responsiveness, and data accuracy within an actual agricultural management environment using the ISO 25010 software quality standard.**

System evaluation is essential to determine whether the proposed platform effectively addresses the identified problems within the livestock sector. This objective focuses on assessing the system's performance against the ISO 25010 standard to ensure high-quality software delivery and operational efficiency. Through evaluations involving livestock technicians and farmers, the study determines whether BreedSmart improves operational efficiency, reduces manual recording errors, and enhances livestock management processes within the Municipality of Oton, Iloilo.

1.4 Significance of the Study

The proposed system is expected to provide significant benefits to various stakeholders involved in livestock management, agricultural operations, academic development, and future technological research. By introducing a centralized web and mobile-based livestock insemination and health management system, the study aims to improve operational efficiency, communication, and data reliability within the Municipality of Oton, Iloilo. The following subsections discuss the significance of the study to specific groups and sectors.

Farmers

The proposed BreedSmart system will greatly benefit cattle farmers by providing a direct, accessible platform for requesting livestock-related services and monitoring animal health activities. Through the mobile application, farmers can submit insemination requests, report animal diseases, monitor breeding schedules, and receive automated reminders regarding vaccination schedules, reheating cycles, and expected calf drop dates.

This functionality reduces the dependence on traditional communication methods such as phone calls, handwritten records, and physical visits to the Municipal Agriculture Office. By improving access to livestock information and agricultural services, the system enables farmers to make more informed decisions regarding breeding management and animal health monitoring. Furthermore, automated reminders help farmers avoid missed schedules, thereby improving breeding efficiency, reducing livestock health risks, and increasing overall productivity within their farming operations.

Livestock Technicians

The study will also provide substantial benefits to livestock technicians responsible for monitoring breeding activities and maintaining livestock records within the municipality. Currently, technicians rely heavily on manual calculations, handwritten logbooks, and fragmented communication methods, which consume considerable time and increase the risk of human error.

Through BreedSmart, livestock technicians will gain access to automated breeding cycle calculations, centralized livestock records, and real-time communication tools. The system minimizes the need for manual estimation of reheating cycles, insemination schedules, and calf drop dates by automatically generating breeding timelines based on breed-specific reproductive logic. This automation reduces administrative burdens, improves monitoring accuracy, and allows technicians to focus more on delivering agricultural services rather than managing paperwork.

Additionally, centralized digital records improve data accessibility and retrieval, enabling technicians to efficiently track livestock histories, vaccination records, mortality reports, and disease monitoring activities. The system also enhances coordination between technicians and farmers by providing faster and more organized communication channels.

Oton Agriculture Office

The Municipal Agriculture Office of Oton will significantly benefit from implementing a centralized digital livestock management system. One of the primary challenges currently faced by the office involves inconsistent livestock records, delayed reporting, and fragmented field data. BreedSmart addresses these issues by providing synchronized and organized digital records accessible to authorized agricultural personnel.

The administrative dashboard enables the Agriculture Office to monitor livestock activities, manage farmer requests, and generate accurate reports for submission to the Provincial Agriculture Office. Through centralized data management, the office can improve the reliability, transparency, and accuracy of agricultural records. This capability strengthens decision-making processes, enhances livestock monitoring programs, and supports data-driven agricultural planning within the municipality.

Furthermore, the digitalization of livestock management operations contributes to the modernization of local agricultural services and aligns with broader government initiatives promoting digital transformation within rural communities.

BSIT Students

The study is also significant to Bachelor of Science in Information Technology (BSIT) students, particularly those interested in software development, database management, mobile applications, and agricultural technologies. BreedSmart serves as a practical application of information technology concepts within a real-world agricultural setting.

The development of the system demonstrates the integration of web development, mobile programming, database management systems, automation logic, and user interface design in solving actual community problems. Through this study, BSIT students may gain insights into the role of technology in agricultural modernization and digital transformation. The project also serves as a reference for future capstone projects involving smart systems, automation, and government service applications.

Additionally, the study highlights the importance of developing socially relevant technologies that address operational challenges within local communities, encouraging students to create innovative solutions that contribute to public service improvement and rural development.

Future Researchers

Future researchers may benefit from this study as a reference for further investigations into digital agriculture, livestock management systems, and agricultural technological innovations. The findings, methodologies, system architecture, and implementation strategies presented in this research may serve as valuable resources for scholars conducting studies involving agricultural automation, mobile information systems, database integration, and smart farming technologies.

The study may also provide a foundation for future enhancements such as integrating Internet of Things (IoT) devices, artificial intelligence-based livestock monitoring, predictive analytics, geographic information systems (GIS), and cloud-based agricultural management platforms. Researchers may expand upon the proposed system to develop more advanced technological solutions that further improve livestock productivity, disease monitoring, and agricultural sustainability.

Ultimately, this study contributes to the growing body of knowledge regarding digital transformation within local government agricultural units and demonstrates the potential of information technology in improving agricultural operations, livestock management efficiency, and rural community development.

1.5 Theoretical Framework

System Development Life Cycle (SDLC)

The System Development Life Cycle (SDLC) is a structured process used in the development of information systems, which involves stages such as planning, analysis, design, development, testing, implementation, and maintenance. This model ensures that systems are developed in an organized and systematic way, reducing errors and improving overall system quality.

In the context of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System, SDLC serves as the main development framework. The study follows each phase to ensure that the system is properly designed and aligned with the actual needs of livestock technicians and cattle farmers in the Municipality of Oton, Iloilo. During the planning and analysis phase, the study identifies problems such as manual record-keeping, inaccurate reporting, and communication gaps. The design and development phases focus on building the system features such as livestock tracking, breed-specific calculations, notification systems, and centralized record management.

Through SDLC, the system is developed in a structured and reliable manner, ensuring that the final output is functional, efficient, and suitable for real-world agricultural use.

Agile Methodology

Agile Methodology is a software development approach that focuses on iterative development, continuous feedback, and flexibility throughout the development process. It allows developers to improve and adjust the system based on user needs and testing results.

In this study, Agile is applied to ensure that BreedSmart is continuously improved during development. The system is built in small increments, allowing features such as insemination tracking, reheating cycle computation, and livestock record management to be tested and refined step by step. Feedback from potential users such as livestock technicians and farmers is considered during each development cycle to ensure that the system meets their actual needs.

Agile is important in this study because livestock management requirements may change depending on field conditions, making flexibility and adaptability essential in system development.

Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), developed by Davis (1989), explains how users decide to accept and use a new technology based on perceived usefulness and perceived ease of use. Perceived usefulness refers to how much a user believes that the system improves their work, while perceived ease of use refers to how easy the system is to learn and operate.

In this study, TAM is relevant because the success of BreedSmart depends on whether farmers, livestock technicians, and agricultural staff are willing to use the system. If users find that the system helps them manage livestock records faster, reduces manual errors, and improves communication, they are more likely to adopt it. Likewise, if the system is simple, easy to navigate, and accessible through mobile devices, users will find it easier to use in their daily livestock management activities.

To support TAM, the system is designed with a simple interface, clear navigation, and mobile accessibility to ensure ease of use and practical usefulness in real farming environments.

Information Systems Success Model

The Information Systems Success Model by DeLone and McLean evaluates system success based on system quality, information quality, service quality, use, user satisfaction, and net benefits. This model is widely used in assessing the effectiveness of information systems in real-world applications.

In this study, the model is used to evaluate the effectiveness of BreedSmart after development. System quality is measured through performance, usability, and reliability of the platform. Information quality is assessed through the accuracy of livestock records, breeding calculations, and generated reports. User satisfaction is evaluated based on feedback from farmers and livestock technicians after system use.

This framework ensures that the developed system is not only functional but also effective in improving livestock management processes in the Municipality of Oton, Iloilo.

1.6 Conceptual Framework

The conceptual framework of this study illustrates the overall structure of how the proposed system, **BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System**, is developed and how it transforms input data into a functional output. The study follows the Input–Process–Output (IPO) model, which is commonly used in system development research to clearly present the flow of information and system operation.

Input

The input phase includes all data and information gathered for the development of the system. This consists of the current manual livestock management practices in the

Municipality of Oton, Iloilo, including paper-based records, breeding logs, vaccination records, and livestock monitoring reports. It also includes user requirements collected from livestock technicians and cattle farmers through interviews, observations, and surveys. These inputs help identify common problems such as inaccurate records, delayed communication, and difficulties in tracking breed-specific breeding cycles.

Process

The process phase involves the actual development of the system using the System Development Life Cycle (SDLC) and Agile methodology. During this phase, the system is designed, developed, tested, and refined based on user requirements. Key processes include designing the system interface, developing the database, implementing breed-specific algorithms, and integrating notification and communication features.

The system also processes livestock data such as insemination records, reheating cycles, pregnancy timelines, and calf drop schedules. Through automated computation and real-time data processing, the system improves the accuracy and efficiency of livestock monitoring and management.

Output

The output of the study is a fully functional **Web and Mobile-Based Livestock Insemination and Health Management System (BreedSmart)**. The system provides centralized livestock record management, automated breeding cycle tracking, real-time communication between farmers and livestock technicians, and accurate reporting for the Municipal Agriculture Office.

The final output aims to reduce manual errors, improve data accuracy, enhance communication, and modernize livestock management practices within the Municipality of Oton, Iloilo.

1.7 Scope and Delimitation

1.7.1 Scope

The primary focus of this study is the comprehensive design, architectural development, field implementation, and software quality evaluation of **BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System**. The operational boundaries, processing modules, and data transit vectors of the software application are systematically organized into distinct functional layers explicitly assigned to three authorized user roles: Farmers (Clients), Livestock Technicians, and Administrators (Municipal Agriculture Office Personnel). Each subsystem is custom-tailored to address specific manual bottlenecks identified within the local government unit's current agricultural workflow.

The Farmer/Client Module is an interface engineered specifically to provide local livestock raisers with a highly responsive, user-friendly mobile application client designed to operate efficiently under volatile cellular network conditions in rural agricultural areas. Within this client module, the processing capabilities, user forms, and data transmission parameters are tightly limited to the following interactions:

- **User Profile and Account Management Subsystem:** Self-registration pipelines, security credential coordination, and digital profile updates, allowing farmers to safely manage their personal contact data, coordinate secure password updates, and pin their physical farm or barangay locations within designated municipal borders.

- **Animal Asset Digital Registry and Identity Module:** Interactive profile creation and inventory management for individual livestock assets, enabling users to digitally log unique physical ear-tag values, specific breed profiles, biological gender, approximate birthdates, body condition scores, current status labels, and maternal lineage associations via parent record mapping (motherId) to preserve genetic tracking.

- **Veterinary Service Request Portal and Ticketing Interface:** Electronic submission of real-time field service requests for artificial insemination or urgent health checks, permitting farmers to describe localized symptoms using text strings, specify preferred calendar windows for field visits, define problem urgency levels, and attach supporting image uploads for remote appraisal by agricultural technicians prior to deployment.

- **Read-Only Information Retrieval Dashboard:** Direct, read-only dashboard access to historical individual health treatment ledgers, vaccination logs, deworming history, medical notes, automated breeding cycle notifications, upcoming appointment alerts, and scheduled veterinary reminders.

The Livestock Technician Module acts as a cross-platform mobile and web client designed for field personnel that prioritizes streamlined, rapid data entry to completely eliminate the vulnerabilities, physical degradation, and inaccuracies of manual, paper-based logbooks. The technical capabilities, input windows, and processing limits are scoped to the following actions:

- **Field Request Management and Dispatch Submodule:** Real-time mobile access, database-driven validation, status modification, and calendar scheduling approval of pending farmer insemination or veterinary service requests submitted from remote barangays.

- **Field Operations Data Logging and Treatment Engine:** Direct digital entry engines for recording artificial insemination procedures, outcome metrics like successful or failed fertilization attempts, precise sire breeds used, sire code tracking, specific vaccination dosages, deworming dates, diagnoses, treatment notes, and field notes containing localized geospatial data and images captured during the visit.

- **Reproductive Surveillance and Lifecycle Monitoring Tool:** Updating active pregnancy diagnostic statuses following field check windows, maintaining reproductive logs, and recording physical calving events, including exact birth dates, the number of offspring delivered, sire details, and historical parity tracking for individual dams.

The Administrative Dashboard Module provides a web-based software platform designed for the central staff at the Municipal Agriculture Office that concentrates on large-scale data aggregation, multi-tier synchronization, and regulatory compliance oversight. Its structural boundaries, database queries, and view spaces cover the following operations:

- **User Identity Management and Role-Based Access Control Subsystem:** System-wide identity management using clerk authentication utilities, role-based routing controls, user validation status flags, secure profile verification systems, and auditable action logs tracking administrative actions across the platform.

- **Analytics Dashboard and Epidemiological Surveillance Engine:** Rendering centralized tracking charts, regional heatmaps, and graphical data dashboards that analyze breeding efficiency percentages, municipal case counts, active dispatch queues, asset counts, and disease distribution trends across various local sectors.

- **Automated Regulatory Report Generation and Data Export Engine:** Compiling active database records to generate UNIP-compliant administrative PDF and Excel data reports, providing accurate, synchronized livestock summaries for formal submission to the Provincial Agriculture Office.

The automated scheduling, lifecycle tracking, and notification layers of the **BreedSmart** engine are strictly governed by hardcoded, breed-specific reproductive calculations. The algorithmic boundaries are confined to managing data changes, calculating target intervals, and firing automated reminders for specific reproductive milestones. The reheating or estrus return calculation module runs an automated background tracking window precisely 18 to 24 days following an active insemination entry to prompt users to watch for signs of fertilization failure. Alongside this, the gestation timeline calculator automatically computes expected delivery dates using specific constants mapped to local livestock breed traits. To manage long-term breeding health, the voluntary waiting period calculator tracks post-calving biological recovery windows before an individual animal asset is marked as eligible for subsequent breeding procedures.

The technology stack selected for building this distributed environment is strictly limited to specific languages, frameworks, and environments. The web application dashboard utilizes React, Vite, JavaScript, and TypeScript for the frontend administration view. The mobile application client utilizes React Native, Expo, Tailwind CSS, and NativeWind to build a responsive interface. The backend architecture implements Node.js and Express.js for the backend server API services. All data schemas, records, and system states are maintained through MongoDB Atlas as the cloud-hosted NoSQL repository for all core data structures.

1.7.2 Delimitation

To maintain an attainable and realistic software engineering scope for an undergraduate capstone study, specific parameters and design restrictions are established to separate the project from commercial, hardware-intensive, or nationwide configurations.

Geographical Boundaries and Local Government Unit Jurisdictional Delimitations

The geographical and local government boundaries strictly localize this study within the municipal borders of Oton, Iloilo. All field evaluation trials, data collection methods, user testing cycles, and synchronization simulations will exclusively involve farmers, field technicians, and administrative officials residing or working within the active barangays of this specific municipality. The system is not designed to support cross-border agricultural logistics, multi-municipal database clustering, or animal data ownership verification from adjacent towns or provinces.

Biological Classification Limits and Animal Breed Core Concentration Boundaries

The biological core concentration optimizes the initial system core to prioritize data models, registries, and reproductive algorithms calibrated for the cattle industry or bovine livestock. This framework highlights commonly managed local breeds such as Brahman, Carabao, and local Bisaya crossbreeds. Features or data inputs for non-livestock animals, companion pets, or domestic poultry are explicitly excluded from this phase of the system. The system does not account for wild or swine, poultry, and goats management records, equine asset registries, or non-bovine commercial livestock tracking.

Linguistic Boundaries and Regional Dialectical Localization Delimitations

This system is strictly limited to Tagalog, English, and Hiligaynon linguistic support. Other languages are not supported, and the system does not account for further regional variations in grammar or localized idioms beyond these three specified languages.

Hardware Infrastructure Exclusions and IoT Architecture Delimitations

The hardware and IoT exclusions dictate that the **BreedSmart** platform is entirely software-reliant and depends completely on human-mediated data logging through web and mobile views. This study explicitly excludes the deployment, procurement, or integration of physical Internet of Things (IoT) sensors, wearable health monitors, GPS animal collars, automated biometric ear-tag readers, or automated drone surveillance networks. All identification practices rely entirely on reading alphanumeric tags manually clipped onto the animals and typing them into user forms.

Network Topology Constraints and Communication Infrastructure Delimitations

The network and infrastructure delimitations clarify that although the mobile client implements an offline caching architecture using AsyncStorage and TanStack Query to let field technicians save records in remote barangays with low signal, the application ultimately requires standard cellular networks (4G/5G) or Wi-Fi to sync local data queues with the cloud server. The development of alternative long-range network infrastructures, custom satellite interfaces, or localized mesh networks is outside the scope of this project.

Database Integration Boundaries and External API Interface Restrictions

The integration and API restrictions limit the database layer and reporting modules to function as a separate system for the Oton Municipal Agriculture Office. The architecture excludes direct API integrations with national livestock health platforms, national agricultural portals, or external provincial databases. All report handovers to higher provincial units must be processed manually by downloading and printing the system's generated PDF or Excel data files.

1.8 Definition of Terms

The following terms are operationally and conceptually defined to provide a clearer understanding of the study and the proposed system:

Insemination Tracking

Refers to the systematic digital documentation and management of reproductive activities, encompassing both artificial and natural breeding procedures. This process involves the comprehensive logging of insemination dates, pregnancy diagnostic results, and reproductive milestones to maintain longitudinal records of herd fertility.

Reheating Cycle

Denotes the biological return of a female bovine to estrus following an unsuccessful conception attempt. The system utilizes automated monitoring algorithms to detect this cycle, alerting stakeholders to facilitate prompt intervention and reduce the intervals between breeding attempts.

Calf Drop

Refers to the parturition event or the act of a cow giving birth to a calf. The system records the precise date and parameters of this event, ensuring high-fidelity data collection for population statistics and administrative reporting within the municipal agricultural framework.

Breed Logic

This refers to the algorithms inside the system. They calculate breeding schedules, how long pregnancies last, and when cattle reach maturity — all based on different breeds like Brahman, Holstein, or Bisaya. It's what makes the automated tracking work.

Livestock Technician

These are government staff who handle breeding, insemination, vaccines, disease checks, and make sure data is correct out in the field, specifically in Oton. They're the main people using and managing BreedSmart.

Precision Livestock Farming (PLF)

An agricultural management approach that integrates data analytics and automated monitoring systems to optimize animal health, reproductive efficiency, and farm productivity, shifting from manual observation to data-driven decision-making.

System Development Life Cycle (SDLC)

A structured process for developing information systems, involving phases such as planning, analysis, design, development, testing, and maintenance to ensure an organized and high-quality system delivery.

Cloud-Synchronized

Refers to the system's ability to maintain real-time data consistency across all user platforms. Any information—such as a new service request or health update—entered via the mobile client is immediately transmitted to and reflected in the central administrative dashboard, ensuring all stakeholders are viewing the same up-to-date data.

Offline-First / Offline Caching

A software design approach implemented in the BreedSmart mobile application that allows field technicians to continue recording data in remote barangays with poor or no cellular connectivity. The system stores these entries locally and automatically "syncs" them with the central database once a stable internet connection is re-established.

Role-Based Access Control (RBAC)

A security framework used in the system to restrict access based on the user's specific function (e.g., Farmer, Technician, or Admin). This ensures that each stakeholder can only interact with modules and data pertinent to their responsibilities, protecting sensitive agricultural records from unauthorized access.

ISO 25010

A global standard for software quality that provides a comprehensive model for evaluating system performance and quality. It categorizes software characteristics into key attributes—such as functionality, performance efficiency, usability, reliability, security, maintainability, and portability—ensuring the system is high-quality, effective, and reliable for operational use.

Sire

Refers to the male parent or male breeding bull whose genetic material (semen) is used during artificial insemination. The system logs sire codes and breed classifications to prevent inbreeding and ensure genetic improvement within municipal farming clusters.

Gestational Period

The biological duration from successful fertilization/conception to parturition (calf drop). In the system's algorithmic calculations, this value varies dynamically according to specific breed constants (e.g., ~279 days for Brahman vs. ~292 days for Carabao).

Sire

Refers to the male parent or male breeding bull whose genetic material (semen) is used during artificial insemination. The system logs sire codes and breed classifications to prevent inbreeding and ensure genetic improvement within municipal farming clusters.

Dystocia

A veterinary term defining abnormal, prolonged, or difficult labor/calving that requires manual or veterinary intervention during the parturition process.

Dam

Refers specifically to the biological mother of a registered livestock asset. The system maintains this relationship via the **motherId** parameter to establish genetic tracking and maternal lineage.

Dystocia A veterinary term defining abnormal, prolonged, or difficult labor/calving that requires manual or veterinary intervention during the parturition process.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Related Literature

Comparative Analysis of Livestock Management Architectures

The transition from legacy paper-based systems to centralized digital platforms represents a fundamental shift in the structural integrity of agricultural data. Traditional manual architectures are characterized by significant "hidden costs," including administrative latency, environmental degradation of physical ledgers, and a high variance in human error. According to the World Bank (2019), manual recording processes create fragmented data silos that prevent real-time analysis and epidemiological surveillance. Digitalizing these workflows through centralized architectures eliminates the physical vulnerability of records while providing an auditable trail for livestock health and productivity. By replacing physical logbooks with cloud-synchronized repositories, municipal agricultural units can mitigate the risks of misplacement and information decay, ensuring that livestock histories remain persistent across the animal's lifecycle.

The Digital Divide in Agricultural Economies

The adoption of smart farming technologies is heavily influenced by the prevailing digital divide, where infrastructural and literacy barriers significantly impact smallholder farmers in developing regions. The International Food Policy Research Institute (IFPRI, 2021) asserts that while technological tools offer immense productivity gains, their implementation is often hindered by limited technical knowledge and a lack of access to stable digital infrastructure. This socio-economic gap creates an "adoption barrier" that favors larger, capital-intensive operations over rural smallholders. To bridge this divide, software solutions must adhere to human-centered design principles that prioritize low-resource deployment and intuitive, mobile-friendly interfaces. By centering technology on the practical capacities of rural users, systems can empower marginalized producers to participate in the digital agricultural economy without requiring prohibitive investments in hardware.

Technological Paradigms in Reproductive Biology

The management of bovine reproductive biology has undergone a paradigm shift from observational human reliance to algorithmic, programmatic validation. Hansen (2020) highlights that the accuracy of estrus detection and gestation tracking is the primary determinant of economic success in cattle raising. Legacy methods dependent on subjective visual observation are prone to missing critical biological windows, leading to extended open periods and financial losses. Modern digital frameworks utilize breed-specific reproductive logic to automate these calculations, providing high-fidelity notifications for insemination intervals and expected calving dates. This programmatic approach reduces the cognitive load on livestock technicians and ensures that interventions are grounded in mathematically verified timelines rather than field approximations.

Socio-Technical Adoption Models in Rural Extension

The integration of digital tools within rural extension services is best understood through the Technology Acceptance Model (TAM) and the Information Systems Success Model. Davis (1989) establishes that "perceived usefulness" and "perceived ease of use" are the dual drivers of technology adoption. In the context of rural Oton, the success of a management system is contingent upon its ability to demonstrate immediate practical value—such as faster service requests—while maintaining an interface that matches the digital literacy levels of the raisers. Furthermore, the Information Systems Success Model emphasizes that system quality and information quality directly correlate with user satisfaction and net benefits. For agricultural extension, this means that the system must provide accurate, real-time data that stakeholders can trust for decision-making, thereby fostering a sustainable culture of digital utilization within the farming cluster.

Global Literature Review

Precision Livestock Farming (PLF) and Agricultural Digitalization

Precision Livestock Farming (PLF) functions as a transformative paradigm within contemporary agricultural engineering, emphasizing the deployment of sophisticated technological frameworks to observe, coordinate, and maximize animal production yields. The Food and Agriculture Organization (FAO, 2023) asserts that PLF infrastructures synthesize digital sensors, analytical engines, and automated surveillance modules to elevate pathological monitoring, reproductive throughput, and overall farm performance. Diverging from legacy methodologies that depend on subjective human observation, PLF frameworks empower agricultural producers to execute data-driven interventions grounded in high-fidelity, real-time datasets.

The incorporation of digital utilities within modern farming ecosystems marks a structural transition away from antiquated legacy protocols. Wolfert et al. (2017) elaborate that the digitalization of agriculture signifies a movement toward information-centric architectures where high-density data serves as the fundamental driver for operational decision-making. This systemic evolution allows producers to maintain rigorous surveillance over individual biological parameters, implement early-stage diagnostic protocols, and optimize reproductive timelines with mathematical precision. Nevertheless, academic discourse highlights that the international adoption of these frameworks remains fragmented, particularly as developing sectors encounter profound infrastructural and economic bottlenecks that impede the implementation of advanced technological infrastructures.

Evaluating these international dynamics within localized municipal settings necessitates the development of specialized software adaptations. The proposed platform, BreedSmart, utilizes the foundational principles of PLF while prioritizing practical accessibility and low-resource deployment. Rather than utilizing cost-prohibitive Internet of Things (IoT) sensors or complex wearable hardware, the system leverages a lightweight distributed web and mobile architecture. This strategic software approach enables stakeholders to record and process vital datasets through structured manual and semi-automated entry points. Consequently, this design ensures that small-scale agricultural clusters in rural sectors, such as Oton, Iloilo, can harness the benefits of digital modernization without requiring excessive capital investments or hardware-intensive maintenance.

Digitalization of Breeding Cycles and Reproductive Oversight

Reproductive efficiency serves as a critical economic pillar in bovine production systems, carrying profound financial consequences for cattle-raising enterprises. Hansen (2020) emphasizes that successful herd management is inherently dependent on the accurate identification of estrus cycles and the timely execution of insemination procedures. Because the biological window for conception is strictly limited, missing even a single cycle initiates a series of operational setbacks, including extended open periods, reduced yields, and significant financial losses for the raiser.

To mitigate these biological vulnerabilities, modern agricultural science relies on specialized digital interventions. Thatcher et al. (2019) establish that modern monitoring platforms substantially enhance conception success by ensuring that procedures are performed at optimal biological windows. These systems minimize human error through the generation of automated alerts and scheduling logic derived from established reproductive benchmarks.

Additionally, centralized record-keeping utilities provide essential support for monitoring vital biological milestones. Digital platforms facilitate the tracking of pregnancy durations and expected delivery dates, which are paramount for strategic planning. In legacy environments lacking such systems, farmers frequently rely on manual estimation, which leads to inaccurate records, mismanaged inventories, and severe operational inefficiencies.

The established body of literature provides a robust academic foundation for the functional modules of the proposed platform. These findings directly support the engineering of BreedSmart's automated breeding module, which is custom-tailored to optimize municipal workflows. The system is engineered to compute reheating schedules, insemination windows, and expected calf drop dates using breed-specific reproductive logic. By replacing unverified estimations with programmatic, hardcoded algorithms, the system minimizes human error and significantly improves the integrity of municipal livestock records.

International Adoption Trends in Smart Farming

The global agricultural landscape reflects an accelerating expansion in the deployment of advanced digital infrastructures. The World Bank (2022) reports that smart farming technologies are being aggressively integrated within developed agricultural sectors to elevate productivity, minimize operational costs, and strengthen food security. These technological assets encompass comprehensive digital registries, automated health monitoring tools, and sophisticated analytical decision support systems.

At the macroeconomic and administrative levels, the integration of these systems yields multi-faceted operational advantages. Nations that implement digital farming infrastructures benefit from superior traceability, enhanced disease containment, and optimized resource allocation. However, widespread implementation remains inconsistent; the World Bank highlights that many developing sectors continue to struggle with adoption due to infrastructural limitations, training gaps, and financial barriers.

This persistent developmental gap is mirrored in localized rural settings across the Philippines, where management processes remain largely anchored in paper-based methodologies. To address this systemic discrepancy, targeted software interventions are essential. BreedSmart resolves this gap by providing a low-cost, accessible digital framework that operates without requiring advanced hardware, making it suitable for municipal government units and localized farming clusters.

Agricultural Digitalization for Smallholder Producers

The socioeconomic conditions of smallholder producers necessitate that technological solutions be meticulously calibrated to local capacities. The International Food Policy Research Institute (IFPRI, 2021) asserts that small-scale farmers often encounter severe barriers to digital adoption due to limited technical literacy and insufficient access to network infrastructure. Despite these structural challenges, evidence confirms that properly designed digital tools significantly improve productivity when aligned with local operational needs.

To achieve sustained utilization, systems must adhere to human-centered design principles. IFPRI emphasizes that effective agricultural technologies must be intuitive, mobile-responsive, and context-specific to ensure adoption within rural demographics. Conversely, complex platforms requiring specialized training or expensive hardware are significantly less likely to achieve success in developing regions.

These insights validate the engineering philosophy of the proposed municipal platform. This principle directly guides the development of BreedSmart, which prioritizes simplicity, mobile accessibility, and user-friendly interfaces for the stakeholders of Oton, Iloilo. By centering the platform around the daily field realities of local users, the system ensures high usability across diverse demographic segments.

Mobile Infrastructures for Veterinary Information

The rapid proliferation of mobile telecommunications has created transformative opportunities for agricultural extension services. Mobile technology has emerged as a fundamental utility for enhancing veterinary services, especially in remote sectors. Aker and Mbiti (2019) state that mobile-based systems strengthen information transit within rural clusters, allowing farmers to report events, receive guidance, and access services in real time. These digital frameworks effectively reduce the dependency on physical travel and manual reporting, which are inherently characterized by labor inefficiencies.

Within specialized veterinary domains, mobile clients deliver significant operational throughput. Tadesse et al. (2022) establish that mobile platforms improve the speed of epidemiological surveillance by enabling farmers to instantly report symptoms. This early notification system functions as a vital biological safeguard, helping contain infectious diseases within vulnerable herds and preventing catastrophic economic losses.

Translating these findings into local practice ensures immediate benefits. In relation to BreedSmart, this literature supports the development of the mobile-based reporting module. Through this dedicated portal, farmers can electronically submit service requests and report health concerns without needing to physically visit the Municipal Agriculture Office, ensuring a significantly faster response from field technicians.

Computational Intelligence in Livestock Surveillance

The frontier of agricultural engineering increasingly utilizes computational intelligence to forecast biological patterns. Neethirajan (2020) highlights that advanced systems utilize physiological data streams to predict estrus cycles and disease risks, enabling producers to transition from reactive troubleshooting toward proactive management. Similarly, Rutten et al. (2019) explain that data-centric management improves decision-making by integrating historical ledgers with real-time field inputs, identifying patterns that are not easily observable through manual monitoring.

However, full-scale implementation presents notable practical barriers. The deployment of comprehensive AI systems often requires high computational resources and advanced hardware, making them largely inaccessible to small-scale producers in developing sectors.

To overcome these technical barriers, BreedSmart implements a simplified predictive framework. Rather than utilizing heavy neural networks, the system utilizes structured, breed-based reproductive logic to estimate milestones such as insemination windows and calving dates. This pragmatic engineering approach delivers operational benefits comparable to advanced systems but in a significantly more accessible and practical format.

Digital Epidemiological Surveillance in Farming

Robust surveillance is an indispensable pillar of municipal management and biosecurity. According to the World Organisation for Animal Health (WOAH, 2023), rapid reporting systems are essential for controlling infectious outbreaks and safeguarding public food security. Research by Del Rio Vilas et al. (2020) highlights that digital surveillance improves accuracy and reduces administrative delays. Conversely, traditional paper systems often suffer from inconsistent reporting and missing entries, which weaken municipal control efforts.

Integrated digital platforms eliminate these systemic vulnerabilities by allowing centralized data collection and real-time monitoring of health conditions. These infrastructures empower authorities to quickly identify disease clusters and implement rapid containment protocols.

Integrating these principles into municipal structures guarantees enhanced oversight. In relation to the proposed study, BreedSmart incorporates a simplified disease reporting module, ensuring faster communication and more accurate surveillance at the municipal government level.

Smart Agriculture and Data-Centric Decision Systems

The modernization of agriculture centers on the systematic elimination of guesswork. Rose and Chilvers (2018) state that smart agriculture improves executive decision-making by providing accurate, granular data regarding livestock conditions. Data-centric frameworks reduce uncertainty by replacing subjective estimation with measurable, verifiable indicators, including the continuous monitoring of reproductive cycles and long-term productivity trends.

Despite these advantages, structural adoption remains limited in rural sectors due to implementation costs and technical complexity. Many small-scale farmers continue to rely on legacy methods, which are inherently less efficient and highly prone to human error.

Bridging this divide requires dedicated, user-centric software engineering. BreedSmart aligns with smart agriculture principles by providing a simplified digital platform that prioritizes practical accessibility. The system utilizes structured data input forms and automated calculations that are easily understood and operated by both farmers and municipal technicians.

Synthesis of Global Literature

The comprehensive review of global literature highlights the paramount importance of digital transformation across the livestock sector. Advanced technological tools—including mobile applications, analytical frameworks, and centralized surveillance networks—have demonstrably improved productivity and reproductive efficiency in developed agricultural economies.

However, a recurring barrier is that many of these sophisticated technologies demand advanced infrastructure and continuous connectivity, which restricts their application within rural, small-scale farming communities. This structural gap underscores the need for simplified, accessible, and localized software solutions. Systems such as BreedSmart deliver comparable operational benefits without requiring complex hardware or advanced technical training.

2.2 Local Literature

National Policy Frameworks and the Digitalization of Philippine Agriculture

The modernization of the Philippine agricultural sector has become a central focus of national development agendas, with the government emphasizing the integration of digital technologies to address long-standing productivity gaps. National roadmaps and strategic policies highlight the potential of digital tools—ranging from mobile applications to cloud-based data repositories—to transform the industry. These initiatives aim to shift agricultural management from reactive, tradition-based methods toward data-centric systems that enhance transparency, accountability, and operational efficiency. Central to these policies is the recognition that digital transformation can facilitate better resource allocation, improve pest and disease surveillance, and streamline the delivery of government services to remote farming clusters.

However, a profound disconnect persists between these high-level national mandates and their actual implementation within local government units. While national agencies conceptualize sophisticated digital ecosystems, the ground-level reality in many rural municipalities remains anchored in antiquated methodologies. This gap is often exacerbated by inconsistent funding, varying levels of technical readiness among local personnel, and the lack of a standardized infrastructure that can support the seamless transit of data from the field to the national database. Consequently, while the policy framework provides a visionary direction for agricultural digitalization, the transition toward a fully integrated digital sector remains a fragmented and ongoing process, requiring localized software solutions that bridge the divide between national ambition and local capacity.

The Socio-Economic Foundation of Livestock as a Rural Safety Net

In the rural Philippines, livestock farming functions as more than a simple agricultural activity; it serves as a fundamental economic safety net and a primary mechanism for household risk management. For smallholder farmers, cattle and other livestock assets represent high-value, liquid capital that provides financial resilience against economic shocks, such as crop failure or health emergencies. Unlike seasonal crops, which are subject to immediate harvest windows and market volatility, livestock assets appreciate over time and can be managed as a form of "living savings" that provides long-term security. The economic impact on rural households is significant, as cattle raising often provides the necessary funds for education, land acquisition, and localized investments that catalyze upward social mobility.

The cultural and economic prominence of cattle raising specifically is driven by its high market value and reliability compared to other livestock sectors. In regions like Oton, Iloilo, cattle represent a prestigious and dependable asset that anchors the rural economy. However, the reliance on these animals as a primary economic cushion makes the health and reproductive success of the herd a critical concern. Any failure in breeding management or a delay in veterinary intervention directly translates into severe financial setbacks for the raiser. This socio-economic reality underscores the necessity of moving toward precise management systems that protect these vital assets, ensuring that the livestock sector continues to fulfill its role as a driver of rural economic stability and national food security.

Structural Barriers and the Digital Divide in Rural Adoption

The adoption of digital tools in rural agricultural communities is hindered by a complex set of barriers that collectively form a persistent "digital divide." Technological literacy remains a primary challenge, particularly among older demographics of farmers who have spent decades relying on traditional observational methods. For these users, the perceived complexity of modern software interfaces can create a psychological barrier to adoption. Language also plays a critical role; many digital platforms are designed in English, which may not align with the primary language of rural producers, such as Hiligaynon or Tagalog. This linguistic mismatch can lead to misunderstandings in data entry or the misinterpretation of critical system notifications, reducing the perceived usefulness of the technology.

Furthermore, the physical infrastructure in remote barangays often fails to provide the stable connectivity required for high-bandwidth digital applications. Frequent network latency, data outages, and the lack of widespread 4G/5G coverage in agricultural hinterlands make the use of cloud-only systems impractical. To overcome these barriers, digital solutions must be engineered with an "offline-first" philosophy, prioritizing intuitive navigation and localized language support. By reducing the technical and infrastructural friction associated with digital tools, systems can become more accessible to smallholder producers, allowing them to benefit from modern data management without requiring prohibitive investments in hardware or specialized training.

Local Governance and the Structural Risks of Paper-Based Systems

Local Government Units (LGUs) in the Philippines are at the forefront of agricultural service delivery, yet many municipal agriculture offices continue to struggle with the structural risks of paper-based data management. The reliance on handwritten logbooks and physical ledgers for tracking livestock inventories, vaccination histories, and breeding records introduces significant administrative vulnerabilities. Physical records are highly susceptible to environmental degradation and are easily misplaced or destroyed during local disasters. Moreover, the manual consolidation of these records is an arduous, error-prone process that leads to significant data silos, preventing administrators from accessing a synchronized view of the municipality's agricultural status.

These administrative challenges directly impact the quality of reports submitted to provincial and national oversight bodies. Inconsistent data collection in the field leads to inaccurate livestock tallies and skewed success rates for government programs. As municipal agriculture offices face increasing pressure to modernize their operations, the transition toward centralized digital registries becomes an administrative imperative. Digital systems eliminate the labor-intensive task of manual data entry and provide a resilient, auditable trail of field activities. By digitizing these workflows, LGUs can improve the efficiency of their service delivery, ensure the long-term preservation of agricultural records, and support more accurate, evidence-based agricultural planning.

Legacy Knowledge vs. Data-Driven Agricultural Modernization

The transition from traditional farming to data-driven practices represents a scholarly shift from observational reliance toward mathematical precision. Legacy knowledge systems, which rely on memory and verbal history, have sustained rural livestock management for generations. However, these methods are inherently limited in their ability to manage the complexities of modern herd production. Subjective estimation of breeding cycles, pregnancy durations, and health metrics often leads to missed windows of opportunity and reproductive failure. In contrast, data-centric systems utilize breed-specific reproductive logic and historical data analysis to provide verified milestones that eliminate the guesswork from livestock management.

While traditional knowledge remains a valuable component of the farming identity, the necessity of modern systems is driven by the need for quantitative accuracy in an increasingly competitive agricultural market. Data-driven practices enable raisers to track individual animal performance over multiple generations, identifying genetic trends and optimizing herd health with surgical precision. By integrating digital tracking into their daily routines, farmers can transition from a reactive approach to livestock care toward a proactive strategy that maximizes biological yields. This modernization does not seek to replace the farmer's intuition but rather to augment it with reliable, high-fidelity information that supports better decision-making and ensures the sustainability of rural agricultural operations.

Synthesizing the Path to Sustainable Digital Transformation

The path to sustainable digitalization in the Philippine livestock sector requires a realistic synthesis of traditional field realities and modern technological capabilities. Bridging the gap between legacy practices and digital infrastructures necessitates localized solutions that prioritize human-centered design and infrastructural resilience. Local governments must move beyond the procurement of generalized hardware and focus on implementing software ecosystems that address the specific operational bottlenecks faced by municipal technicians and rural raisers. This involves the creation of platforms that are not only technically robust but also socially inclusive, respecting the digital literacy levels and linguistic preferences of the community.

Ultimately, the success of agricultural digitalization depends on its ability to demonstrate immediate practical value to all stakeholders. When farmers experience faster service responses and technicians see a reduction in their administrative burden, the culture of digital utilization will naturally expand. By replacing vulnerable manual records with synchronized, cloud-backed systems, local governments can build a resilient foundation for agricultural growth. This transformation is essential for enhancing the competitiveness of the domestic cattle industry, supporting regional food security, and ensuring that rural communities are equipped with the tools necessary to thrive in a data-driven global economy.

2.2.1 Foreign Studies

Mobile Systems for Epidemiological Surveillance

Investigations provide robust validation for mobile field tools. A specialized study conducted by Gizaw et al. (2023) in Ethiopia revealed that utilizing mobile applications significantly improved both the speed and accuracy of disease reporting compared to traditional methods. Field workers were able to transmit real-time reports, empowering authorities to mount faster containment responses and preventing the dissemination of animal diseases. The study highlighted that institutional delays in manual reporting frequently culminate in uncontrolled outbreaks, while modern mobile systems successfully eliminated communication gaps between farmers and veterinary services.

In relation to BreedSmart, this research supports the development of a mobile-based reporting module, allowing farmers in Oton, Iloilo to instantly notify technicians of health concerns, guaranteeing faster response times and improved disease control.

Reproductive Oversight and Conception Efficiency

Advanced monitoring technologies have been the subject of rigorous international field evaluations. Research conducted in Kazakhstan (2026) established that automated detection algorithms significantly increased conception rates. The evaluated system delivered timely, automated alerts identifying optimal insemination periods, thereby minimizing missed opportunities. The investigation concluded that reproductive efficiency improved because producers no longer depended on subjective observation; instead, digital platforms supplied reliable predictive indicators for critical breeding decisions. This finding reinforces BreedSmart's automated tracking feature, which computes milestones based on breed-specific reproductive data.

Mobile Veterinary Extension Support

Bridging the service gap between experts and remote producers represents a primary objective of extension research. An evaluation in Nigeria (2025) demonstrated that mobile platforms dramatically improved access to professional assistance in regions where technicians are limited in number. Participating farmers reported faster communication channels and improved outcomes resulting from real-time messaging and reporting functionalities. This study is highly relevant to BreedSmart, which is engineered to improve coordination between farmers and technicians through a unified digital platform.

Hardware-Based IoT Surveillance Architectures

Advanced research in India and China examining IoT monitoring demonstrates that sensor-based tracking significantly improves productivity and reduces economic losses. Kumar et al. (2022) state that IoT systems possess the capacity to monitor physiological parameters in real time. However, the literature notes that these systems are exceptionally expensive and require advanced infrastructure, making them inaccessible to small-scale producers. BreedSmart addresses this economic limitation by offering an accessible software-based alternative that delivers structured monitoring through automated calculations and user-generated data inputs.

2.2.2 Local Studies

Breeding Success in Central Luzon Agriculture

Domestic studies provide critical regional context for breeding interventions. Research conducted by the Department of Agriculture Region III (2021) established that accurate record-keeping significantly improves insemination success rates. Farmers utilizing structured records achieved markedly higher conception rates compared to those depending on traditional manual logs. The study emphasized that seamless data synchronization is an absolute prerequisite for optimizing reproductive outcomes, providing direct validation for BreedSmart's centralized database and automated tracking system.

Reproductive Performance and Record Integrity in Philippine Livestock Systems

Research published by Flores et al. (2019) in the ***Philippine Journal of Veterinary and Animal Sciences*** evaluated artificial insemination success rates and reproductive efficiency across smallholder livestock raisers. The study established that accurate, structured tracking of estrus cycles and post-service dates directly correlates with higher conception rates and reduced calving intervals. However, the researchers emphasized that rural breeding programs are severely bottlenecked by missing or degraded paper-based breeding logs maintained by raisers and field technicians. While Flores et al. focused on identifying the biological and administrative causes of failed conception, **BreedSmart** provides the functional technological solution by replacing paper logs with automated, breed-specific reproductive tracking algorithms that notify users of exact reheating and calving windows. .

Digital Transformation Gaps in Philippine Agricultural Value Chains

A comprehensive national assessment published by Briones, Galang, and Latigar (PIDS, 2023) evaluated the adoption patterns and structural bottlenecks of digital extension platforms across Philippine smallholder farming communities. The authors identified that while mobile tools significantly improve service coordination between farmers and local government units, primary barriers to sustained adoption include complex software interfaces, lack of regional dialect support, and frequent network disruptions in rural areas. While PIDS highlights these systemic gaps nationwide, **BreedSmart** directly resolves them at the municipal level by offering a simplified interface designed for varying digital literacy levels, multi-dialect support (English, Tagalog, and Hiligaynon), and an offline caching mechanism (AsyncStorage and TanStack Query) to overcome rural network failures. .

Digital Extension Systems in the Philippine Context

Domestic research focusing on mobile platforms provides essential guidelines for user design. Studies examining systems across the Philippines demonstrate that mobile platforms successfully improve communication channels between rural households and agricultural offices. However, persistent challenges such as low digital literacy and inconsistent connectivity remain formidable barriers. This reality reinforces the importance of designing BreedSmart as a lightweight, highly mobile-friendly system capable of functioning reliably even within constrained operational environments.

Synthesis of Research

The reviewed studies collectively demonstrate that digital systems produce measurable improvements across livestock management, particularly regarding disease reporting, reproductive monitoring, and stakeholder coordination. International research highlights advanced frameworks such as AI and IoT, while domestic studies emphasize the unique adoption barriers encountered within Philippine communities. Despite these advancements, a structural gap persists regarding the absence of simple, municipal-level management software. BreedSmart addresses this niche by providing a centralized software solution engineered specifically for administrative use in Oton, Iloilo.

2.3 Analytical Synthesis

The exhaustive review of literature and investigations confirms that livestock management is undergoing a structural evolution driven by the integration of modern digital technologies. Academic investigations consistently verify that digital transformation elevates productivity, strengthens biosecurity, and enhances the auditability of health records. Global literature highlights cutting-edge innovations such as PLF and AI-driven forecasting, which facilitate real-time tracking and superior decision-making. Authoritative publications confirm that these systems yield higher operational efficiency and superior outcomes in modern agricultural settings.

However, the vast majority of these high-tech international solutions demand prohibitive financial capital and advanced infrastructure, rendering them largely unsuitable for smallholder farmers and rural communities in developing nations. Local literature and domestic studies further reveal that Philippine management continues to depend heavily on legacy paper systems, suffering from chronic data inaccuracies and reporting delays. Smallholder raisers in rural municipalities routinely depend on handwritten logs and fragmented communication, which elevates the risk of data loss and transcription errors.

Related studies prove that mobile-based information systems, even in simplified formats, deliver massive performance enhancements across monitoring and reproductive tracking.

Digital platforms successfully streamline communication, accelerate response times during outbreaks, and elevate success rates. Nevertheless, prevailing systems are frequently too technologically complex or misaligned with the specific workflows of local government offices. This exposes a clear operational gap: a profound disconnect exists between advanced international technologies and the grounded needs of rural communities in the Philippines.

This identified gap defines the core purpose of the proposed capstone project. BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System aims to resolve this disconnect by successfully integrating the most vital functional capabilities—including automated breeding computations, secure record-keeping, and real-time communication—into a simplified software platform. Unlike capital-intensive IoT architectures, BreedSmart focuses on intelligent software-based automation that eliminates the need for expensive hardware, ensuring high usability for end-users with minimal technical backgrounds.

Ultimately, the exhaustive synthesis of related research confirms the absolute operational necessity of a localized, highly user-friendly, and centralized municipal management system. BreedSmart successfully bridges the structural divide between advanced technologies and rural grassroots implementation by delivering an innovative digital solution that is simultaneously functional, secure, and fully accessible for routine government agricultural operations.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study adopts a **developmental research design** integrated with an **applied software engineering framework** to govern the complete lifecycle of the proposed web and mobile platform. Developmental research is an iterative, structured methodology explicitly engineered for technology-driven capstone projects that focus on the conceptualization, architectural design, programmatic development, field deployment, and systematic evaluation of software solutions. This design bridges theoretical information technology concepts—such as distributed cloud architectures, offline-first data caching, and automated algorithmic calculations—with practical agricultural management in local government units.

Through this approach, the research structures the platform's creation into continuous, manageable iterations rather than relying purely on descriptive or observational methods. By embedding feedback loops throughout the development lifecycle, the methodology ensures that the software is systematically refined to fit the operational realities of cattle farmers, field technicians, and Municipal Agriculture Office personnel in Oton, Iloilo. This systematic process ensures that the platform functions as a resilient, real-world tool capable of replacing legacy paper-based workflows.

Finally, the developmental research design provides a rigorous framework for empirical validation and quality assurance. The system's performance and practical usability are continuously evaluated against the ISO 25010 software quality standard within an actual rural agricultural setting. This evaluation assesses critical attributes—including functionality, usability, reliability, security, and performance efficiency—confirming that the final platform delivers an accurate, secure, and accessible solution for municipal livestock management.

3.2 System Development Methodology

The study adopts a **developmental research design** using the **Agile Software Development Model** combined with the **System Development Life Cycle (SDLC)** approach. This methodology is suitable for the iterative development of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System, allowing continuous improvement based on user feedback from farmers and livestock technicians in the Municipality of Oton, Iloilo.

The SDLC framework ensures that the system is developed in a structured and systematic process, from planning up to deployment. Each phase focuses on improving system quality, usability, and functionality to meet the actual needs of the end users.

Requirements Gathering Phase

In this phase, the system requirements are gathered and analyzed based on the needs of the target users, which include livestock technicians, farmers, and the Municipal Agriculture Office.

Key requirements identified include:

- Digital recording of livestock information
- Automated tracking of insemination, reheating, and calf drop dates
- Easy communication between farmers and technicians
- Centralized storage of livestock records
- Generation of accurate reports for administrative use

This phase ensures that the system is aligned with actual user needs rather than assumptions, improving its effectiveness and usability

System Planning Phase

The planning phase involves identifying the overall scope of the project and understanding the livestock management situation in Oton, Iloilo. Preliminary site visits and informal interviews with livestock technicians and farmers are conducted to gather initial insights.

During this phase, the researchers also study common issues in cattle management such as manual record-keeping, difficulty in tracking breeding cycles, and delays in reporting livestock health conditions. These observations help define the objectives of the system and establish its relevance to real-world agricultural problems.

System Design Phase

The system design phase focuses on creating the blueprint of the BreedSmart system. This includes designing the user interface (UI), user experience (UX), system architecture, and database structure.

The UI design prioritizes simplicity and accessibility, especially for users with limited technical experience. The system is designed to be mobile-responsive to support farmers who primarily use smartphones.

The database design uses MongoDB to store livestock records, breeding history, vaccination data, and user accounts. The system architecture is structured to separate frontend and backend processes for better scalability and maintainability.

Software Development Phase

The development phase involves the actual coding and implementation of the system. The frontend is developed using **React.js**, while the backend is built using **Express.js** with **Node.js**, and **MongoDB** is used as the database management system.

During this phase, key system modules are developed, including:

- Farmer request module (insemination and disease reporting)
- Technician management module (breeding and health monitoring)
- Automated breeding cycle calculator based on cattle breed logic
- Notification system for reminders and updates
- Administrative dashboard for the Municipal Agriculture Office

This phase follows Agile principles, allowing iterative improvements based on continuous evaluation.

Testing Phase

The testing phase ensures that the system functions correctly and meets user requirements. Different levels of testing are performed, including:

- **Unit Testing** – testing individual system components
- **System Testing** – testing the complete integrated system

- **User Acceptance Testing (UAT)** – involving actual users such as livestock technicians and selected farmers

This phase helps identify errors, bugs, and usability issues before full deployment. Feedback from users is also collected to improve system performance and interface design.

Deployment Phase

In this phase, the system is prepared for deployment and presentation. All documentation is completed, including user manuals and technical documentation.

The system is finalized based on feedback gathered during testing. After validation, BreedSmart is prepared for implementation in a controlled environment within the Municipality of Oton, Iloilo.

This phase ensures that the system is stable, functional, and ready for real-world use in livestock management operations.

7. Maintenance Phase

The Maintenance Phase is an ongoing stage dedicated to ensuring the system's long-term sustainability through iterative improvements driven by user feedback. This phase focuses on refining system performance, addressing technical issues, and adapting to evolving agricultural workflows reported by farmers and technicians. By conducting regular security audits, backups, and providing continuous user support, this phase ensures that BreedSmart remains a reliable, secure, and effective tool that evolves alongside the needs of the Municipality of Oton.

Presentation Layer (Edge Layer - Top Panel):

- Livestock Technicians (Cross-Platform Flexibility): Clearly visualizes that technicians access the system through both a Laptop (Web Portal for office bulk entry/admin) and a Mobile Device (Tablet/Smartphone for field logging).
- Offline-First Caching: Explicitly details AsyncStorage and TanStack Query for local data persistence on the technician's mobile client during signal dropouts in remote barangays.
- Admins & Farmers: Retains the Municipal Administrator (Web Dashboard on Desktop/Laptop) and Cattle Farmer (Mobile App on Smartphone)**.**

Application Layer (Logic Layer - Middle Panel):

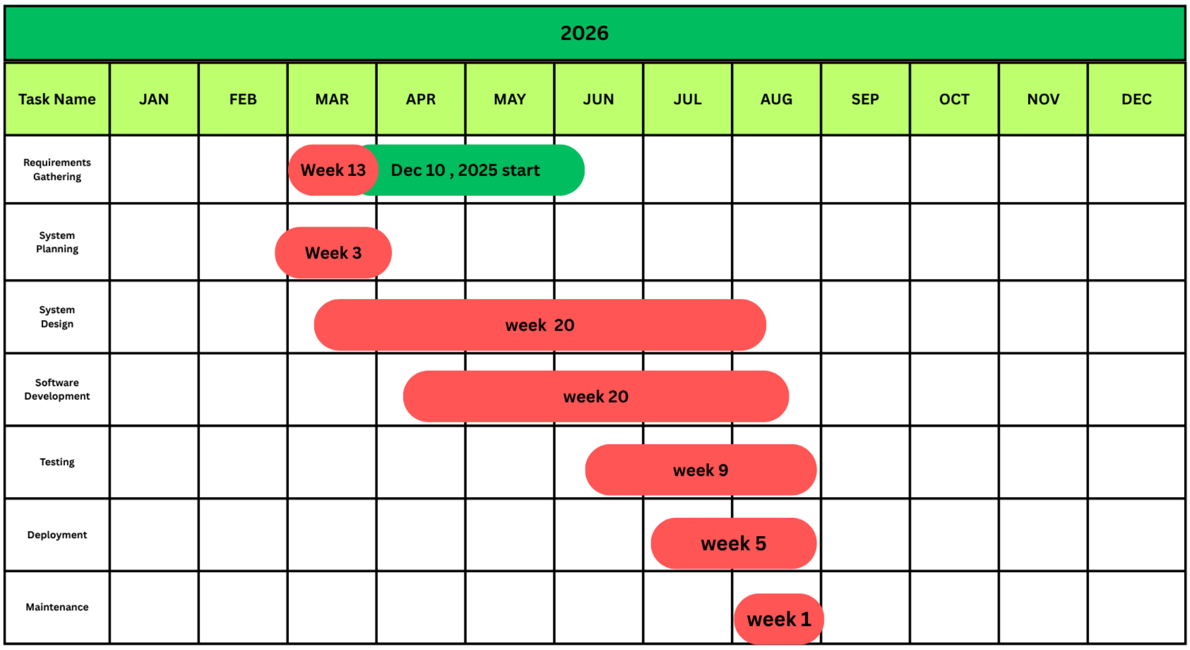
- MERN Backend API Cluster: Highlights the continuous Express.js/Node.js server hosted on Ubuntu Linux.
- Core Business Logic Nodes: Includes explicit sub-modules for Algorithmic Breed Calculations (estrus return & calving dates), Role-Based Access Control (RBAC) via Clerk, and Offline Sync Queue Processing.

Data Layer (Persistence Layer - Bottom Panel):

- MongoDB Atlas NoSQL Cloud Database: Shows unified collections for Animals Registry (with motherId genetic tracking), User Accounts, and Insemination/Medical History.
- Disaster Recovery: Includes automated daily and weekly backups for high availability.

3.4 Project Gantt Chart (Weekly Estimation)

The timeline spans 21 weeks, from December 10, 2025, to May 2, 2026.



3.5 Comparative Benchmark: BreedSmart System Features

To contextualize the proposed Livestock Management and Veterinary Service System for Oton, Iloilo, this study examines BreedSmart, a contemporary agricultural and veterinary services platform. BreedSmart demonstrates how localized systems integrate advanced features such as real-time livestock tracking, offline-first sync engines, veterinary gestation calculators, multilingual AI assistant support, and administrative dashboards. By analyzing its architecture, the study identifies both strengths and limitations that inform the design of the localized Iloilo-focused system.

Complete Feature List

Frontend (User-Facing Modules)

● Livestock Registry: Profile creation, digital ear-tag registration, breed classification (Beef Cattle, Dairy Cattle, Carabao, Swine, Goat), dynamic age tracking, status labels, and image attachments.

● Service Request Portal: Quick-actions for requesting Artificial Insemination (AI) or veterinary health checks, symptoms reporting, and image attachments.

● Breeding & Gestation Calendar: Interactive heat-cycle calendars, voluntary waiting period (VWP) tracking, diagnostic check windows, and expected calving calculators.

● Technician Dashboard: Task actions, walk-in registration sheets, queue logs, and dynamic expected diagnostic dates visualization.

● Reports & Analytics: Graph charts displaying breeding efficiency, municipal case counts, and automated UNIP-compliant PDF exports for record-keeping.

● User Accounts: Clerk authentication, OTP verification, secure password reset, profile editing, and role-based permissions (Admin, Technician, Farmer).

● Admin Features: Role assignment, user verification, system-wide analytics, data exports, log management, and system settings.

● Landing Page: General information, service overviews, public reports, and download links for the mobile client.

Backend (System Services)

● Activity & Sync Tracking: Queues offline actions using client-side storage, with automatic resolution and server-side synchronization upon reconnection.

● Security & Protection: Encrypted sessions, validation schema guards, API rate limiting, and role-based security checks.

● Notifications: Real-time web socket alerts, automated mobile pushes, status update reminders.

● Image Uploads: Cloud storage integration for animal verification and health report photos.

● Queue System: Background task handler managing sync retries, notifications, and scheduled calving updates.

● Location Features: Barangay-level coordinates caching, regional heatmaps, route optimization coordinates.

● Soft Delete System: Cascading soft deletes using timestamp indicators, preserving historical breeding ledgers for auditing.

● API Versioning: Supports multiple versions for feature expansion.

● Multi-Environment Support: Local development, staging, and production databases.

● Language detection and localized Hiligaynon, Tagalog, and English greetings.

● Symptom diagnostics and first-aid recommendations.

● Breeding cycle guidance (step-by-step).

● Interactive help menus and FAQs.

● Follow-up question handling with context memory.

Security Summary

● JWT tokens, role-based routing middleware, Clerk Identity protection, API rate limiting, auditable action logs, and validation schemas.

Technology Stack

● Frontend: React, Vite, React Native, Expo, Tailwind CSS, NativeWind, DaisyUI, Leaflet maps.

● Backend: Node.js, Express, Clerk.

● Real-time: Socket.io.

● Database & Caching: MongoDB, Mongoose, AsyncStorage, TanStack Query.

● Queue: Inngest.

● Storage: Cloudinary.

3.6 Requirement Modeling

3.6.1 Context Diagram

The **Context Diagram** serves as a high-level representation of the entire BreedSmart ecosystem, distilling the platform into a singular process to clearly define the external entities and their primary interactions with the system.

- **Farmer / Client**: Responsible for submitting personal profiles, livestock specifications, and service requests for veterinary or field visits; receives automated registration confirmations and synchronized visit schedules in return.
- **Livestock Technician**: Acts on real-time alerts for service requests and performs direct digital entry of health treatments, vaccinations, and insemination records to eliminate manual logbook dependency.
- **Agriculture Officer / Admin**: Oversees health status summaries and total livestock populations while utilizing the system to generate comprehensive analytical reports for municipal and provincial administrative oversight.

3.7 Tools and Technologies Used

The development of BreedSmart utilizes modern technologies to ensure system responsiveness, scalability, and maintainability.

● **Frontend:** React, JavaScript, TypeScript, Tailwind CSS, NativeWind, Vite, and Expo are used to develop responsive and user-friendly web and mobile interfaces.

● **Backend:** Express.js and Node.js are used for server-side processing, API development, and system logic implementation.

● **Database:** MongoDB serves as the database management system for storing livestock records, breeding information, and user data.

● **Development Tools:** Visual Studio Code is used for coding and debugging, MongoDB Compass for database management, and GitHub for version control and collaboration.

These technologies were selected for their compatibility with Agile development practices, open-source availability, and reliability in modern application development.

3.6 Data Gathering Techniques

The researchers utilized several data gathering techniques to collect relevant information needed for the development of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System.

● **Surveys/Questionnaires:** Structured questionnaires were distributed to selected farmers and livestock personnel to assess current challenges related to manual record-keeping, communication delays, livestock monitoring, and accessibility of agricultural services within different barangays in Oton, Iloilo.

● **Interviews:** Direct interviews and consultations were conducted with the livestock technician and selected farmers to gather detailed information about breeding cycles, insemination procedures, livestock health monitoring, and existing reporting practices.

● **Observations:** The researchers conducted physical observations and reviews of existing hardbound logbooks, folders, and manual livestock records to identify issues such as data degradation, record inconsistencies, and inefficiencies in the current documentation process.

3.8 Testing Procedures

The researchers conducted several testing procedures to ensure the functionality, reliability, and overall performance of the BreedSmart system. These testing methods were performed to identify system errors, validate features, and confirm that the system meets the requirements of the intended users.

● Unit Testing – Unit testing was conducted to evaluate individual components and modules of the system separately. Each function, including livestock record management, breeding cycle calculations, notification features, and authentication processes, was tested to ensure proper operation and accurate outputs.

● System Testing – System testing was performed to evaluate the integrated functionality of the entire BreedSmart system. This testing ensured that all modules work together correctly, including communication between the frontend, backend, and database. The researchers also tested system responsiveness, data synchronization, and overall workflow performance.

● User Acceptance Testing (UAT) – User Acceptance Testing was conducted with selected farmers, livestock technicians, and intended users to determine whether the system meets their operational needs and expectations. Feedback gathered during this phase was used to identify usability issues, improve user experience, and validate the practicality of the system in a real-world agricultural environment.

3.9 Evaluation Criteria

The BreedSmart system will be evaluated based on selected software quality characteristics from the ISO 25010 standard to ensure that the system is reliable, effective, and suitable for livestock management operations in the Municipality of Oton, Iloilo.

● **Functionality** – Evaluates the ability of the system to perform its intended functions accurately and efficiently. This includes livestock record management, breeding cycle monitoring, notification services, and report generation.

● **Usability** – Measures how easy and user-friendly the system is for farmers, livestock technicians, and administrators. This includes interface design, accessibility, navigation, and overall user experience.

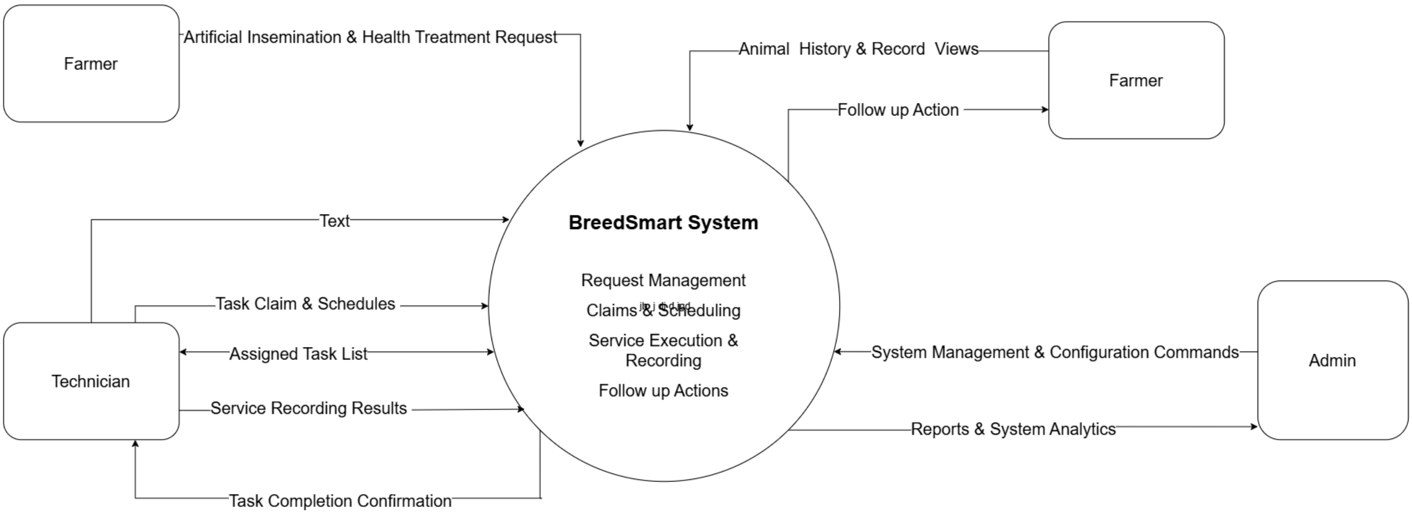
● **Accuracy** – Assesses the correctness and reliability of automated breeding calculations, reheating schedules, calf drop estimations, and livestock monitoring records generated by the system.

● **Security** – Evaluates the system's ability to protect user accounts, livestock records, and sensitive information from unauthorized access and data loss through authentication and secure data handling.

● **Efficiency** – Measures the responsiveness and performance of the system during data processing, record retrieval, notification handling, and overall system operations across web and mobile platforms.

3.10 Supporting Diagrams

Data Flow Diagram (DFD – Level 0)



Farmer Registration & Animal Management

Farmer / Client → BreedSmart → Database

- Farmer submits personal profile, farm location, and animal details
- System processes the registration and links the animals to the farmer's profile
- Data is stored securely in the database and a confirmation is returned to the Farmer

Health Record & Treatment Logging

Technician → BreedSmart → Database

- System associates the treatment data with the correct farm and animal records
- Database updates the animal's health status and saves the historical health record
- Technician records health treatments, vaccinations, Insemination and diagnoses for specific animals

Visit Request & Scheduling

Farmer / Client → BreedSmart → Technician

- Farmer submits a request for a veterinary or farm visit through the system
- System processes the request and alerts the assigned Technician
- The technician reviews the request, schedules the visit, and the system sends the scheduled date back to the Farmer.

Analytics and Reporting

Agriculture Officer / Admin → Agriculture Office System → Database

- Officer or Admin requests a summary of health statuses or total livestock population
- System queries the database for aggregated animal and health data
- System compiles the findings, generates analytical reports, and displays them to the Officer/Admin

3.11 Use Case Diagram

3.12.1 Interface Diagram (Wireframe Design)

Website Design

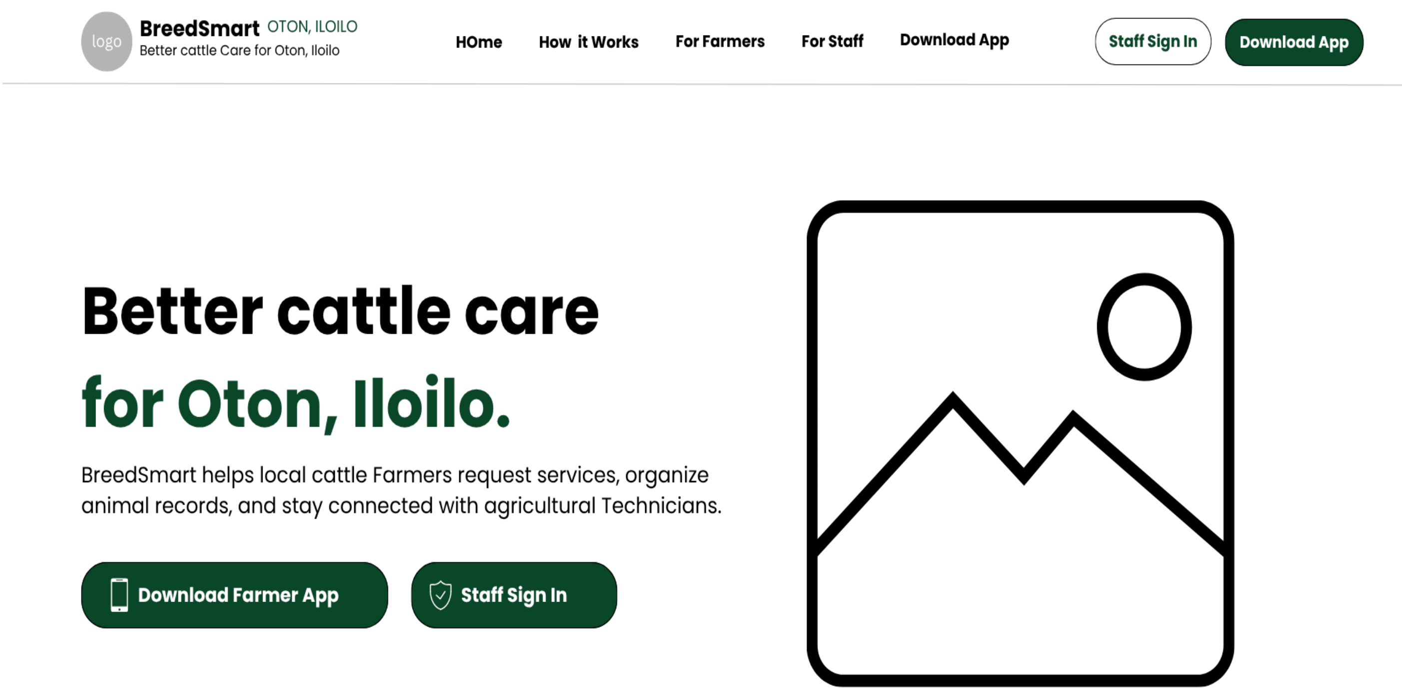


Figure 1:

Represent the BreedSmart Website Homepage, serves as the main interface of the system. It provides information about BreedSmart and its purpose of improving cattle care in Oton, Iloilo.The homepage includes navigation options such as Home, Home It Works, F Farmers, For Stuff, and Download Farmer App.

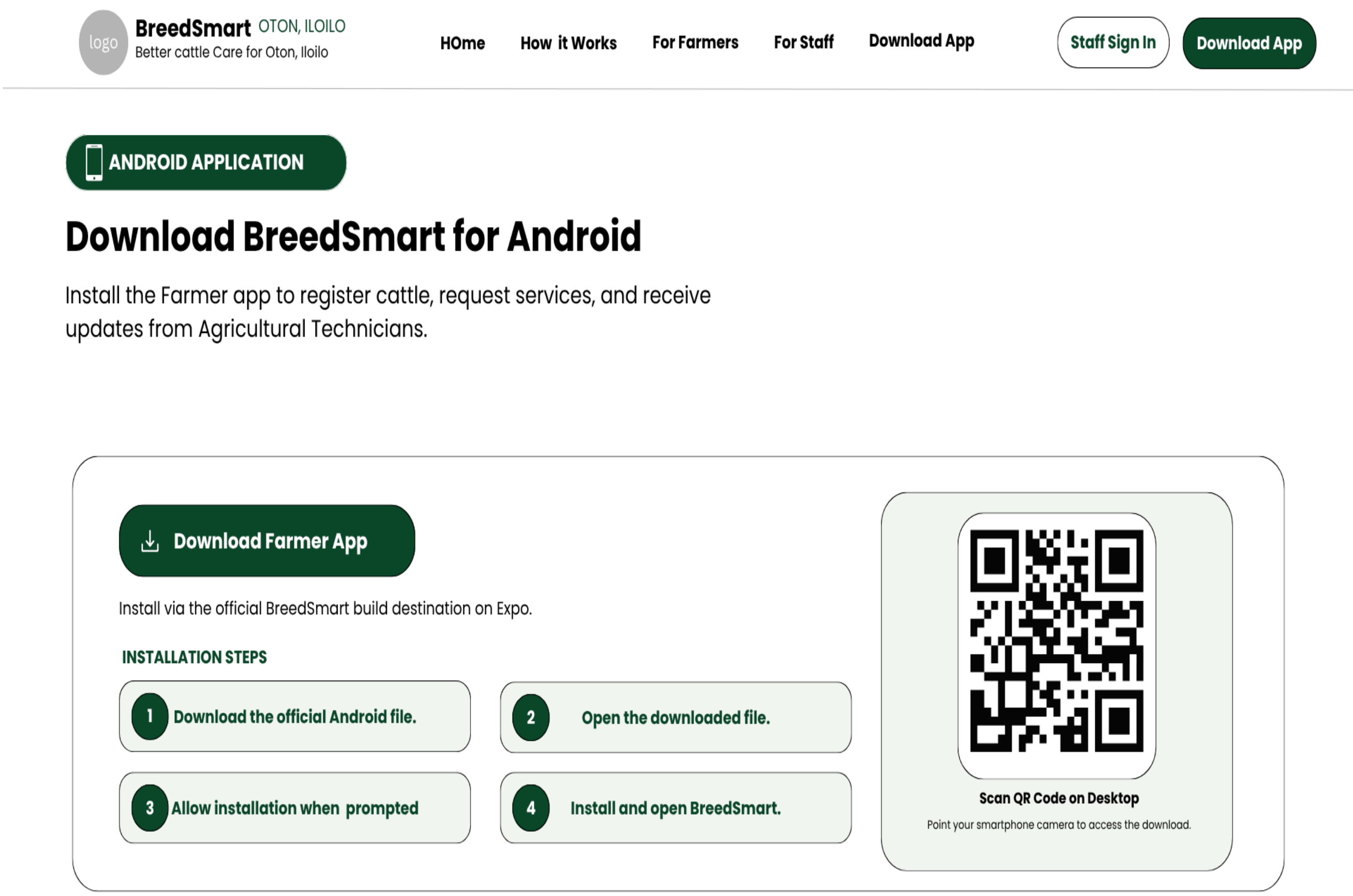


Figure 2:

Represent the BreedSmart Android Application installation instruction, a Download Farmer App button, and a QR code for convenient access to the application. The mobile application allows farmers to register cattle, request livestock services, monitor breeding activities, and receive updates from technicians.

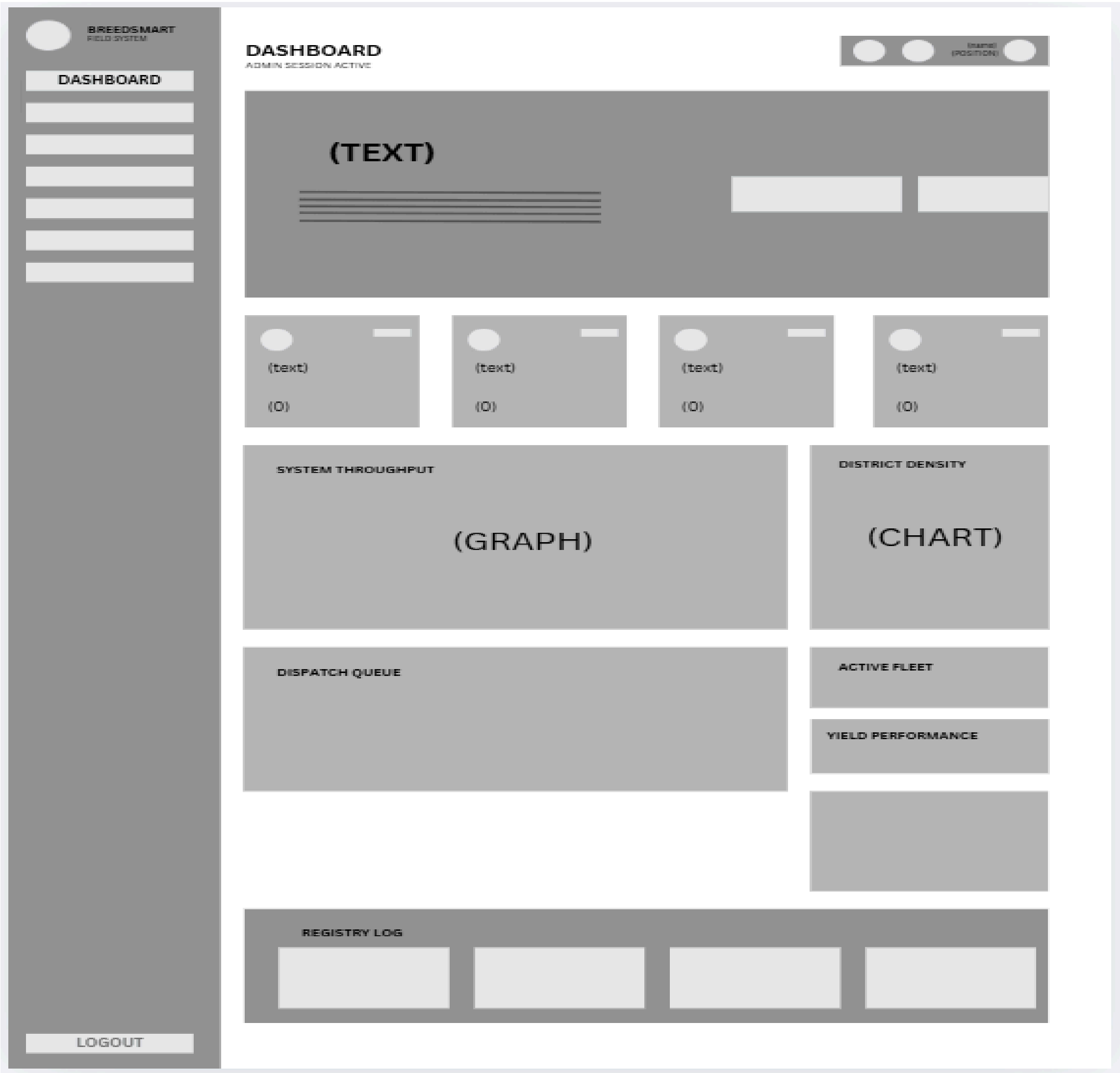


Figure 1:

Represents the primary landing interface for Municipal Agriculture Office personnel upon logging into an active session. It features a fixed left-hand navigation sidebar and top header indicating administrator session details. The main view integrates high-level municipal statistics tiles, a system throughput performance graph, district density charts, dispatch queue monitors, active fleet counters, and real-time registry activity logs.

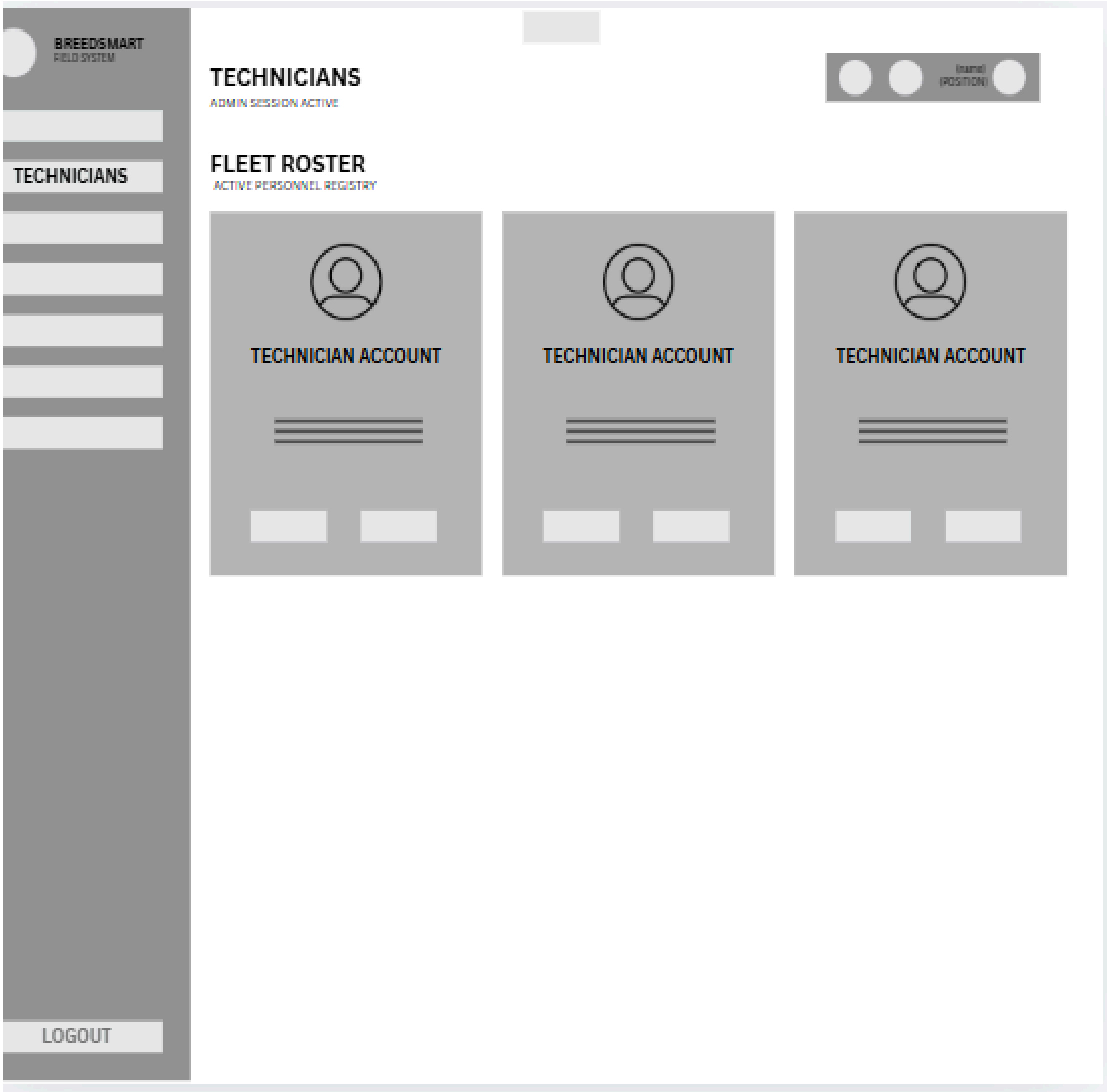


Figure 2:

Outlines the management portal for overseeing registered field technicians within the municipality. The workspace displays grid cards representing individual technician accounts, detailing active field status, contact information, assigned dispatch workloads, and quick-action management triggers.

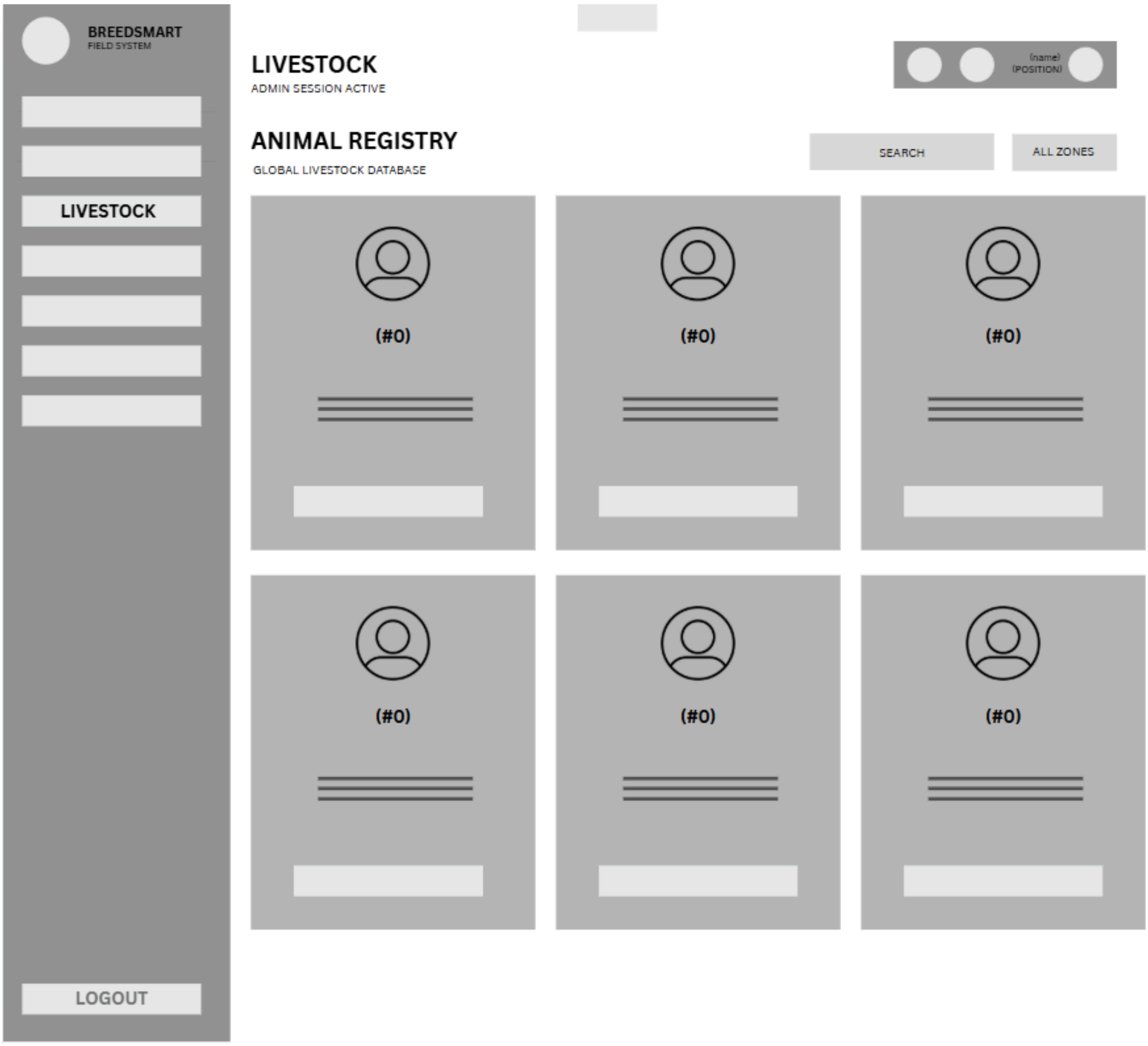


Figure 3:

Illustrates the master database view for all livestock registered across municipal zones. It includes global search and zone-filtering controls at the top, alongside structured card views for individual cattle profiles detailing ear-tag identification numbers, breed classifications, ownership mappings, and health status indicators.

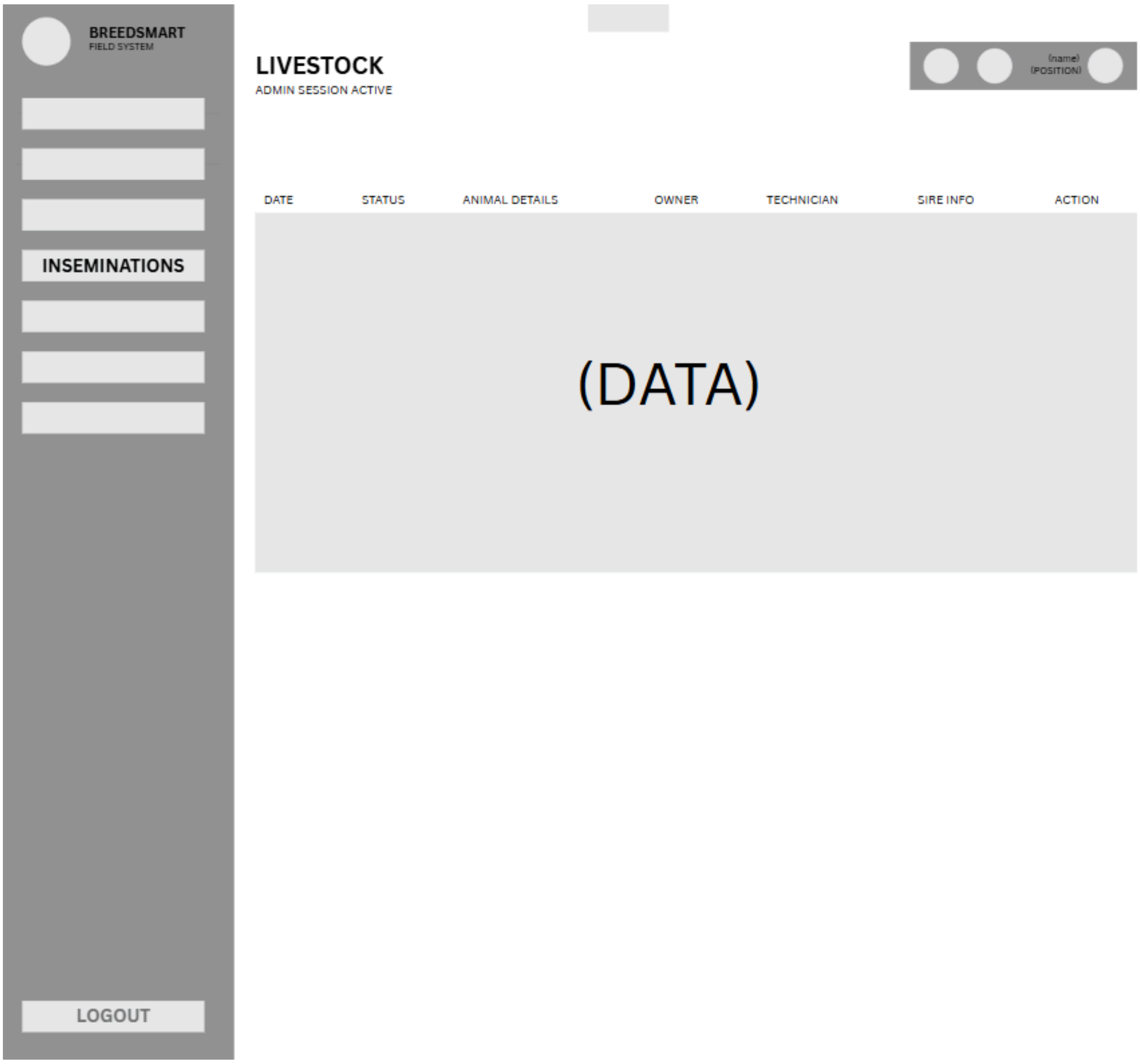


Figure 4:

Displays the centralized tabular ledger for monitoring artificial insemination activities. The data table categorizes records by procedure date, current status (Pending, Approved, Done, Failed), animal details, registered raiser/owner, assigned technician, sire code/breed details, and administrative action buttons.

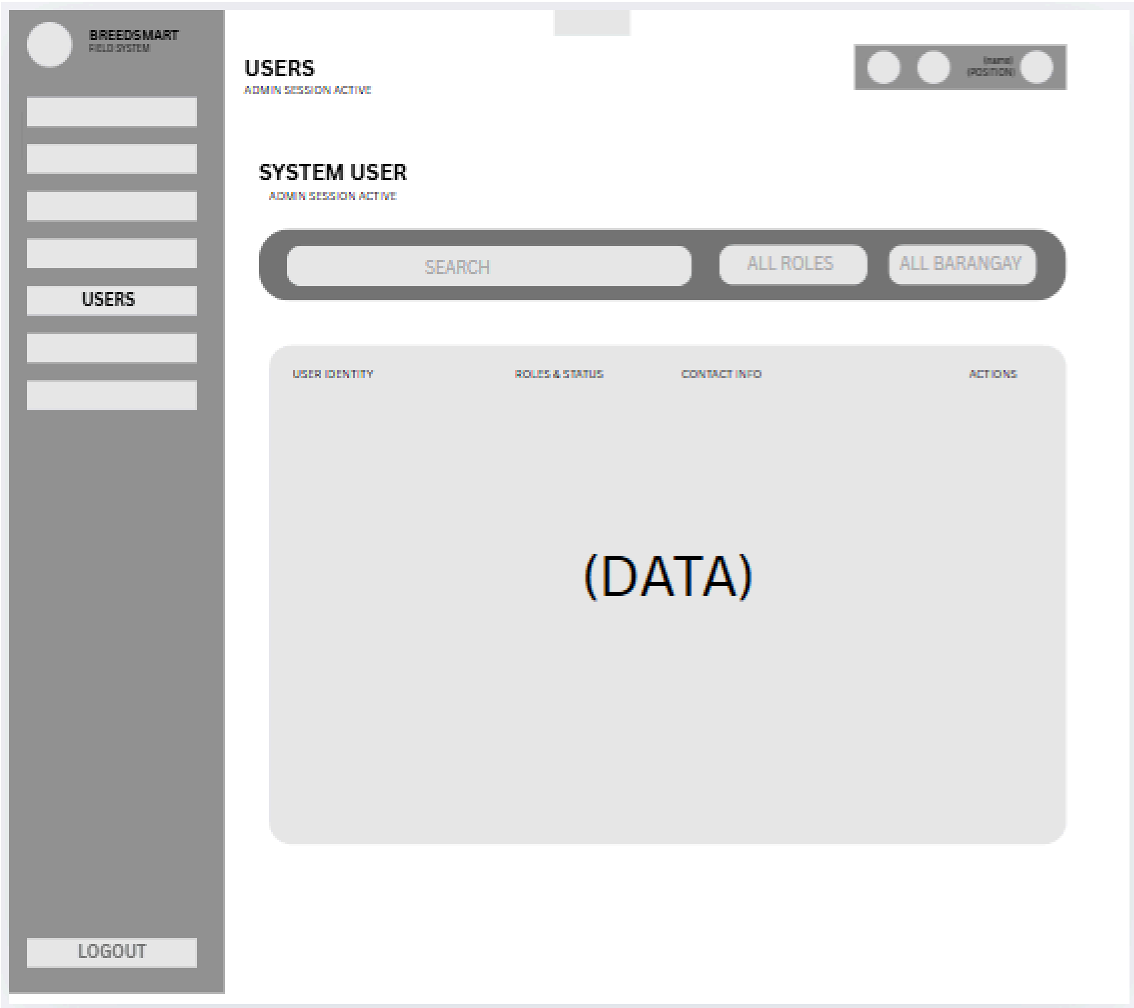


Figure 5:

Depicts the central user directory for managing platform accounts. It incorporates search bars and filtering dropdowns (All Roles, All Barangay) to audit user identities, assigned access levels (Admin, Technician, Farmer), verification flags, contact details, and account permission controls

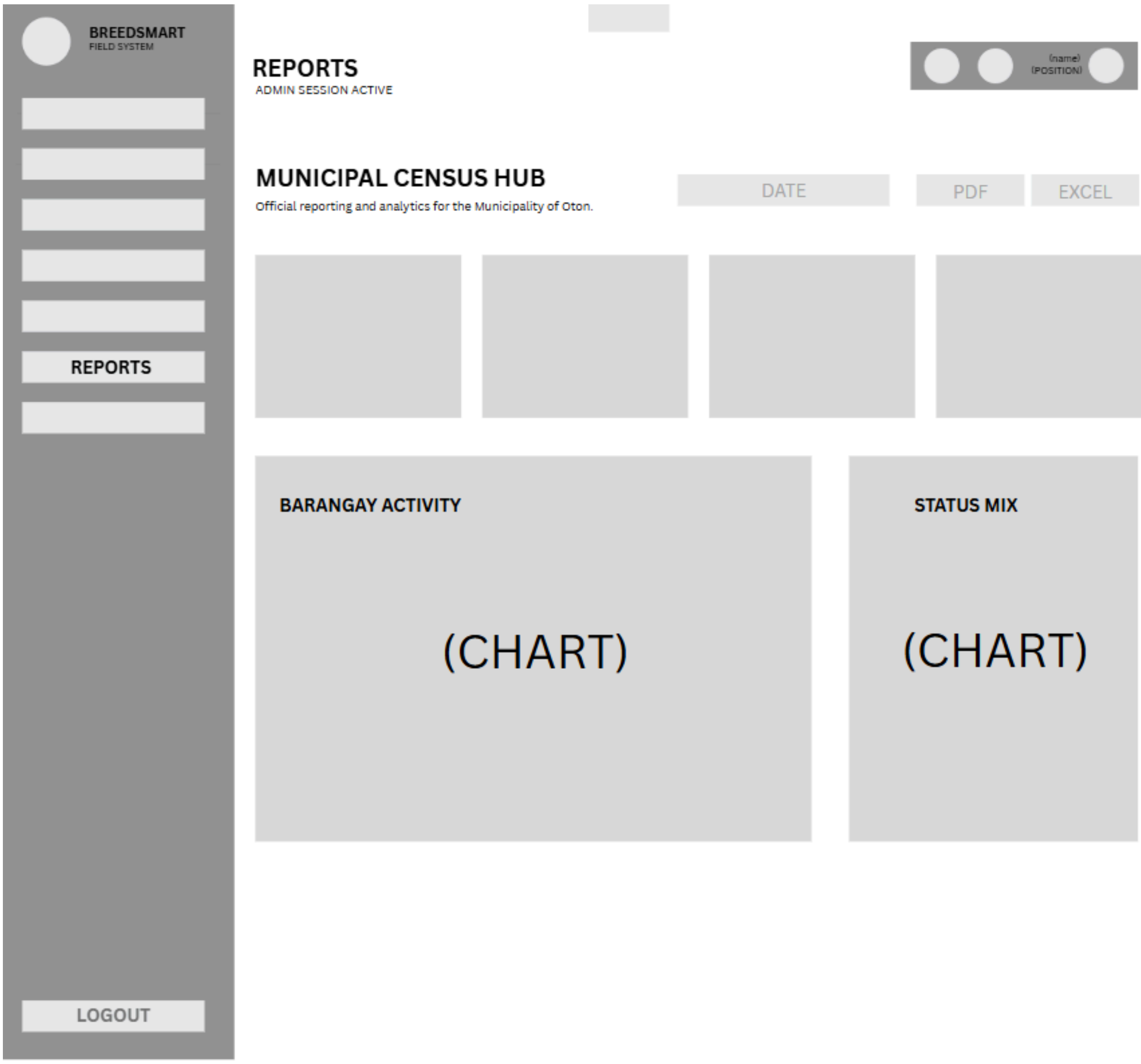


Figure 6:

Shows the analytics and report generation dashboard designed for municipal and provincial reporting. The top area features date-range filters and export buttons (PDF, EXCEL) to compile UNIP-compliant data, while the main canvas renders comparative charts for barangay-level field activities and municipal livestock population status mixes.

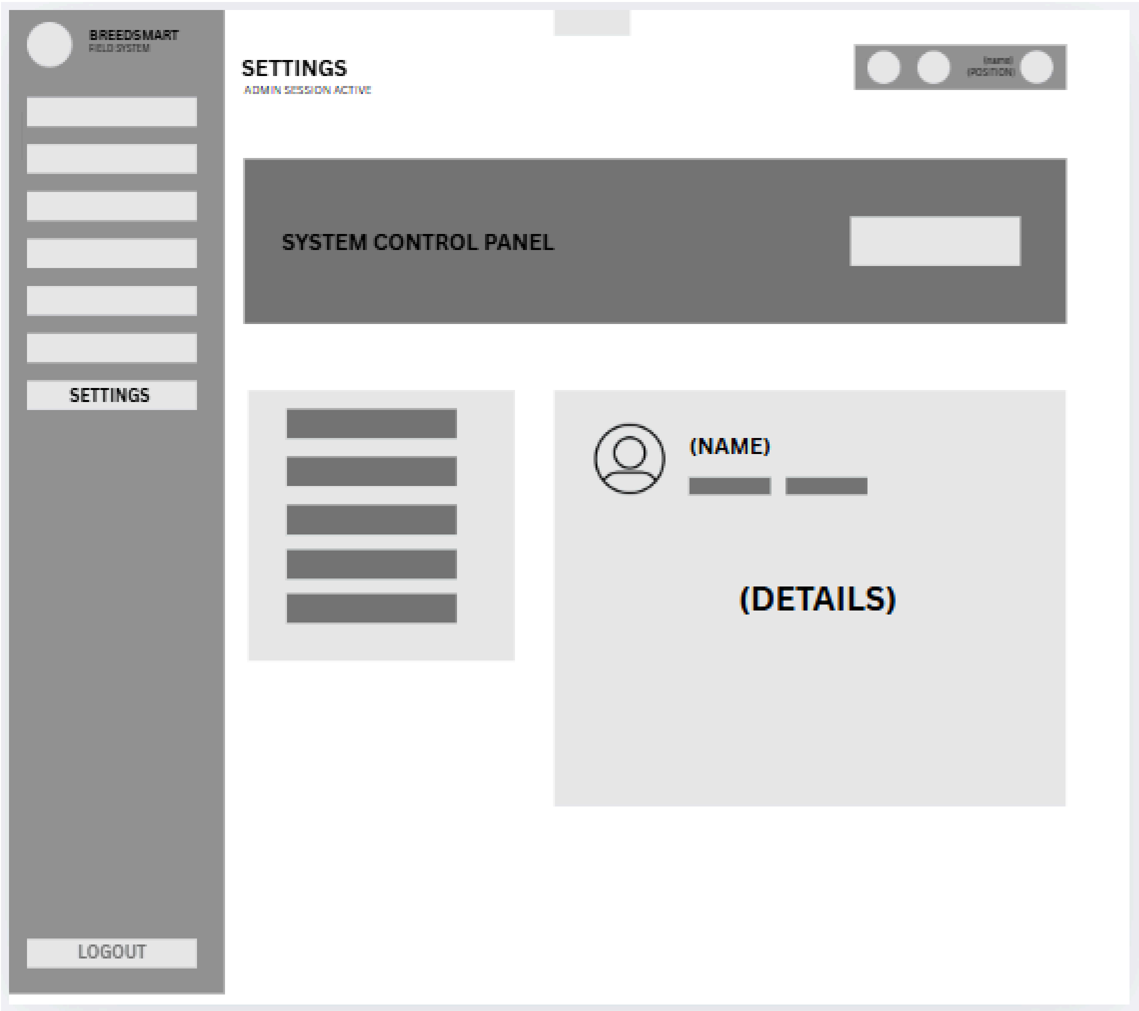


Figure 7:

Outlines the system control panel for administrative preferences and account management. It includes a navigation list for configuring system-wide parameters alongside a detailed administrator profile card displaying active credentials, security settings, and session controls.

Mobile Design

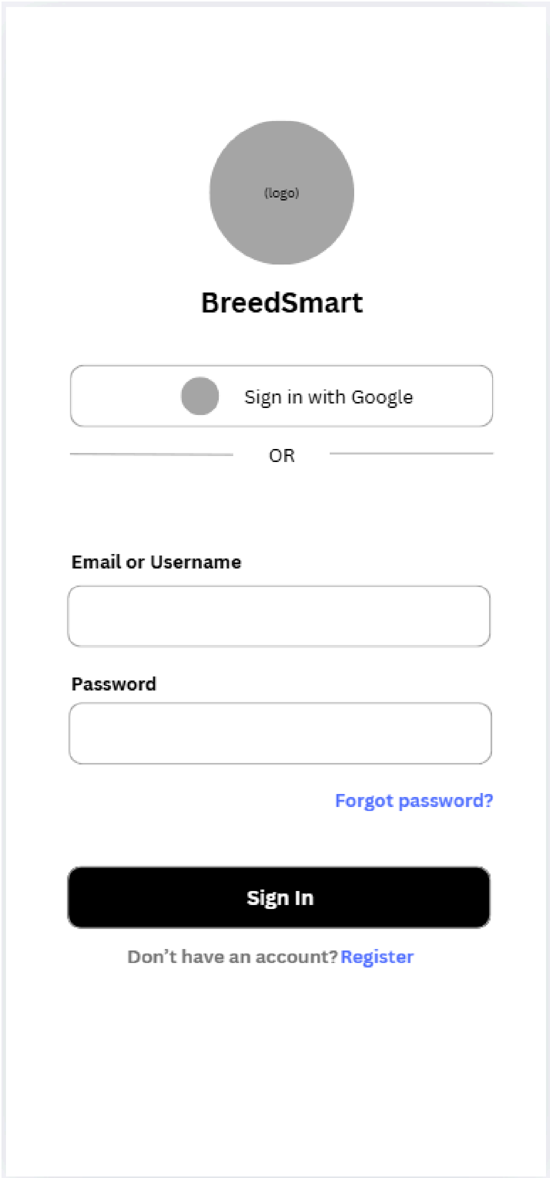


Figure 1:

Represents the primary gateway to the BreedSmart mobile application. It features secure single sign-on via Google authentication, alongside standard credential entry fields (Email/Username and Password). The screen includes account recovery access via "Forgot password?"

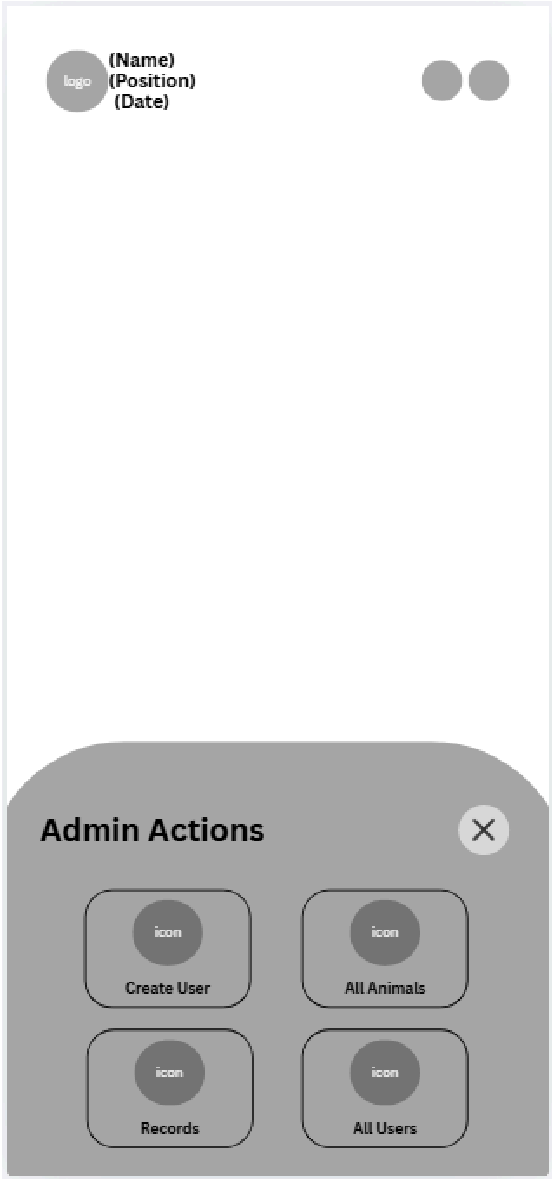


Figure 2:

Description: Illustrates the overlay modal accessible to authorized personnel. It provides rapid navigation triggers to create new user accounts, browse complete livestock profiles (All Animals), view global system records, and manage active system user registries (All Users).

figure 3:

Description: Displays the primary home workspace for field personnel. It highlights active task counters (Missions, AI Cycle, and Clinical), accompanied by quick-action tiles for rapid field logging, including recording artificial insemination,logging health interventions, adding clients,registering livestock, and recording pregnancy and calving milestones

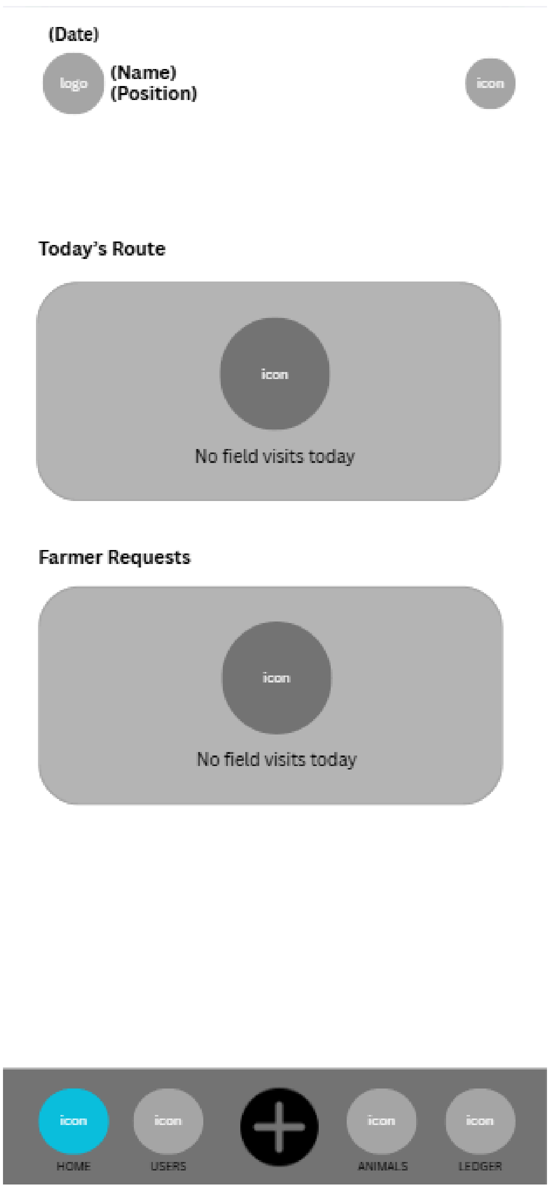


Figure 4:

Outlines the daily dispatch queue for field technicians. It displays two primary content blocks: Today's Route, showing assigned field visits, and Farmer Requests, listing real-time service submissions requiring technician review and scheduling.

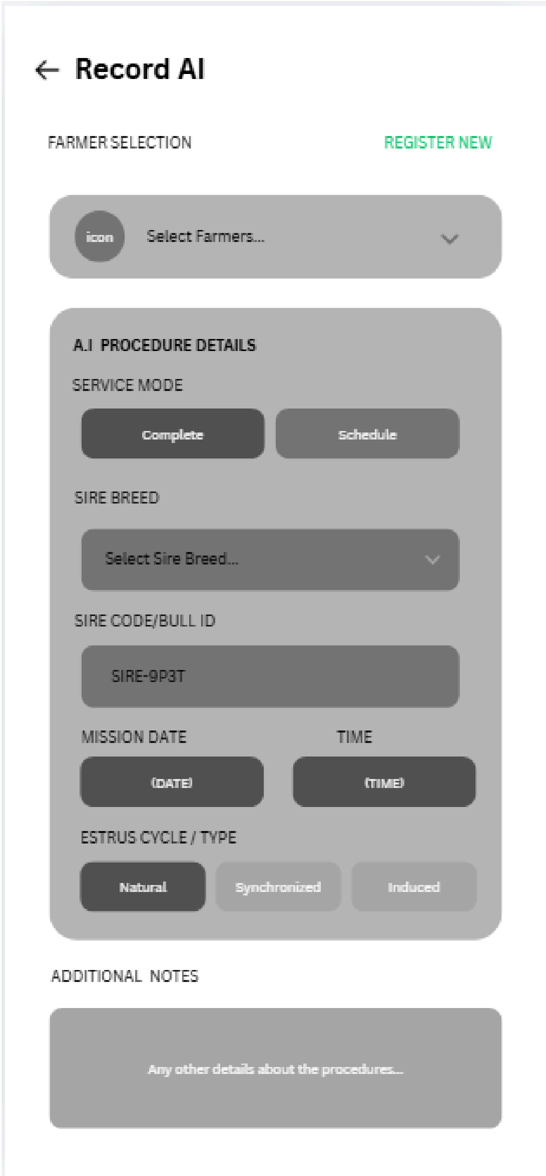


Figure 5:

Shows the form used by technicians to record AI field procedures. It captures farmer selection, service modes (Complete or Schedule), sire breed classifications, sire identification codes, procedure date/time, estrus cycle types (Natural, Synchronized, or Induced), and additional field notes.

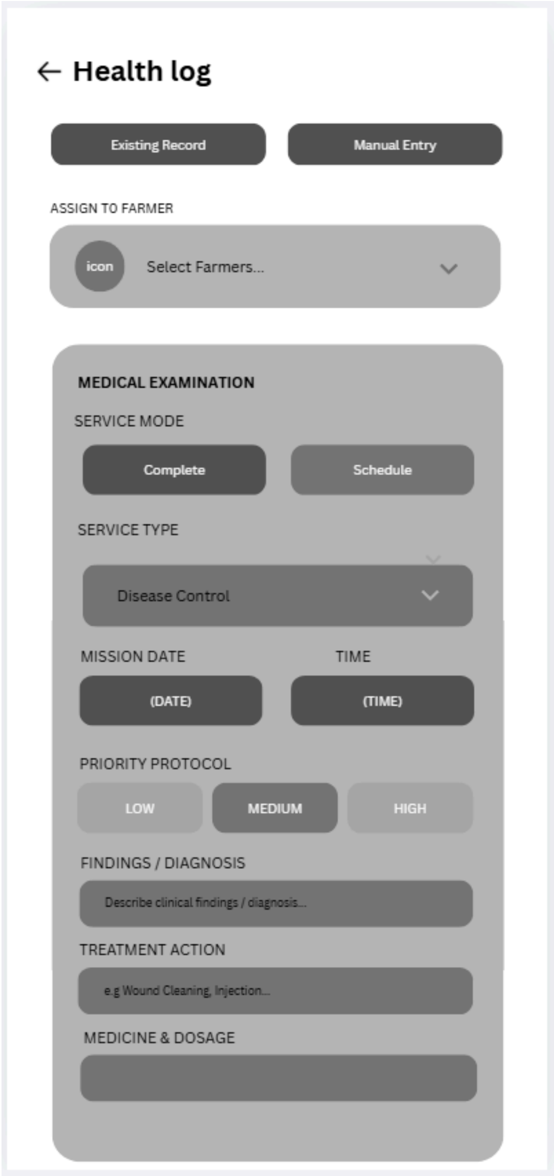


Figure 6:

Depicts the form used to capture animal health treatments. Options allow toggling between existing health records and manual entry, defining priority protocols (Low, Medium, High), logging clinical diagnoses, specifying treatment actions, and encoding exact medicine and dosage parameters.

Figure 7:

Represents the mobile form allowing technicians to manually register new raisers in remote barangays. It captures personal details (First Name, Last Name), contact parameters (Phone Number, Optional Email), and location coordinates down to the specific barangay level.

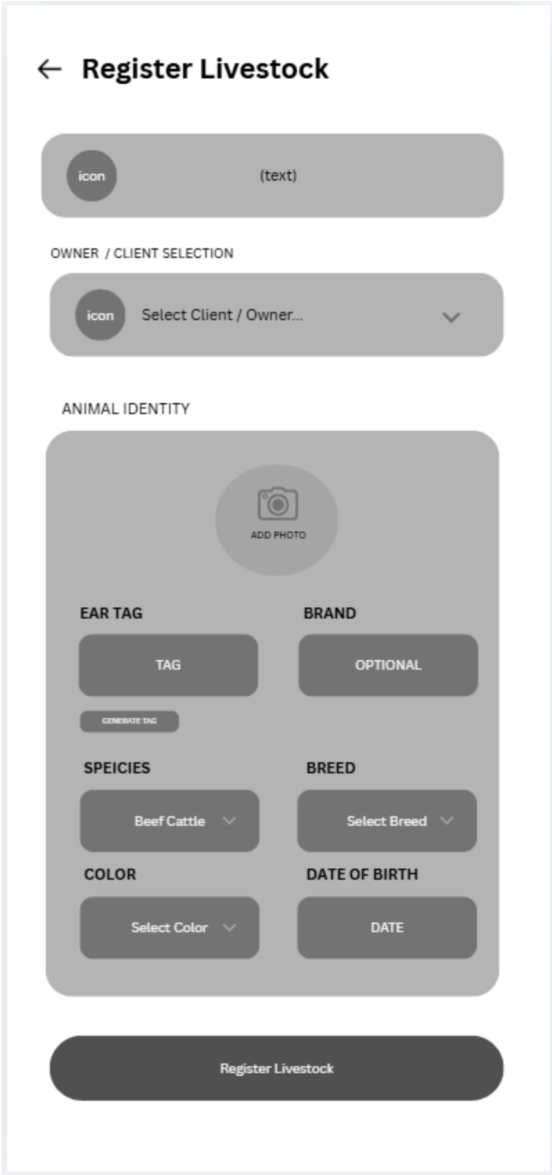


Figure 8:

Illustrates the form for adding new animal assets to a client profile. It includes options for attaching visual asset photos, auto-generating or manually entering ear-tag values, selecting species and breed, specifying color markings, and setting birth dates.

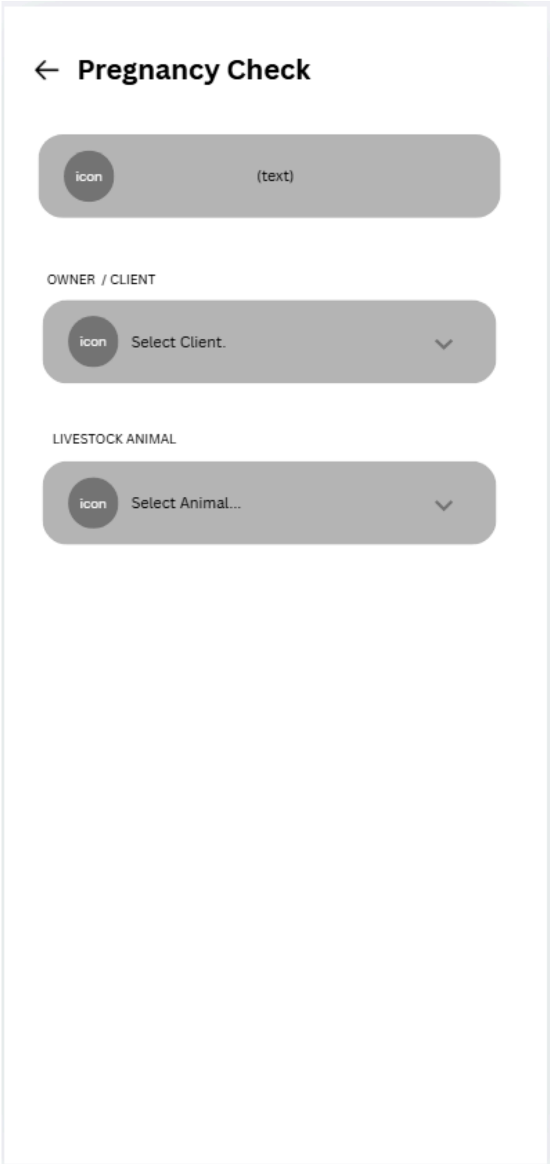


Figure 9:

Shows the workflow for logging follow-up pregnancy checks. Technicians can select the specific owner and animal asset from a dynamic dropdown menu to record physical diagnostic outcomes and update the animal's reproductive status.

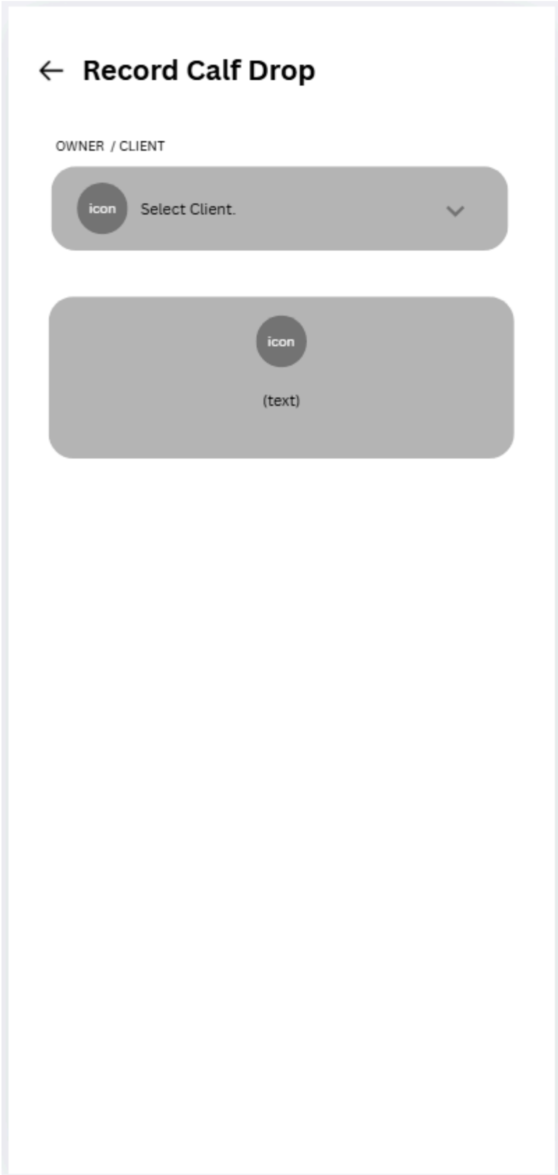


Figure 10:

Displays the screen for recording calving events. It allows field staff to associate a newly delivered calf with its mother (dam), updating municipal birth records and starting automated post-calving recovery trackers.

Figure 11:

Illustrates the mobile client directory for field personnel. It includes a search bar for looking up raisers by name or email, location filters, and verification status indicators for individual client profiles.

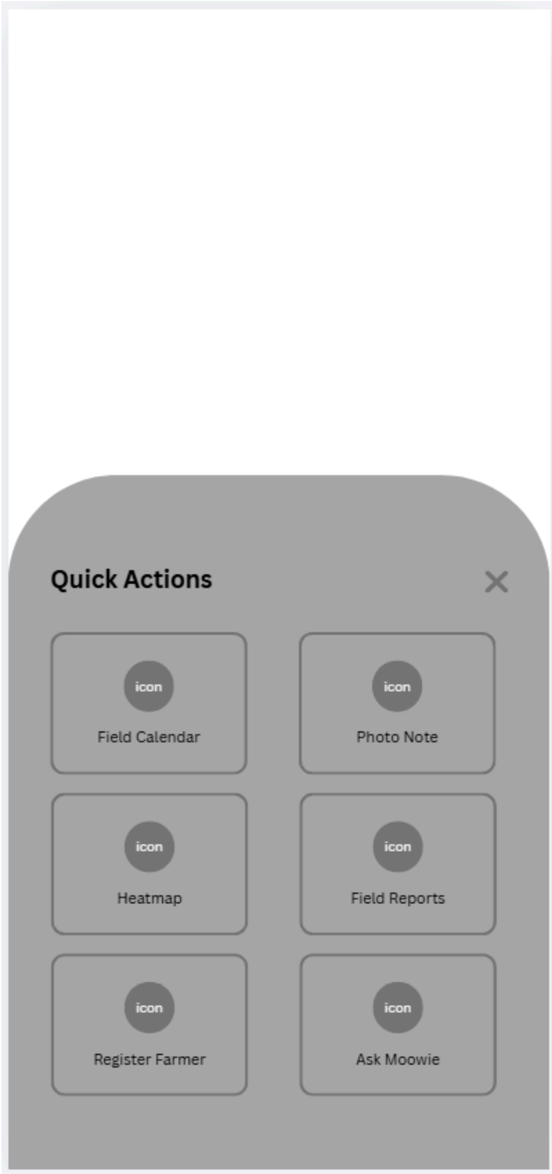


Figure 12:

Shows the secondary quick-actions modal for technicians. It gives rapid access to the field calendar, photo note camera tool, regional epidemiological heatmaps, field summary reports, farmer registration forms.

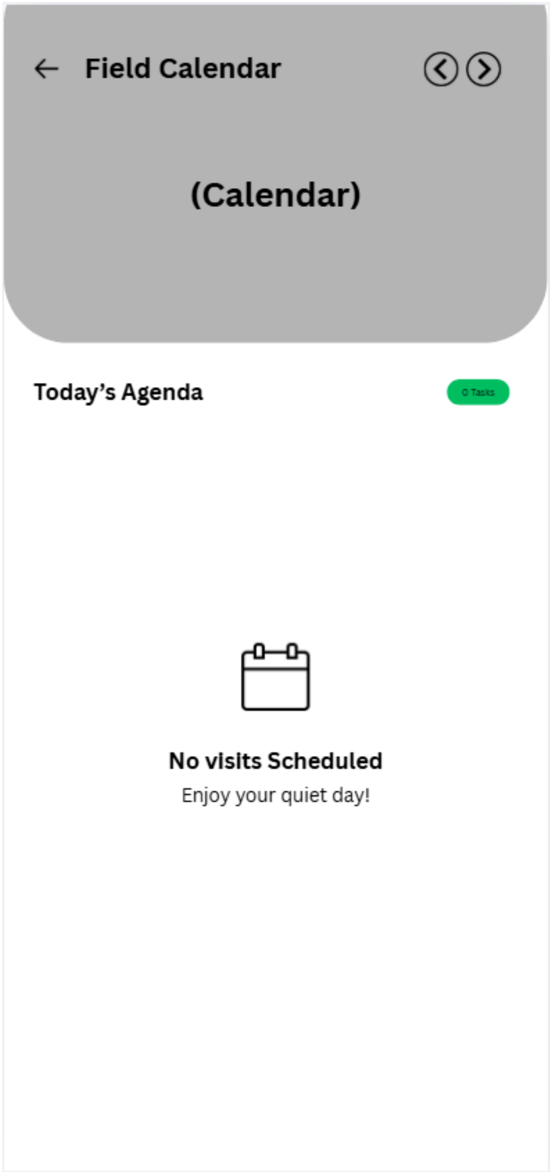


Figure 13:

Displays the interactive calendar interface for technicians. The top area provides date selection, while the lower pane renders Today’s Agenda, detailing scheduled farm visits and veterinary appointments.

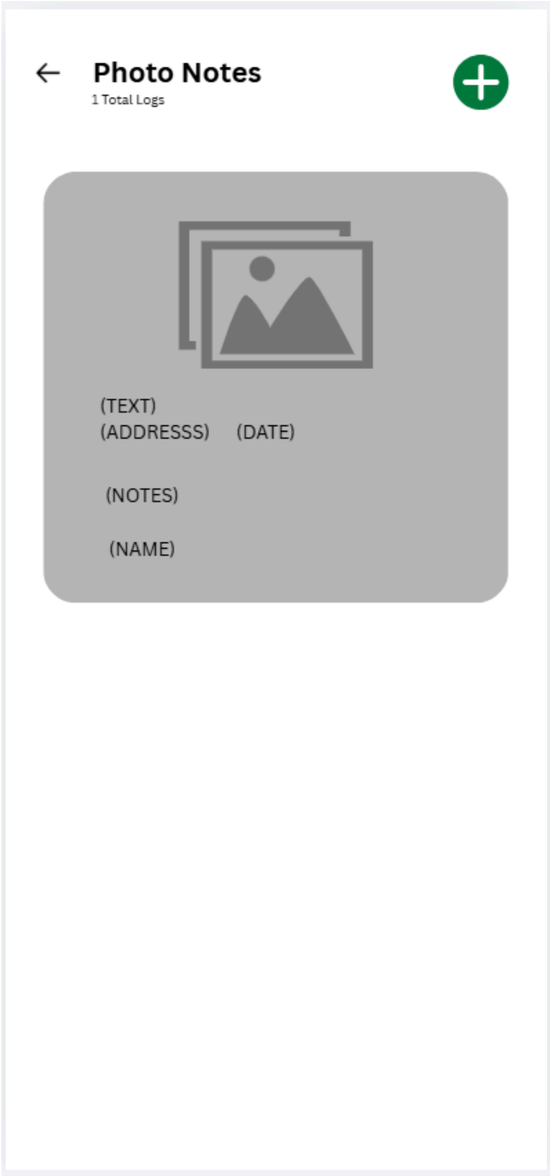


Figure 14:

Depicts the multimedia logging screen where technicians can upload farm photos. Each entry automatically binds image data with address notes, timestamp markers, technician notes, and client identification details.

Figure 15:

Represents the livestock management list accessible to field staff. It features search functionality by tag ID or breed, along with structured card views detailing ear-tag codes, breed classifications, and owner profiles.

Figure 16:

Represents the livestock management list accessible to field staff. It features search functionality by tag ID or breed, along with structured card views detailing ear-tag codes, breed classifications, and owner profiles.

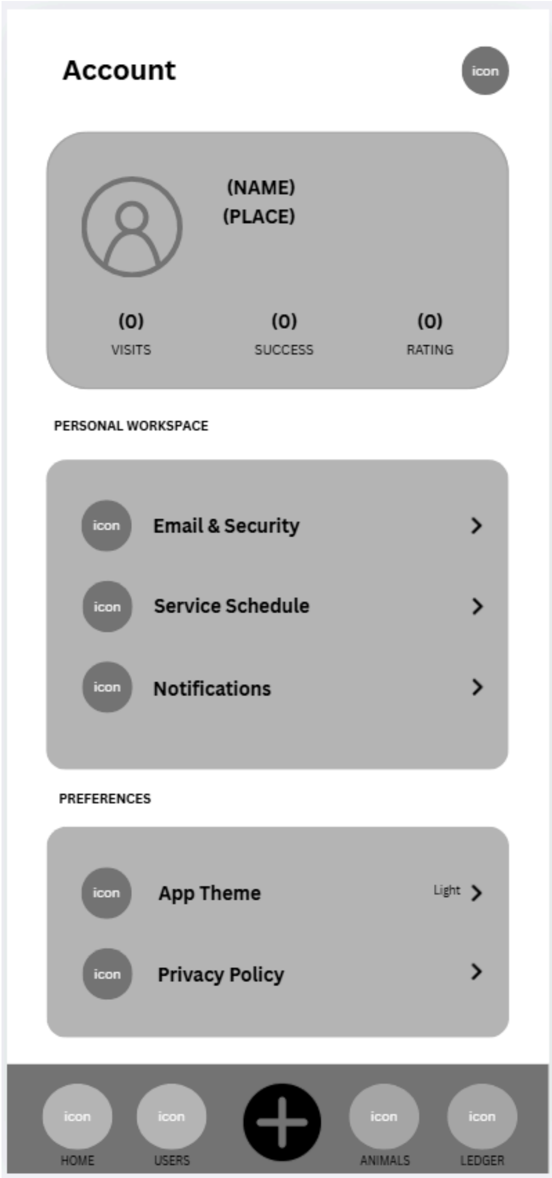


Figure 17:

Outlines the field technician profile page. It displays performance stats (Visits, Success, Rating), along with access to security preferences, service schedules, notification settings, app themes, and privacy policies.

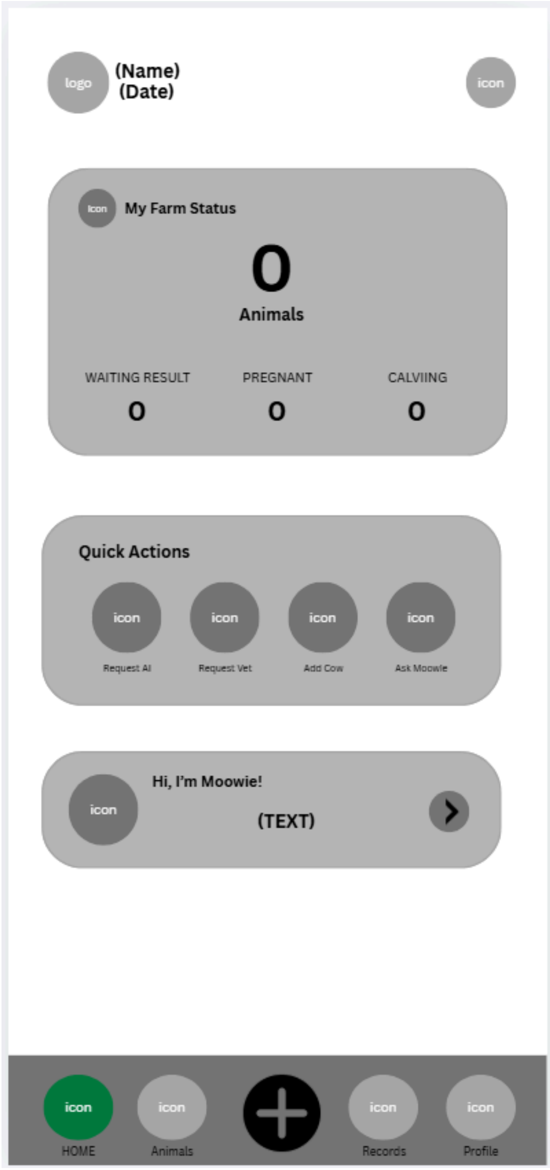


Figure 18:

Represents the home dashboard for livestock raisers. It displays total registered animals, real-time counters (Waiting Result, Pregnant, Calving), core service request and shortcuts

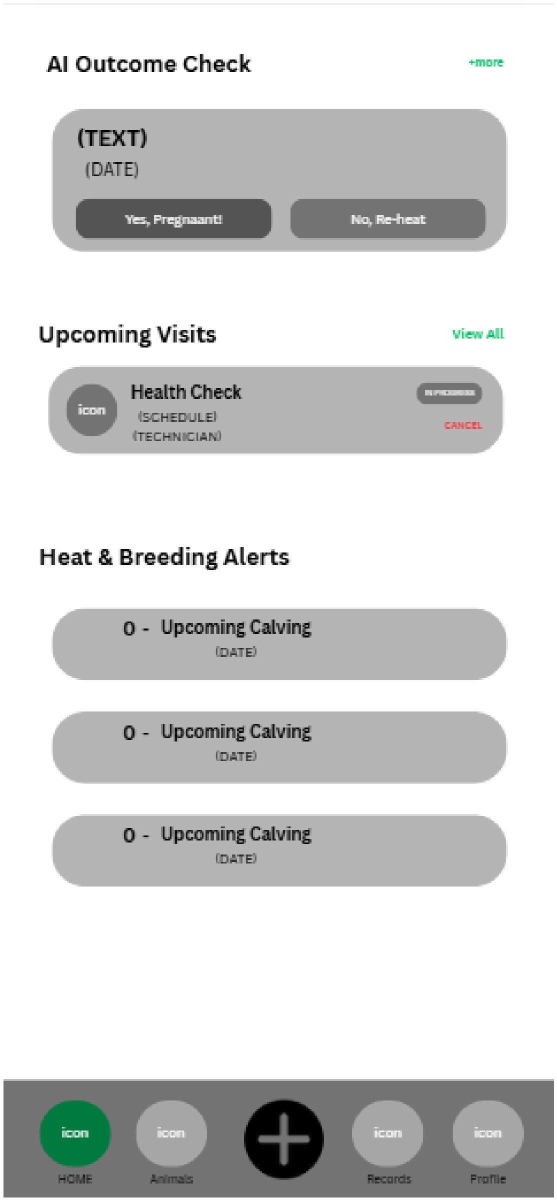


Figure 19:

Illustrates the notification hub for farmers. It allows farmers to confirm artificial insemination outcomes (Yes, Pregnant! or No, Re-heat), track upcoming scheduled visits, and review automated alerts for expected calving dates.

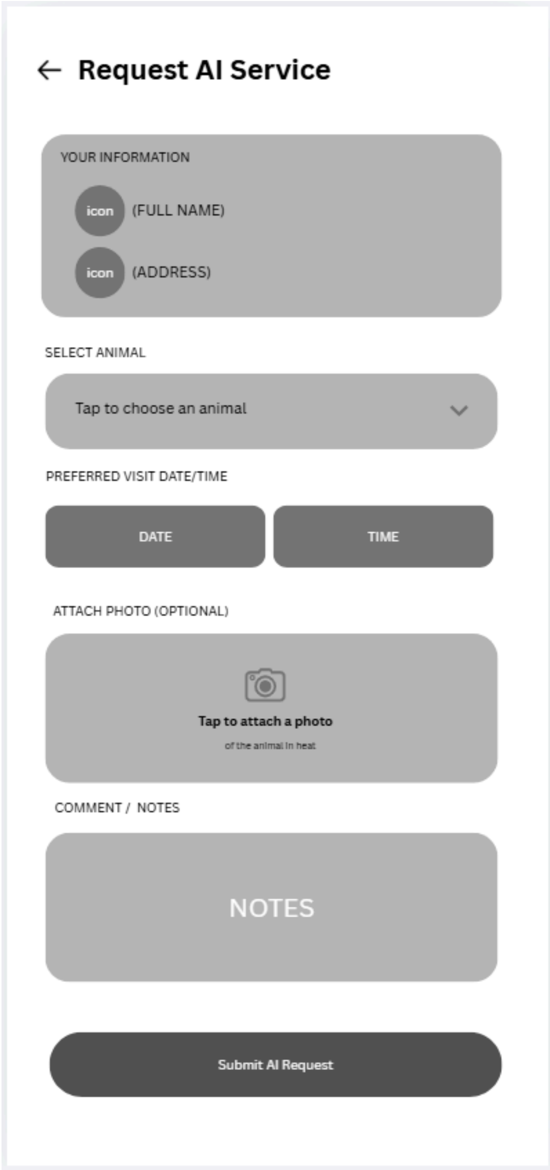


Figure 20:

Depicts the submission form where farmers can request breeding services. It auto-fills user profile information, enables animal selection, captures preferred visit dates/times, lets users attach photos of cows in heat, and accepts custom notes.

Figure 21:

Displays the form used by farmers to report animal health issues. It captures affected animal profiles, request categories (Disease/Infection, Medicine, Checkup, Injury), urgency levels (Low, Medium, Urgent), and preferred visit dates.

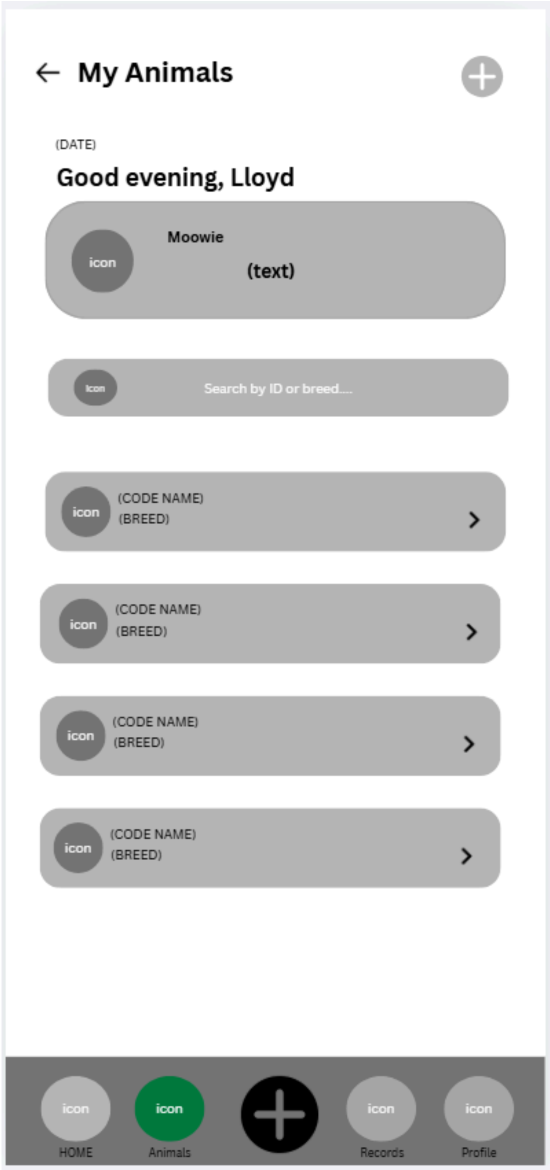


Figure 22:

Shows the personal livestock registry for a registered raiser. It includes a search bar to filter animals by ID or breed, list views of registered cattle cards, and a quick-action button to add new livestock.

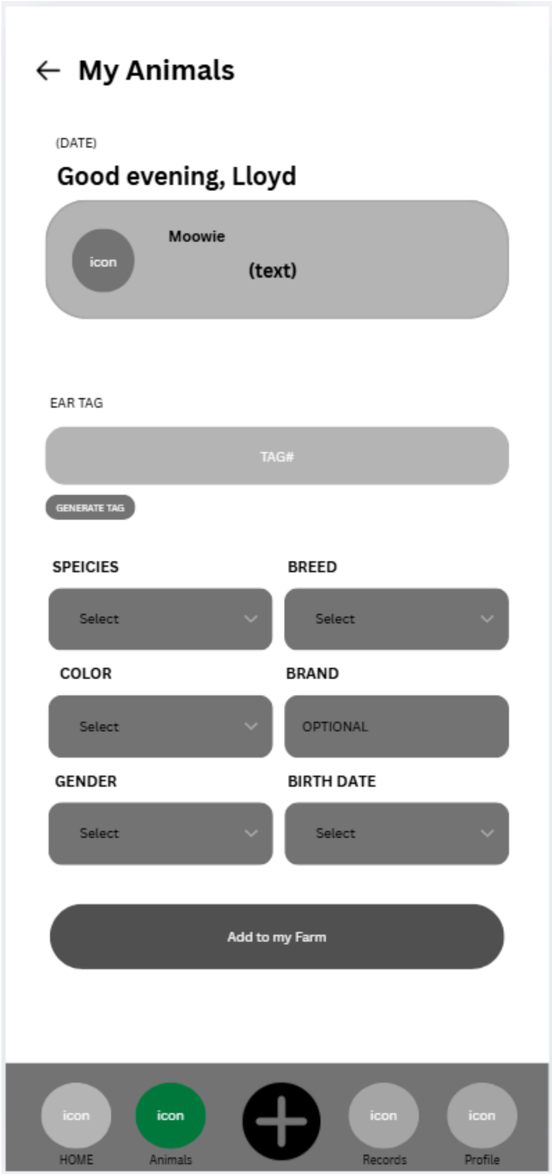


Figure 23:

Represents the form allowing raisers to register a new animal to their farm. Fields include ear-tag ID generation, species, breed, color, brand markings, gender, and date of birth.

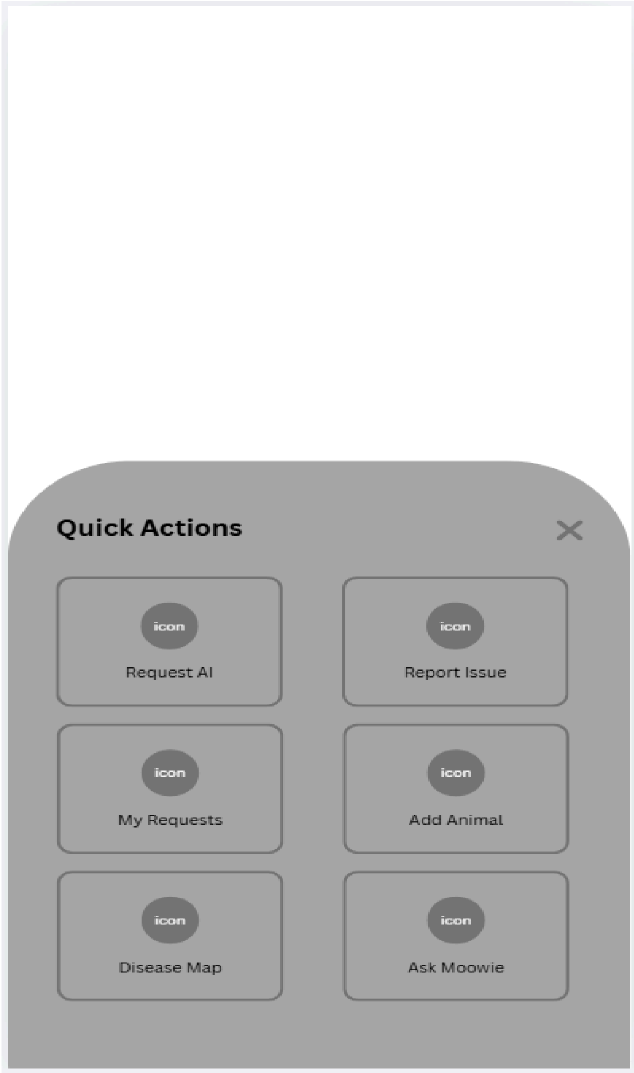


Figure 24:

Illustrates the shortcut menu accessible from the client navigation bar. It provides rapid access to request AI services, report health issues, review active service requests, add animals, and view disease distribution maps.

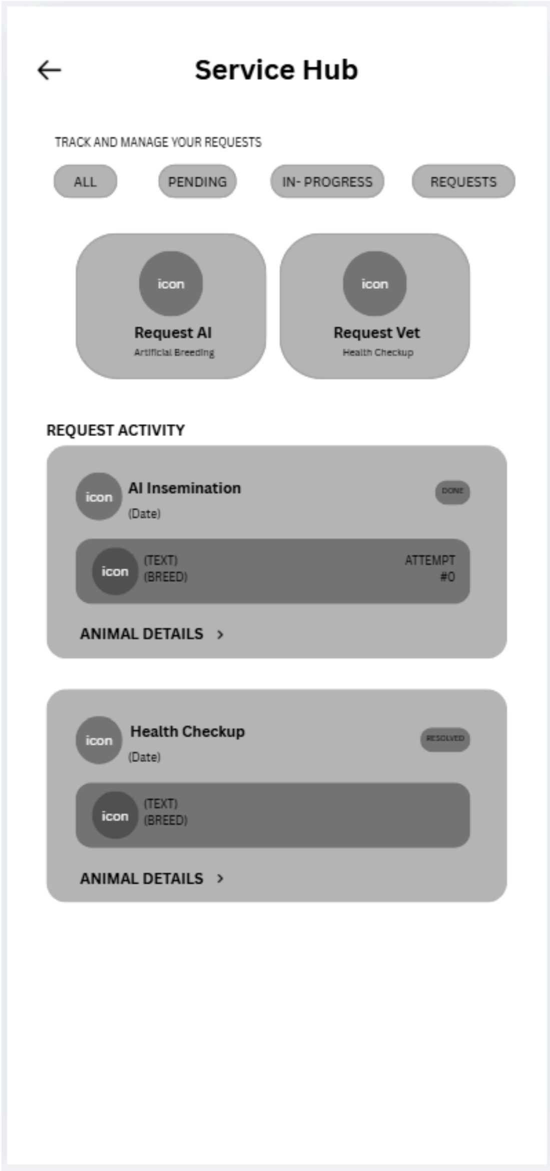


Figure 25:

Shows the service hub where farmers monitor submitted requests. Filter tabs (All, Pending, In-Progress, Requests) let users track artificial breeding and health check statuses in real time.

Figure 26:

Represents the production timeline screen. It displays countdown badges for upcoming milestones, such as Upcoming Calving (e.g., Overdue or Days Left), Heat Watch return notifications, and due dates for pregnancy diagnosis checks.

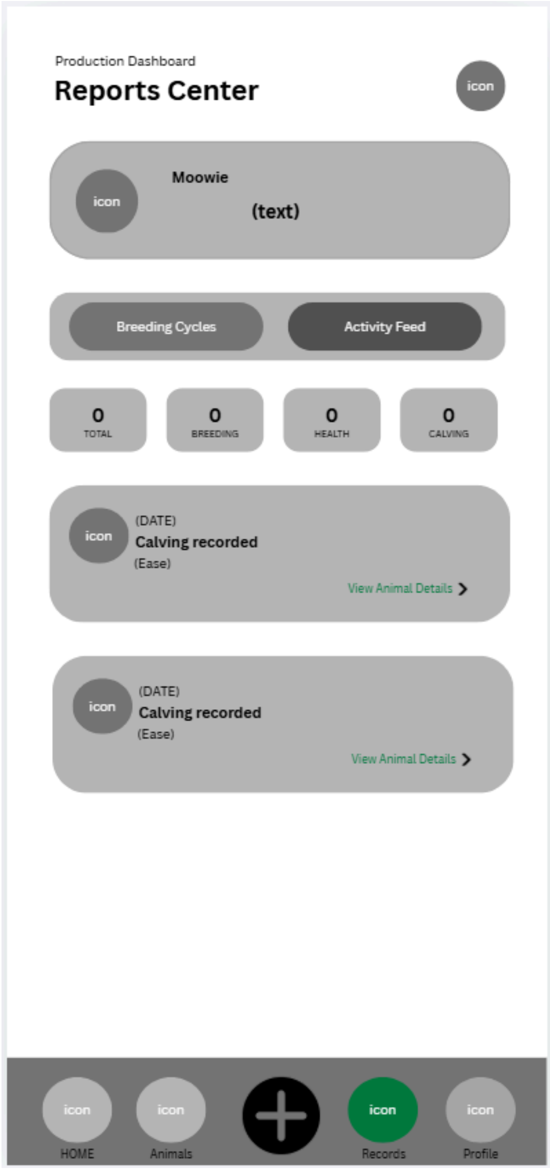


Figure 27:

Illustrates the activity feed tab within the Reports Center. It summarizes total historical counts across Breeding, Health, and Calving events, alongside individual logs detailing recorded birth events and ease of calving.

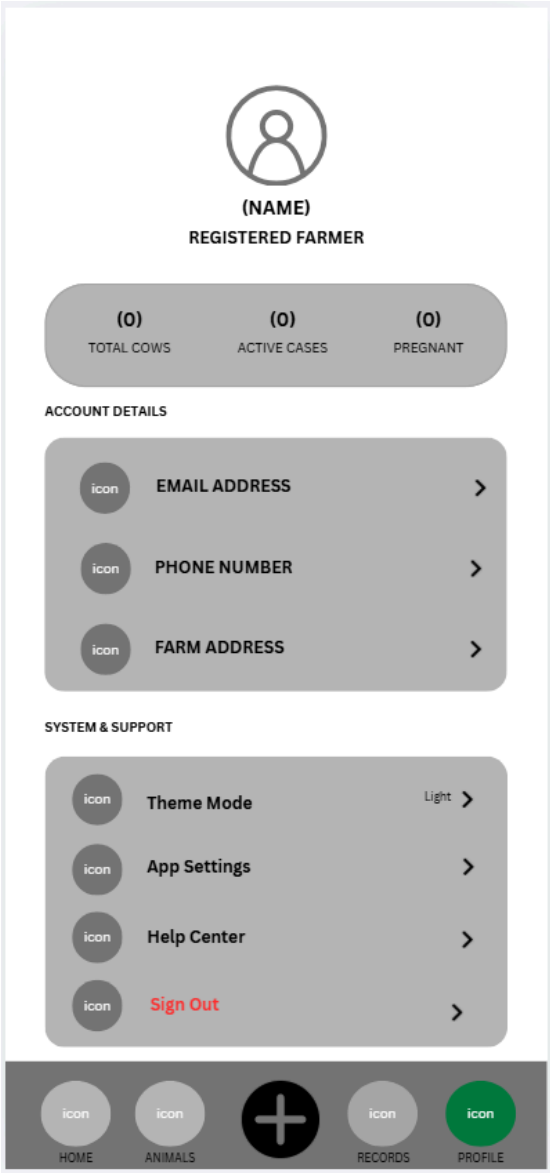


Figure 28:

Shows the client settings screen. It displays overall farm counts (Total Cows, Active Cases, Pregnant), contact details (Email, Phone, Farm Address), app theme toggles, support help center links, and account logout options.

3.13 Network Plan: Technical Architecture

The BreedSmart system utilizes a Cloud-Synchronized Client-Server Architecture designed to support efficient communication, centralized data management, and real-time information exchange between farmers, livestock technicians, and the Municipal Agriculture Office of Oton, Iloilo. The architecture was developed to address the existing challenges experienced in manual livestock monitoring, such as delayed reporting, inaccurate record-keeping, paperwork duplication, and difficulties in monitoring breeding and health activities across different barangays.

The network plan enables users to access the system through both mobile and web platforms while ensuring that livestock records remain synchronized, organized, and accessible within the municipal agricultural environment. Through cloud-based infrastructure and centralized database management, the system improves operational efficiency, strengthens communication between stakeholders, and supports accurate agricultural reporting.

3.13.1 Network Infrastructure & Topology

TThe network infrastructure of BreedSmart follows a multi-layered client-server topology to support secure and reliable data communication throughout the municipality. The architecture is divided into three major layers: the Edge Layer, Transport Layer, and Core Layer.

Edge Layer (Field Access Layer)

The Edge Layer serves as the primary point of interaction between users and the system. Farmers and livestock technicians access the platform through smartphones, tablets, laptops, or desktop computers connected through cellular networks such as 4G and 5G or through available wireless internet connections within Oton, Iloilo.

The mobile application allows farmers to submit insemination requests, report livestock diseases, monitor breeding schedules, and review livestock health records in real time. Livestock technicians use the application during field visits to update insemination procedures, vaccination activities, pregnancy diagnoses, and calving information directly from the farm location. This eliminates the dependence on handwritten records and minimizes delays in updating livestock information.

The web-based dashboard is primarily used by administrators and agricultural personnel within the Municipal Agriculture Office. Through this interface, authorized personnel can monitor livestock records, review technician activities, validate reports, and generate official documentation for submission to the Provincial Agriculture Office.

Transport Layer (Communication and Data Transmission Layer)

The Transport Layer is responsible for transmitting information securely between client devices and the cloud server. BreedSmart utilizes RESTful API architecture combined with HTTPS encryption protocols to ensure safe and reliable communication across the network.

Whenever users submit livestock information, requests, or health updates, the data is securely transmitted to the backend server for processing and storage. The use of encrypted communication protects sensitive agricultural information such as livestock ownership records, insemination histories, vaccination schedules, and reproductive monitoring data from unauthorized access or data interception.

This layer also ensures stable synchronization between the mobile application, web dashboard, and centralized database. By using API-based communication, the system allows real-time updating of livestock records, enabling technicians and administrators to immediately access newly submitted information without requiring manual consolidation of reports.

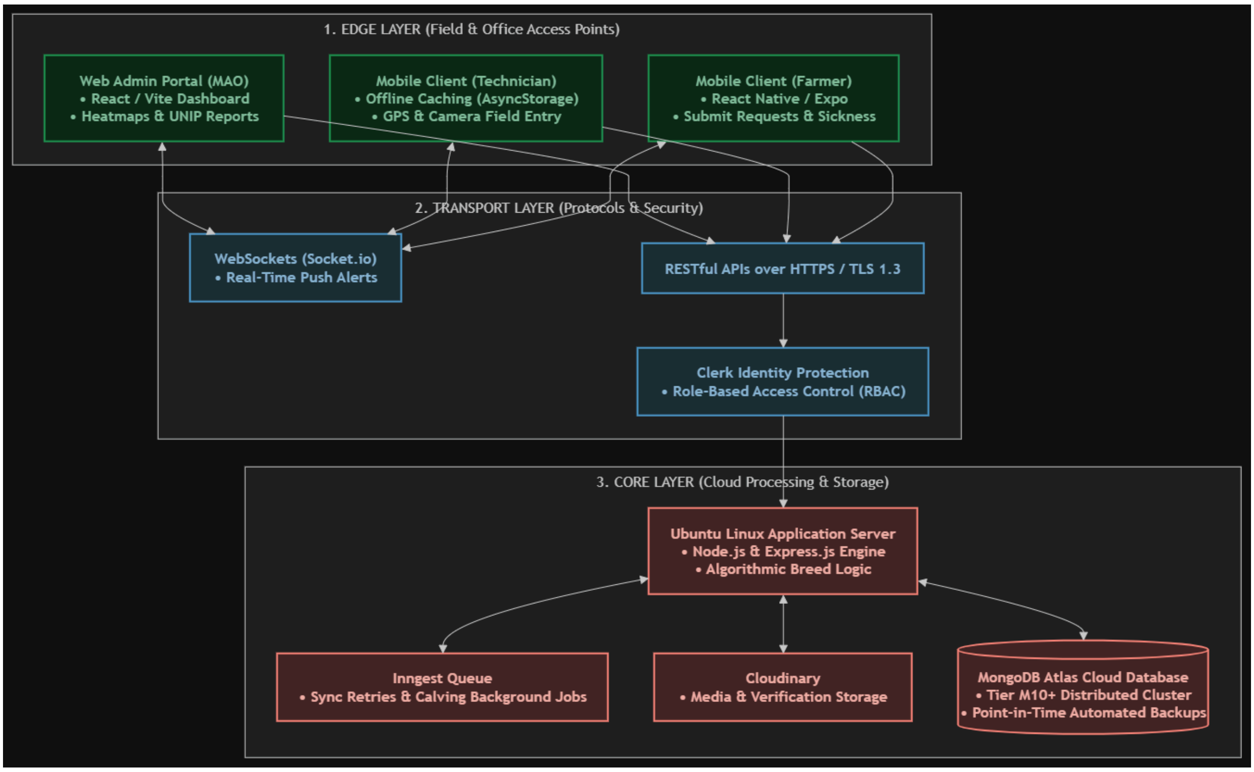
Core Layer (Cloud Server and Database Layer)

The Core Layer functions as the central processing and storage component of the BreedSmart system. The backend infrastructure is developed using Express.js and Node.js, while MongoDB Atlas serves as the cloud-hosted database management system.

The cloud server processes all system operations, including user authentication, livestock management, insemination scheduling, notification handling, pregnancy monitoring, and report generation. It also handles the automated logic used for breed-specific monitoring and date calculations.

MongoDB Atlas stores all livestock-related information, including animal profiles, insemination records, pregnancy data, health treatment histories, health reports, treatment records, calving information, and administrative logs. The use of a cloud database ensures centralized data storage, allowing authorized users to access updated records regardless of their location.

The cloud-based architecture also improves scalability, allowing the system to accommodate increasing numbers of users, livestock records, and agricultural transactions as the municipality continues to expand its digital livestock management operations.



3.13.2 Operational Synchronization Logic

To maintain absolute data integrity and operational consistency across all connected devices and user tiers, BreedSmart implements a sophisticated, multi-stage synchronization workflow. This mechanism is designed not only to automate routine data entry but also to ensure that the "single source of truth" is continuously maintained across the municipal agricultural ecosystem.

The lifecycle begins at the client-side edge, where farmers submit insemination requests, health concerns, or general livestock status updates through the mobile application. The system enforces strict input validation protocols at this point, capturing essential metadata—such as unique animal identification, breed-specific reproductive data, symptoms, preferred service windows, and multimedia documentation like symptom images. For field-based livestock technicians, the application facilitates rapid data encoding during farm visits, covering critical inputs such as vaccination logs, pregnancy diagnostic outcomes, insemination success metrics, and calving details. By digitizing these inputs at the point of origin, the system significantly reduces the "reporting lag" and transcription errors characteristic of manual recording.

Technicians may also directly encode field data during farm visits, including vaccination records, pregnancy diagnoses, insemination outcomes, and calving details.

Upon transmission, the cloud-based server acts as the central validation engine. Before any record is committed to the database, the server-side logic subjects the incoming data to rigorous consistency checks. The system dynamically identifies the breed—such as Brahman, Holstein, or Bisaya—and applies context-aware business logic to cross-reference the animal's reproductive history. This server-side intelligence automatically computes and schedules critical milestones, including:

Automated calculations are then generated for important livestock management schedules, including:

- Reheating or estrus return schedules
- Expected insemination intervals
- Pregnancy monitoring timelines
- Vaccination follow-up schedules
- Expected calf drop dates

By centralizing these computations, the system removes the variability of manual technician estimation, ensuring that every schedule is mathematically precise and aligned with biological breed constants.

Once validated, updated records are immediately propagated across the BreedSmart ecosystem. The system leverages WebSocket protocols to push real-time notifications to relevant stakeholders, ensuring that data availability is synchronized globally.

- Farmers through the mobile application
- Livestock technicians for scheduling and field operations
- Administrators through the web dashboard

This automated flow creates a seamless information pipeline, significantly reducing administrative overhead, eliminating manual reporting delays, and ensuring that municipal decisions are based on the most current data available.

3.13.3 Reliability and Disaster Recovery

The BreedSmart system is engineered with a multi-layered approach to reliability, explicitly designed to transcend the limitations of traditional, paper-based agricultural record-keeping.

Cloud-Based Reliability and Continuity

Unlike traditional logbooks, which are physical artifacts prone to environmental hazards like moisture, thermal degradation, fire, and biological damage, BreedSmart utilizes the robust infrastructure of MongoDB Atlas. This cloud-native approach ensures that data is not merely stored but preserved within a highly available, globally distributed environment. This design guarantees that authorized agricultural personnel maintain secure, 24/7 access to critical livestock records, regardless of whether they are working from a municipal office, a remote farm, or a field site, ensuring business continuity despite local facility disruptions.

The cloud environment provides continuous system availability, allowing authorized users to access livestock records anytime and from different locations within the municipality.

Automatic Backup Mechanisms

Data persistence is safeguarded through a regimented automated backup schedule. These systematic backups create point-in-time recovery capabilities, allowing the system to revert to a previous state in the event of accidental deletion, catastrophic hardware failure, or unintended system interruptions. This robust backup strategy acts as a critical fail-safe for preserving years of accumulated agricultural intelligence, including historical breeding performance, long-term health monitoring, and vaccination trend analysis.

Data Redundancy and High Availability

BreedSmart leverages MongoDB Atlas's distributed clustering model to maintain multiple concurrent copies of the database across different physical nodes. This redundancy ensures that the system architecture remains inherently fault-tolerant; if a single server node encounters technical failure or connectivity loss, the infrastructure automatically fails over to a secondary node. This seamless transition minimizes downtime and ensures that service remains operational for the municipal office and field workers at all times.

Security and Access Control

Security is enforced through a strict role-based access control (RBAC) framework. The system implements granular permissions where farmers, livestock technicians, and administrators interact only with data relevant to their specific roles and authorized jurisdictions. All communication between the mobile client, the web dashboard, and the central database is secured using industry-standard HTTPS protocols, utilizing TLS encryption for data in transit and encryption at rest. This architecture ensures that sensitive agricultural records, user credentials, and municipal statistics are shielded from unauthorized access, cyber threats, and data breaches.

3.13.4 Network Performance and Scalability

The network architecture of BreedSmart is engineered with a modular design philosophy, ensuring it is prepared for both the current operational needs of the Municipality of Oton and the inevitable growth of digital agricultural demands. By utilizing cloud-native infrastructure, the system decouples processing power from local hardware, allowing it to scale dynamically to handle increased user volume, expanding livestock records, and a higher density of agricultural transactions without performance degradation.

Future-Proofing and Technological Integration

- GPS-based livestock monitoring
- IoT-enabled animal health sensors
- GIS-based livestock mapping
- Expanded reporting and analytics modules

By prioritizing a scalable architectural design, BreedSmart ensures that the investment made in today's system provides a resilient and expandable base, capable of evolving alongside the municipality's broader objectives for agricultural modernization and digital transformation.

Chapter 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1. Overview of the Developed System

The BreedSmart System is a web-based and mobile-responsive livestock insemination and health management application engineered to modernize and centralize cattle breeding, health surveillance, and record-keeping operations for the Municipality of Oton, Iloilo. The platform assists the Municipal Agriculture Office (MAO), livestock technicians, and local cattle raisers in tracking artificial insemination schedules, estrus return cycles, pregnancy timelines, vaccination histories, and disease reporting in real time. By transforming traditional paper-based ledgers into synchronized digital repositories, the application enables data-driven decision-making and enhances service delivery across local barangays.

Unlike conventional livestock tracking tools that rely on manual observation or hardware-heavy Internet of Things (IoT) sensors, BreedSmart integrates specialized breed-specific reproductive logic directly into its software architecture. The application processes historical breeding and health inputs to calculate exact reheating windows, expected calving dates, and voluntary waiting periods for breeds such as Brahman, Holstein, and local Bisaya crossbreeds. Unlike conventional livestock tracking tools that rely on manual observation or hardware-heavy Internet of Things (IoT) sensors, BreedSmart integrates specialized breed-specific reproductive logic directly into its software architecture. The application processes historical breeding and health inputs to calculate exact reheating windows, expected calving dates, and voluntary waiting periods for breeds such as Brahman, Holstein, and local Bisaya crossbreeds. Furthermore, the system aims to provide local government units (LGUs) and smallholder cattle farmers with an accessible, low-cost digital solution that improves reproductive efficiency, strengthens biosecurity, and streamlines official reporting for provincial oversight bodies.

The major objectives of the developed system are as follows:

- Automate the recording, storage, and retrieval of municipal livestock records.
- Calculate reheating schedules, pregnancy timelines, and expected calf drop dates usin g automated breed-specific logic.
- Establish a real-time service request and notification between farmers and field technicians.
- Enable online data logging for field technicians in remote farm locations with low cellular connectivity.
- Generate standardized, administrative PDF and Excel analytical reports for the Municipal Agriculture Office.
- Support agricultural personnel and cattle farmers in making data-driven breeding and health management decisions.

The development of the BreedSmart System follows modern software engineering principles by incorporating responsive cross-platform interface design, role-based security access control (RBAC), cloud-backed NoSQL database management, and asynchronous data synchronization. The completed platform consists of three interconnected user layers—Farmers (Clients), Livestock Technicians, and Municipal Administrators—that work together to capture, process, analyze, and visualize agricultural data.

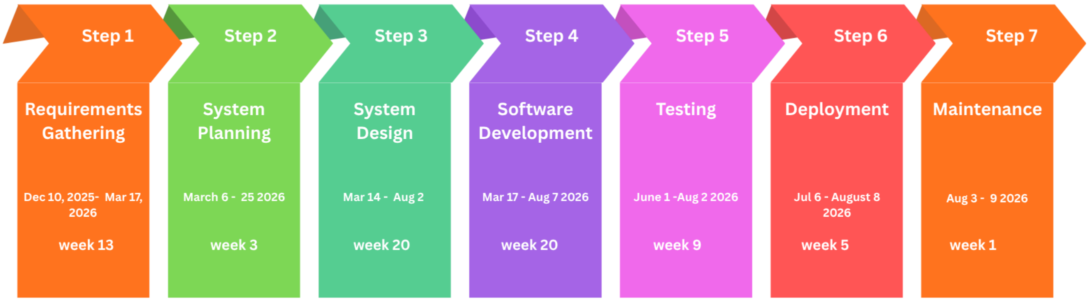


Figure 4.1: System Workflow

4.2. System Development Methodology

Requirements Gathering

The planning phase involves identifying the overall scope of the project and understanding the livestock management situation in Oton, Iloilo. Preliminary site visits and informal interviews with livestock technicians and farmers are conducted to gather initial insights.

During this phase, the researchers also study common issues in cattle management such as manual record-keeping, difficulty in tracking breeding cycles, and delays in reporting livestock health conditions. These observations help define the objectives of the system and establish its relevance to real-world agricultural problems

System Planning

Requirements Analysis Phase

In this phase, the system requirements are gathered and analyzed based on the needs of the target users, which include livestock technicians, farmers, and the Municipal Agriculture Office.

Key requirements identified include:

- Digital recording of livestock information
- Automated tracking of insemination, reheating, and calf drop dates
- Easy communication between farmers and technicians
- Centralized storage of livestock records
- Generation of accurate reports for administrative use

This phase ensures that the system is aligned with actual user needs rather than assumptions, improving its effectiveness and usability.

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System Design Phase

The system design phase focuses on creating the blueprint of the BreedSmart system. This includes designing the user interface (UI), user experience (UX), system architecture, and database structure.

The UI design prioritizes simplicity and accessibility, especially for users with limited technical experience. The system is designed to be mobile-responsive to support farmers who primarily use smartphones.

The database design uses MongoDB to store livestock records, breeding history, vaccination data, and user accounts. The system architecture is structured to separate frontend and backend processes for better scalability and maintainability.

Software Development Phase

The development phase involves the actual coding and implementation of the system. The frontend is developed using **React.js**, while the backend is built using **Express.js** with **Node.js**, and **MongoDB** is used as the database management system.

During this phase, key system modules are developed, including:

- Farmer request module (insemination and disease reporting)
- Technician management module (breeding and health monitoring)
- Automated breeding cycle calculator based on cattle breed logic
- Notification system for reminders and updates
- Administrative dashboard for the Municipal Agriculture Office

This phase follows Agile principles, allowing iterative improvements based on continuous evaluation.

Testing Phase

The testing phase ensures that the system functions correctly and meets user requirements. Different levels of testing are performed, including:

- **Unit Testing** – testing individual system components
- **System Testing** – testing the complete integrated system
- **User Acceptance Testing (UAT)** – involving actual users such as livestock technicians and selected farmers

This phase helps identify errors, bugs, and usability issues before full deployment. Feedback from users is also collected to improve system performance and interface design.

Deployment Phase

In this phase, the system is prepared for deployment and presentation. All documentation is completed, including user manuals and technical documentation.

The system is finalized based on feedback gathered during testing. After validation, BreedSmart is prepared for implementation in a controlled environment within the Municipality of Oton, Iloilo.

This phase ensures that the system is stable, functional, and ready for real-world use in livestock management operations.

7. Maintenance Phase

The Maintenance Phase is an ongoing stage dedicated to ensuring the system's long-term sustainability through iterative improvements driven by user feedback. This phase focuses on refining system performance, addressing technical issues, and adapting to evolving agricultural workflows reported by farmers and technicians. By conducting regular security audits, backups, and providing continuous user support, this phase ensures that BreedSmart remains a reliable, secure, and effective tool that evolves alongside the needs of the Municipality of Oton.

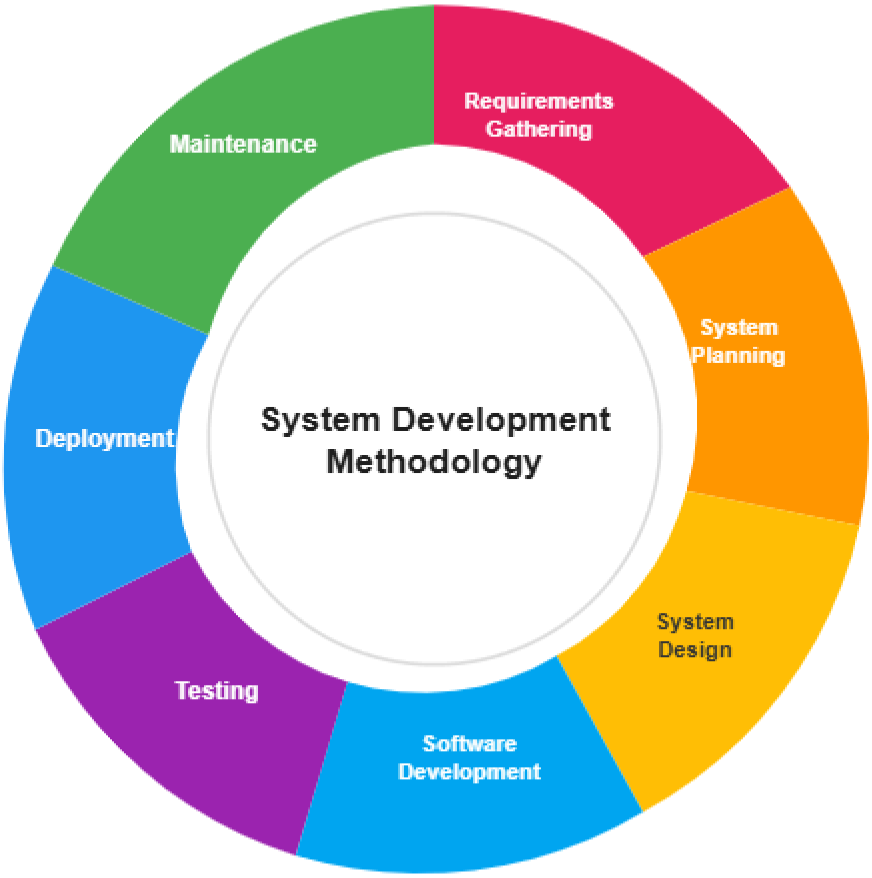
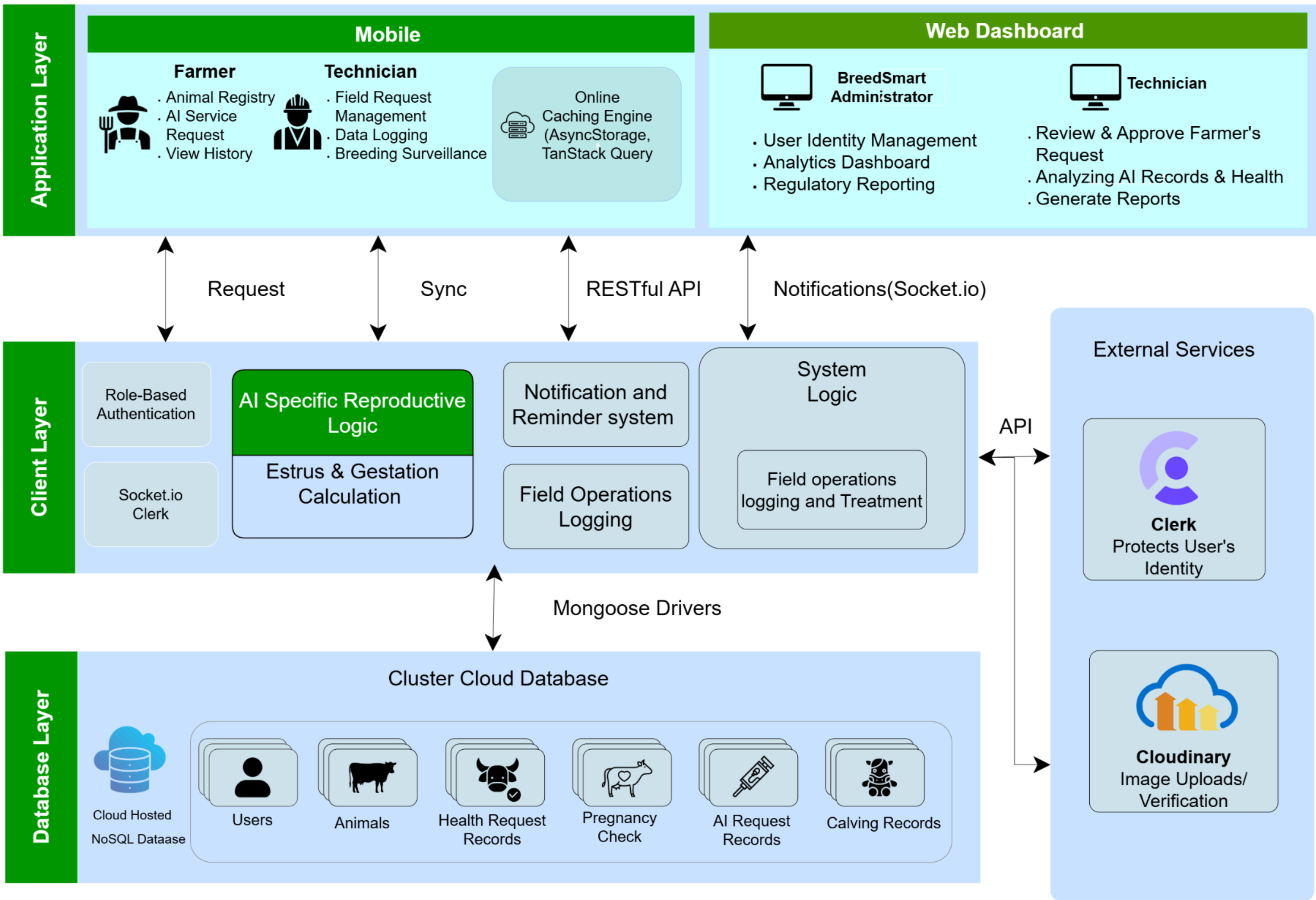


Figure 4.2: System Development Methodology

The process consists of requirements gathering, system planning, system design, software development, deployment, and maintenance to ensure the system meets the needs of farmers, livestock technicians, and administrator

4.3 System Architecture



4.5 Hardware and Software Requirements

4.5.1 Hardware Requirements

- The system requires a desktop or laptop computer for the Municipal Agriculture Office to access the administrative dashboard, featuring a reliable internet connection for cloud synchronization.
- For field technicians and farmers, the system requires smartphones with stable mobile data (4G/5G) or Wi-Fi connectivity to access the mobile application, record data, and receive real-time notifications.
- The system is platform-independent regarding specific hardware models, provided the devices support modern web browsers (Chrome, Firefox, Edge) or the operating systems (Android/iOS) required for the mobile client.

4.5.2 Software Requirements

- The server-side infrastructure requires a cloud-hosting environment (such as MongoDB Atlas) to manage the NoSQL database and host the backend services.
- The web-based administrative dashboard requires modern web browsers (e.g., Google Chrome, Mozilla Firefox, or Microsoft Edge) for full functionality and responsive rendering.
- The mobile application requires Android (version 10 or higher) or iOS (version 14 or higher) to ensure compatibility with React Native and Expo frameworks used for development.
- The development environment requires Node.js and Express.js for API services, and Git for version control.

4.5.1 Hardware Requirements

Components	Minimum Specification	Recommended Specification
Laptop/Desktop Processor	Intel Core i3/ AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5 or higher
Memory /RAM	4 GB	8 GB or higher
Storage	1366 x 768 Resolution	Full HD Display (1920 x 1080)
Monitor	10 Mbps	50 Mbps
Internet Connection	10 Mbps	50 Mbps
Mobile	Android 9.0 (API 28)	Android 13 / iOS 14 or higher

4.5.2 Software Requirements

Software	Purpose
Windows 11 / Ubuntu Linux	Operating System
Visual Studio Code	Source Code Editor
MongoDB Atlas	Database Management System
Node.js / Express.js	Backend Development
ReactJS / React Native (Expo)	Frontend Development
Google Chrome / Edge	Web Browser
Git / GitHub	Version Control
Clerk / Socket.io	Authentication & Real-Time Server Services

4.6 Database Design

Entity Relationship Diagram(ERD)

```
%% ===========================================
%% CENTRAL DATABASE ROOT & COLLECTIONS
%% ===========================================

users {
  ObjectId _id PK
  string clerkId
  string name
  string email
  string phoneNumber
  string role "admin | technician | farmer"
  string status "active | on-site | on-leave"
  boolean isVerified
  string pushToken
  object address
  date createdAt
  date updatedAt
}

animals {
  ObjectId _id PK
  ObjectId farmerId FK "Ref: users"
  ObjectId motherId FK "Ref: animals (Dam)"
  string animalId
  string earTag
```

```
    string species "Beef | Dairy | Carabao"
    string breed
    string gender "Male | Female"
    string reproductiveStatus
    date birthDate
    number parity
    object sireDetails
    array bcsHistory
    array activityLogs
    boolean isVerified
}

inseminations {
    ObjectId _id PK
    ObjectId farmerId FK "Ref: users"
    ObjectId animalId FK "Ref: animals"
    ObjectId technicianId FK "Ref: users"
    ObjectId approvedBy FK "Ref: users"
    ObjectId pregnancyId FK "Ref: pregnancies"
    date inseminationDate
    date preferredDate
    date scheduledDate
    string sireBreed
    string sireCode
    string estrus "Natural | Sync | Induced"
    string status "pending | approved | done | rejected"
    string outcome "Pending | Pregnant | Failed | Re-heat"
    number attemptNumber
```

```
    string technicianNote
}

pregnancies {
    ObjectId _id PK
    ObjectId animalId FK "Ref: animals"
    ObjectId farmerId FK "Ref: users"
    ObjectId inseminationId FK "Ref: inseminations"
    object pregnancyDiagnosis
    date targetCalvingDate
    string technicianNote
}

calvings {
    ObjectId _id PK
    ObjectId animalId FK "Ref: animals"
    ObjectId farmerId FK "Ref: users"
    ObjectId pregnancyId FK "Ref: pregnancies"
    ObjectId technicianId FK "Ref: users"
    date calvingDate
    number numberOfCalves
    string easeOfCalving "Normal | Assisted | Dystocia"
    string technicianNote
}

healthrequests {
    ObjectId _id PK
    ObjectId farmerId FK "Ref: users"
```

```
    ObjectId animalId FK "Ref: animals"
    ObjectId handledBy FK "Ref: users"
    string requestType "disease | medicine | checkup | injury"
    string symptoms
    string urgency "low | medium | high"
    string status "pending | approved | resolved | rejected"
    date preferredDate
    date scheduledDate
    string diagnosis
    string treatment
    string advice
    string technicianNote
}

medicalrecords {
    ObjectId _id PK
    ObjectId animalId FK "Ref: animals"
    ObjectId technicianId FK "Ref: users"
    ObjectId farmerId FK "Ref: users"
    string type "vaccination | deworming | treatment | disease"
    string diagnosis
    string treatment
    string medicine
    string dosage
    date recordDate
    date nextDueDate
}
```

```
tasks {
  ObjectId _id PK
  ObjectId technicianId FK "Ref: users"
  ObjectId farmerId FK "Ref: users"
  array animalIds FK "Ref: animals"
  string taskType "AI | PD | CD | Vaccination | Other"
  string category "Urgent | Routine | Follow-up"
  number priority
  string status "Pending | In Progress | Done"
  string notes
}

fieldnotes {
  ObjectId _id PK
  ObjectId technicianId FK "Ref: users"
  ObjectId farmerId FK "Ref: users"
  string title
  string description
  string farmerName
  string latitude
  string longitude
  string locationName
  string imageUrl
}

notifications {
  ObjectId _id PK
  ObjectId recipientId FK "Ref: users"
```

```
    ObjectId senderId FK "Ref: users"
    string title
    string message
    string type "request_status | reminder | system"
    boolean read
}

configs {
    ObjectId _id PK
    string key
    object value
    string description
    ObjectId updatedBy FK "Ref: users"
}

%% ==========================================
%% DATABASE RELATIONSHIPS / LINKAGES
%% ==========================================

users ||--o{ animals : "owns"
animals ||--o{ animals : "motherId (Dam)"
users ||--o{ inseminations : "farmerId / technicianId"
animals ||--o{ inseminations : "animalId"
animals ||--o{ pregnancies : "animalId"
inseminations ||--o| pregnancies : "pregnancyId"
pregnancies ||--o| calvings : "pregnancyId"
animals ||--o{ calvings : "animalId"
users ||--o{ healthrequests : "farmerId / handledBy"
```

animals ||--o{ healthrequests : "animalId"

animals ||--o{ medicalrecords : "animalId"

users ||--o{ medicalrecords : "farmerId / technicianId"

users ||--o{ tasks : "assignedTo"

animals ||--o{ tasks : "animalIds"

users ||--o{ fieldnotes : "technicianId"

users ||--o{ notifications : "recipientId"

users ||--o{ configs : "updatedBy"

Chapter 5

Conclusion and Recommendations

5.1 Introduction

This Chapter provides all the comprehensive solutions and other necessities mostly required to the development of Breedsmart System. It combines the project results, assesses whether the research objectives were achieved,and offers recommendations for future enhancement and implementation techniques to fully maximize the system's usability for small and medium-size agricultural community and local government units(LGU).

5.2 Summary of Findings

The development and evaluation of the BreedSmart App provides a significant results aligned with the research objectives:

- The system successfully automates the records and storing breeding and health transactions, eliminating manual data entry errors as possible.
- Analytical sections effectively processed the historical data to identify reheating cycles, allowing agricultural officers to easily predict the calving timelines with higher accuracy.
- Visual dashboards provide clear, actionable insights that improve pregnancy, calving and health status by distributing all its necessary details of the animal to the system, directly supporting data-driven decision-making.
- User acceptance testing confirmed that the interface is intuitive, reducing the learning flaw for field operations between technicians and farmers.

5.3 Conclusion

The BreedSmart App has proven to be a viable and effective solution for local government units in Oton, Iloilo seeking to optimize their agricultural efficiency. By transforming raw field records into functional insights, the system empowers municipal officers to make informed strategic decisions regarding artificial insemination scheduling, Health check campaigns, and farmer relationship management. The project addresses the need for an affordable, responsive, and data-centric tool, demonstrating that sophisticated analytics can be made accessible to small-scale farming sectors. The research questions were answered affirmatively, confirming that the system enhances operational efficiency and competitive advantage.

5.4 Recommendations

To ensure the continued relevance and efficacy of the BreedSmart App, the following recommendations are proposed:

System Improvements

- Integrate automated notification alerts for upcoming reheating windows and vaccination dates to further improve field response times.
- Enhance security protocols by implementing two-factor authentication to better protect sensitive municipal and livestock data.
- Develop advanced export features allowing reports to be generated in multiple formats (e.g., PDF, CSV, Excel) for easier sharing and integration with provincial agricultural software.

Future Research Directions

- Explore the integration of machine learning algorithms to enable predictive analytics, such as forecasting future calving success rates based on seasonal data.
- Conduct a longitudinal study to analyze the long-term impact of the system on livestock yield growth and profitability for local farmers.
- Investigate the feasibility of expanding native offline functionality for field technicians operating in areas with prolonged lack of internet connectivity.

Implementation Strategies

- Agriculture officers should conduct training sessions for staff and farmers to ensure consistent data entry habits, which are critical for accurate analytics.
- Local Government Units are encouraged to perform regular data backups and system maintenance to ensure security and prevent data loss.

5.5 Impact of the Study

The BreedSmart App contributes to the IT field by demonstrating the practical application of data analytics in a resource-constrained environment. By providing local agricultural units with accessible business intelligence, this study bridges the gap between complex data processing technologies and everyday farming operations. While the project faced limitations—such as the need for comprehensive historical data to generate accurate behavioral patterns—the results underscore the significant potential for technology to level the playing field for rural communities in an increasingly data-driven market.

5.6 Summary

In summary, the BreedSmart App represents a successful integration of software engineering principles and data analytics. It serves as a testament to the power of tailored technology solutions in addressing real-world agricultural challenges. This research provides a solid foundation for future developers to build upon, contributing to the broader goal of digital transformation among small and medium-sized agricultural communities.

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APPENDICES

APPENDIX A: Comprehensive Software Architecture and System Dependency Documentation

The systematic configuration, deployment, and long-term maintenance of the **BreedSmart** platform necessitate the rigorous documentation of all software dependencies, runtime environments, and framework iterations to guarantee peak performance within the municipality of Oton, Iloilo. In strict accordance with the developmental research framework, this section delineates the technical system requirements essential for the execution of the web-based administrative portal and mobile client applications.

The Mobile Application Subsystem is engineered upon a cross-platform React Native codebase, bundled and compiled utilizing the Expo application framework Version 50.0.0+. To ensure optimal compatibility with the standard smartphone assets utilized by local livestock raisers in rural sectors, the application identifies Google Android Version 9.0 (API Level 28) as the minimum deployment baseline. For high-fidelity client rendering, the system further supports devices operating on Apple iOS Version 14.0 or above. Localized data operations on the client-side utilize the native **AsyncStorage** interface synchronized with **TanStack Query** lifecycle states. This configuration efficiently manages transient memory tasks and the offline transaction queue, enabling resilient field operations even during temporary cellular signal volatility in remote agricultural zones.

The Central Administrative Dashboard handles high-density data aggregation and municipal synchronization within a web-based environment, necessitating specific runtime engines. The web interface leverages React.js built upon a Node.js LTS engine Version 20.x, compiled via Vite automation tools to ensure rapid interface state transitions. The styling architecture utilizes the Tailwind CSS compiler Version 3.4 integrated with NativeWind configurations to render views accurately across mobile and desktop displays. To mitigate visualization discrepancies, office terminals must access the platform through modern V8 JavaScript engine clients, such as Google Chrome Enterprise Version 118+, Mozilla Firefox Extended Support Release Version 115+, or Microsoft Edge Chromium Version 119+.

The Server-Side Cloud Infrastructure is hosted on an Ubuntu Server Enterprise Linux LTS Build 22.04 environment, which facilitates main system services and processes application requests. The API routing layer implements the Express.js framework Version 4.19+ built upon the Node.js runtime core. This backend configuration processes validation logic, executes algorithmic breeding cycle computations, and interacts with the NoSQL database layer through Mongoose drivers Version 8.x. The centralized storage environment resides on a MongoDB Atlas Cloud Enterprise instance Server Cluster Tier M10+, ensuring real-time data synchronization and protection for critical agricultural documentation. External security is managed via Clerk Identity Protection services Version 4.x.

APPENDIX B: Detailed Engineering Specifications for Physical Hardware Components and Devices

To sustain day-to-day operations and prevent database bottlenecks during intensive administrative tasks, the hardware infrastructure must adhere to precise technical specifications. This section describes the minimum and recommended physical equipment requirements necessary for local government workstations and mobile field operations.

The Office Server and Desktop Station is designed to handle heavy administrative workloads, including the compilation of municipal summaries, rendering of complex data tables, and the display of tracking heatmaps for the Oton Agriculture Office. The mandated configuration includes an Intel Core i5 or AMD Ryzen 5 Hexa-Core 64-bit multi-threaded microprocessor with a baseline frequency of 3.20 GHz. The motherboard assembly must accommodate 16 Gigabytes (GB) of DDR4 dual-channel random-access memory (RAM) to support simultaneous server processes. Persistent data storage is facilitated by a 512 Gigabyte (GB) NVMe M.2 Solid-State Disk Drive (SSD) to ensure rapid read-write cycles during large-scale database queries. Visual output necessitates an LED backlit desktop terminal with a minimum screen diameter of 19.5 inches and a native resolution of 1920 by 1080 pixels for optimal dashboard viewing.

The Mobile Field Smart Device represents the primary physical tool utilized by livestock technicians to record insemination data and update health files directly from the field. The smartphone or tablet hardware must utilize an ARM Cortex Octa-Core system-on-chip (SoC) with a minimum processing speed of 2.0 GHz. System logic requires at least 4 Gigabytes (GB) of LPDDR4X system memory (RAM) to maintain background data synchronization. Internal storage necessitates a minimum of 64 Gigabytes (GB) of non-volatile flash memory for application indices, images, and field records. For registration and health reporting, devices must feature a 13-Megapixel digital camera sensor with autofocus capabilities, enabling technicians to capture high-fidelity verification photos of alphanumeric ear tags and individual livestock assets.

APPENDIX C: Detailed Network Topology, Communication Protocols, and Infrastructure Architecture

As **BreedSmart** is engineered as a distributed client-server application, its operational throughput depends heavily on the network framework connecting user touchpoints with the cloud server. This section describes the multi-layered communication pipeline designed to ensure data transit security and synchronization reliability across all barangays in Oton, Iloilo.

The Field Access Layer, or Edge Layer, manages the initial connection points where stakeholders interact with the platform via cellular data. Technicians and raisers connect using 4G Long-Term Evolution (LTE) or 5G New Radio (NR) networks operated by regional telecommunication providers. Mobile data packets are transmitted through the Transport Layer utilizing secure Hypertext Transfer Protocol Secure (HTTPS) formats. This traffic is protected with Transport Layer Security (TLS) Version 1.3 encryption to safeguard sensitive data from interception during transit. For real-time messaging and dashboard updates, the system utilizes the Socket.io WebSocket protocol to maintain persistent connections for automated push notifications.

The Processing and Storage Core Layer facilitates data delivery to the backend API hosted on the cloud server. Upon reaching the server layer, Node.js and Express.js routing middleware validate JSON inputs before executing matching database functions. The storage architecture utilizes MongoDB Atlas, which duplicates records across multiple cloud data centers to ensure backup access during technical anomalies. To maintain optimal system usability, the network infrastructure requires a minimum upload and download bandwidth of 2.0 Megabits per second (Mbps). Packet latency should be maintained under 150 milliseconds (ms) to ensure the accuracy of data tables and the proper synchronization of offline records.

APPENDIX D: Formal Research Transmittals and Project Administrative Correspondence Documents

This appendix contains official communications, request forms, and authorization documents filed by the research team to secure access to local records and finalize the development study. These transmittals serve as formal records of coordination between the academic institution and the local government office of Oton, Iloilo.

The primary transmittal document is the official Request for Field Access and Pilot Implementation Support, submitted to the Municipal Agricultural Officer of Oton on March 11, 2026. Signed by the project researchers, this document formally defines the goals of the capstone project. It outlines the operational boundaries of the pilot testing phase and requests authorization to interview field staff, inspect physical paper ledgers, and conduct evaluation sessions with local livestock raisers.

The document further provides the agricultural agency with a written data privacy guarantee. It specifies that all information collected during field testing—including livestock tallies, ownership registries, and health files—will adhere to strict encryption protocols. This documentation certifies that the **BreedSmart** platform functions as a secure tool for community development and public service enhancement while maintaining institutional research ethics.

APPENDIX E: Software Quality Testing Protocols and User Evaluation Instrument Records

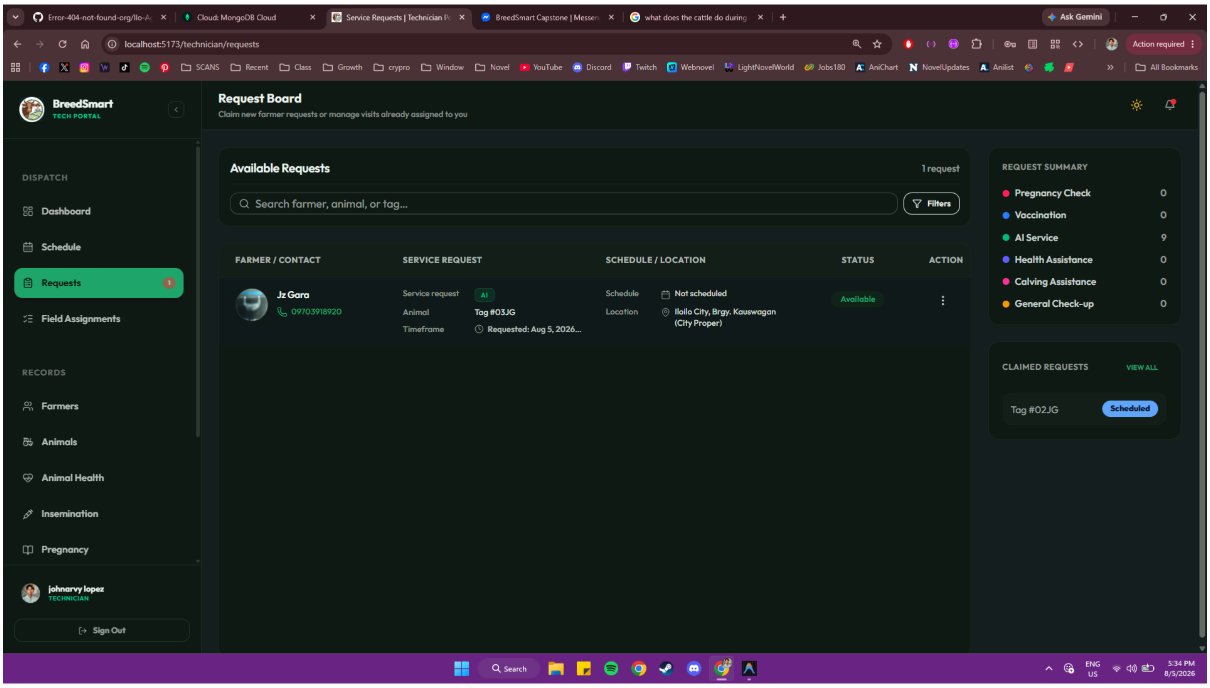
This section contains the official software quality testing forms, verification checklist templates, and evaluation tools utilized to analyze application performance. These instruments measure the system against software quality benchmarks during user acceptance testing (UAT) cycles.

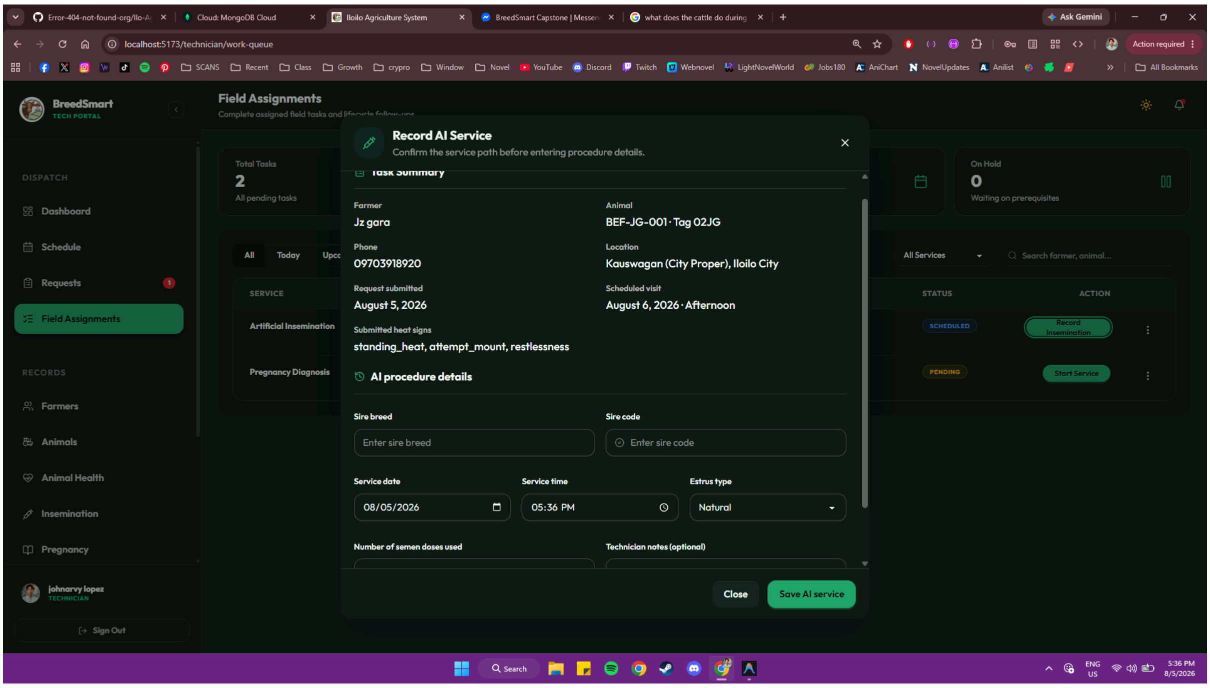
The primary form template is the User Acceptance Testing (UAT) Sign-Off and System Verification Log, used to collect feedback from participants following each testing iteration. The form categorizes reviewers into three explicit roles: Office Administrators, Livestock

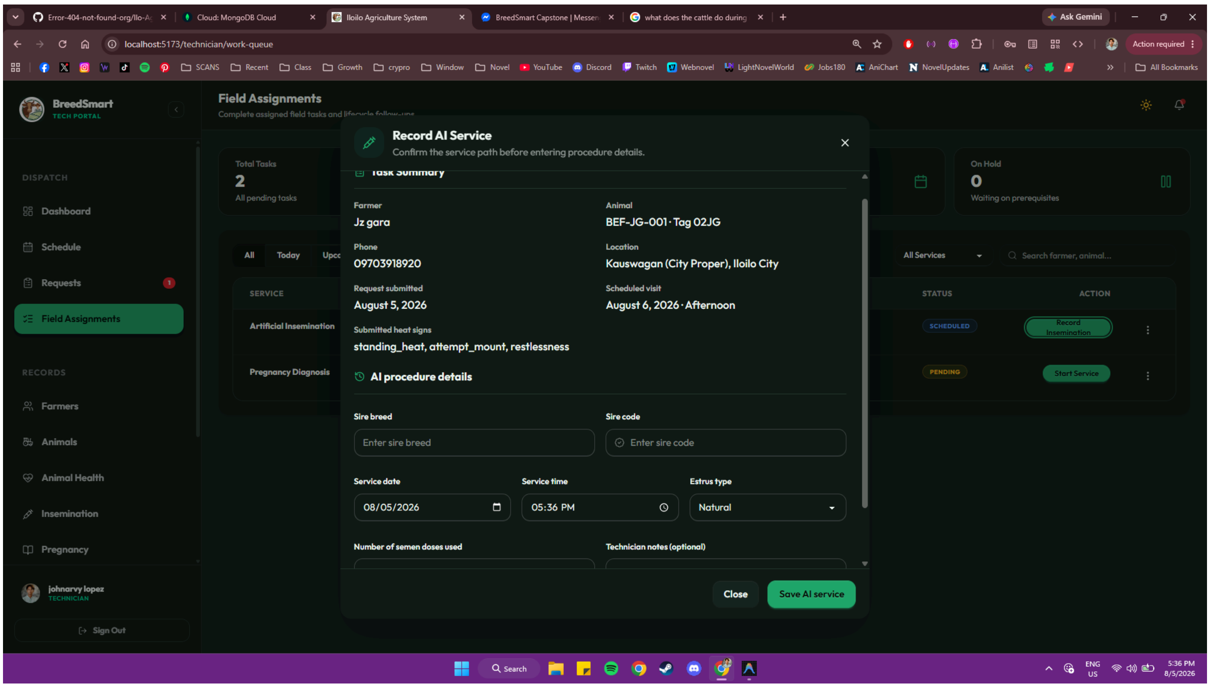
Technicians, and Registered Farmers, ensuring a multi-dimensional evaluation of the user experience. The evaluation table delineates multiple testing scenarios, including user registration, animal identity entry, automated gestation computation, and offline synchronization tests.

Beneath these functional tables, the form includes open-ended sections for user commentary and interface suggestions. This feedback is essential for refining features during iterative development cycles. The form concludes with a signed system confirmation statement, verifying that the application workflows satisfy technical objectives and comply with required ISO 25010 software quality benchmarks prior to final deployment.

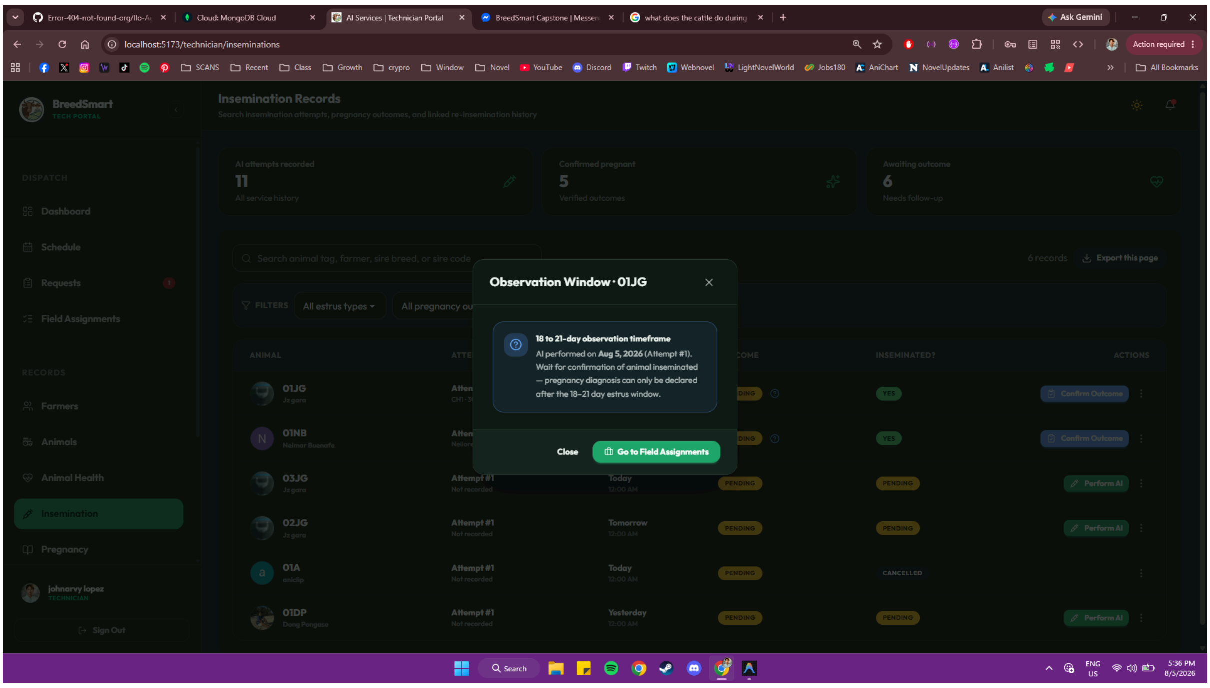
APPENDIX F: Screenshot images



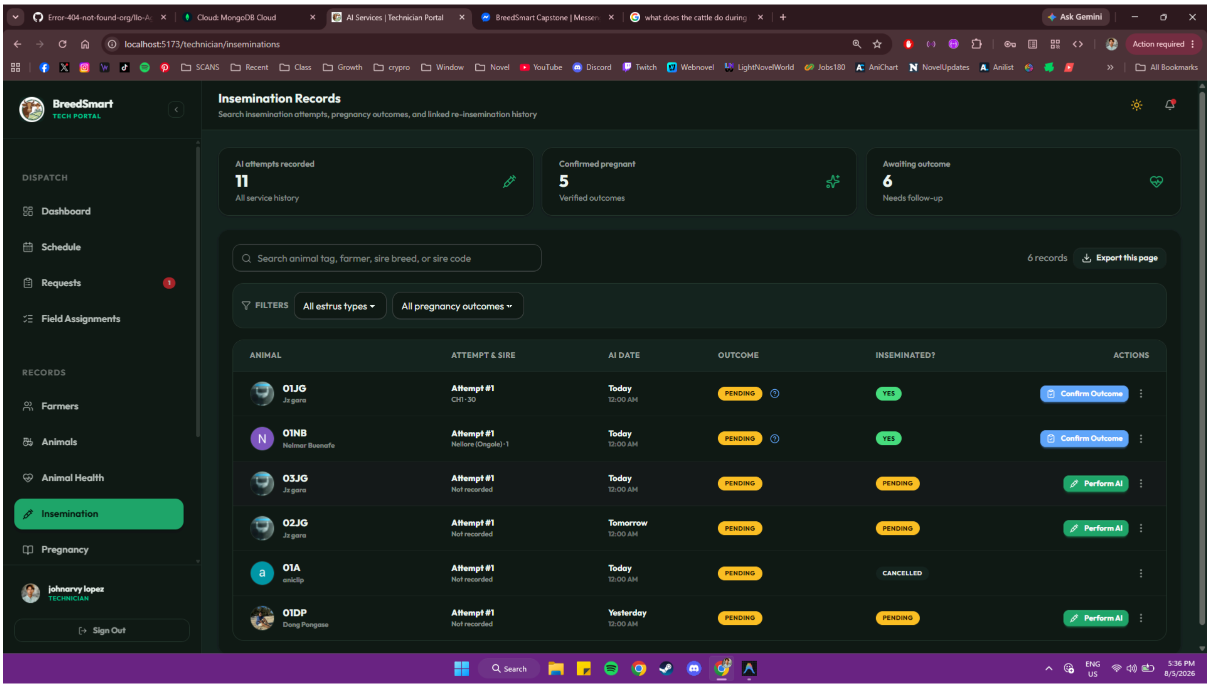




localhost:5173/technician/work-queue
BreedSmart
TECH PORTAL
Field Assignments
Record AI Service
Confirm the service path before entering procedure details.
Farmer
Jz gara
Animal
BEF-JG-001 · Tag 02JG
Phone
09703918920
Location
Kauswagan (City Proper), Iloilo City
Request submitted
August 5, 2026
Scheduled visit
August 6, 2026 · Afternoon
Submitted heat signs
standing_heat, attempt_mount, restlessness
AI procedure details
Sire breed
Enter sire breed
Sire code
Enter sire code
Service date
08/05/2026
Service time
05:36 PM
Estrus type
Natural
Number of semen doses used
Technician notes (optional)
Close
Save AI service
DISPATCH
Dashboard
Schedule
Requests
Field Assignments
RECORDS
Farmers
Animals
Animal Health
Insemination
Pregnancy
johnarvy lopez
TECHNICIAN
Sign Out
Total Tasks
2
All pending tasks
On Hold
0
Waiting on prerequisites
Artificial Insemination
Pregnancy Diagnosis
SCHEDULED
PENDING
Start Service



localhost:5173/technician/inseminations
Insemination Records
Observation Window · 01JG
18 to 21-day observation timeframe
AI performed on Aug 5, 2026 (Attempt #1).
Wait for confirmation of animal inseminated — pregnancy diagnosis can only be declared after the 18–21 day estrus window.
Close
Go to Field Assignments



APPENDIX G:Curriculum Vitae

This section presents the profiles of the five Information Technology candidates specializing in the System Development Track who facilitated the design, analytical modeling, program execution, and technical documentation of the **BreedSmart** ecosystem for the Municipality of Oton, Iloilo. Each curriculum vitae index delineates academic trajectories, specialized technical competencies, and the precise structural responsibilities assigned to each researcher throughout the comprehensive project lifecycle at PHINMA University of Iloilo.

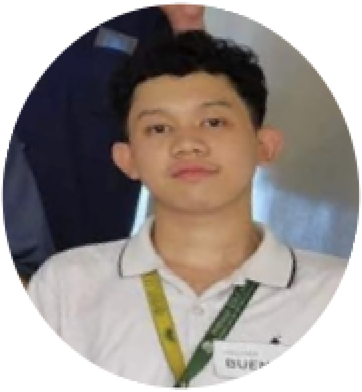
Name: John Lloyd Cabanig

Address: Jaro, Iloilo City, Iloilo

Email: jopo.cabanig.ui@phinmaed.com

Github Link: https://github.com/Johncodexx28

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.
- **Specialization Track Focus:** System Development Track.
- **Core Systems Competencies:** prioritizing project lifecycle synchronization, managing team collaboration, distributed system architectures, cross-functional project scope enforcement, and MERN stack architectural coordination.
- **Project Assignment Role:** Project Manager and Backend Associate; structurally responsible for the strategic maintenance of the project Gantt chart, supervising milestone throughput across the research cluster, and providing technical support in the development of backend services utilizing the MERN stack.



Name: Nelmar Buenafe

Address: Sarah, Iloilo

Email: neba.buenafe.ui@phinmaed.com

Github Link:

https://github.com/NelmarBuenafe

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.

- **Specialization Track Focus:** System Development Track, concentrating on user interface architectural design, experience mapping for localized agricultural environments, and responsive style compilation.

- **Core Systems Competencies:** Tailwind CSS interface configuration, NativeWind mobile property styling, figma model design, and responsive UI prototyping.

- **Project Assignment Role:** Lead UI/UX Interface Layout Architect; structurally responsible for blueprint modeling, designing Low-fidelity wireframes for the administrative web portal and mobile client, mapping navigation logic, and evaluating interface metrics against ISO 25010 usability benchmarks.

Name: John Arvy Lopez

Address: Iloilo City, Iloilo

Email: joza.lopez.ui@phinmaed.com

Github Link: https://github.com/Khaos-s

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.

- **Specialization Track Focus:** System Development Track, specialized in full-stack engineering, database driver logic, cloud architecture scaling, and asynchronous state synchronization.

- **Core Systems Competencies:** React.js/Vite component composition, Node.js/Express.js API services, React Native mobile client engineering, and cloud-synchronized MongoDB Atlas NoSQL querying.

- **Project Assignment Role:** Lead Programmer; structurally responsible for the execution of core system logic, developing algorithmic breed-specific calculation matrices, implementing the offline caching engine, establishing API integration layers, and managing secure database connectivity.

Name: Gian Rovik Somes

Address: Bagacay,San Dionisio, Iloilo

Email: gibu.somes.ui@phinmaed.com

Github Link:
https://github.com/Eunech-ballist23

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.
- **Core Systems Competencies:** Technical research standards, academic writing methodologies, collecting survey reviews, and observation tracking.
- **Project Assignment Role:** Lead Documentor I, responsible for the primary management and synchronization of the core research manuscript, aggregating field research datasets, conducting technical reviews of analytical segments, and supervising the foundational documentation lifecycle.

Name: Justine Balmores

Address: Pototan, Iloilo

Email: usap.balmores.ui@phinmaed.com

Github Link:

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.
- **Specialization Track:** System Development Track / Business Informatics
- **Core Systems Competencies:** Collaborating to research standards, academic writings , technical manuscript synthesis, and longitudinal project tracking.
- **Project Assignment Role:** Lead Documentor II, responsible for the collaborative review and management of the research paper in synchronization with Lead Documentor I, including the maintenance of system flows, executing structural manuscript audits, drafting transmittal templates, and organizing final documentation blocks..